Bi-weekly Report

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Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

Team members

Name	Roll No.	Name	Roll No.
Nithin T M	230709	Shubham Singh	230999
Aditya Nitin Patil	230074	Shivam Dongare	230971
Nithin D H	230708	Sajag Masane	230896

Objectives for this report

1. Process identification and selection of the synthesis route.
2. Specification of the product and the feed streams.
3. Preparation of the block diagram.
4. Input–output cost analysis and preliminary economic feasibility test.
5. Selection and validation of the thermodynamic model.
6. Tuning of the thermodynamic model parameters by regression.

1 Process Identification and Route Selection

1.1 Why aniline has been chosen

Aniline ($C_6H_5NH_2$) is a bulk aromatic amine and it is one of the most important intermediates of the organic chemical industry; about 85 per cent of the world's production is used in the manufacture of MDI (methylene diphenyl diisocyanate), which is further used for rigid polyurethane foam, insulation panels and adhesives. The rest is allocated to rubber processing chemicals, agrochemicals, dyes and pharmaceutical intermediates.

In 2026 the global aniline market had a value of about USD 8.2 billion and it is projected to reach approximately USD 12.2 billion by 2033, with a CAGR of nearly 5.8 %. Asia–Pacific is holding the largest share, close to 39 % of the market, which makes an Indian production facility a reasonable suggestion; India also possesses an existing nitrobenzene manufacturing base, so the feedstock is available domestically.

1.2 The chemistry

The route selected is the vapour-phase catalytic hydrogenation of nitrobenzene over a supported copper or nickel catalyst. The nitro group is reduced in stages to the amine group and water is released as the only main by-product.

Main reaction

$$C_6H_5NO_2 + 3\,H_2 \longrightarrow C_6H_5NH_2 + 2\,H_2O \qquad \Delta H_r \approx -443\ \text{kJ/mol} \tag{1}$$

Side reactions

$$C_6H_5NH_2 + 3\,H_2 \longrightarrow C_6H_{11}NH_2 \qquad \text{over-hydrogenation to cyclohexylamine} \tag{2}$$

$$C_6H_5NH_2 + H_2 \longrightarrow C_6H_6 + NH_3 \qquad \text{hydrogenolysis of the C–N bond} \tag{3}$$

Both the side reactions are proceeding through aniline itself, therefore the overall consumption of nitrobenzene for each side product is one mole, and the water formation stays at two moles per mole of nitrobenzene reacted in each instance; this point was later made use of when closing the material balance.

The reaction mainly releases a large amount of heat. This is the most important feature of the process from the standpoint of design. Our capacity yields a heat release amounting to 17.6 MW (computed in Section 3.5), which is enough to raise a significant quantity of steam, and at the same time it creates a genuine hot-spot and runaway risk within the reactor. These points will be dealt with in the later reports.

1.3 Comparison of the available routes

Route	Catalyst / medium	Selectivity	Remarks
Vapour-phase *(selected)*	Cu or Ni on support, H_2 gas	95–98 %	Continuous, clean by-products, heat recoverable. Hydrogen handling and catalyst coking are the concerns.
Liquid-phase	Pd/C, Pt or Raney Ni in solvent	95–97 %	Easier temperature control, but solvent handling, catalyst separation and more side products.
Béchamp	Iron filings + HCl	85–95 %	Cheap reagents but massive Fe_3O_4 sludge; largely superseded.
Phenol amination	NH_3 at high pressure	—	Economical only where phenol is cheap; not suited to Indian conditions.

We went with the vapour-phase route. It runs continuously, gives the highest selectivity of the four options, produces only water as the bulk by-product, and the reaction heat can be recovered as steam rather than just dumped.

The battery limits start from purchased nitrobenzene, so nitration of benzene is not part of this project. This keeps the mixed acid system, the electrolyte thermodynamics and the spent acid reconcentration outside our scope. Section 5 discusses what the decision costs us economically.

2 Feed and Product Specifications

Stream	Composition / purity	Temp.	Pressure
Aniline product	$\geq$ 99.9 wt%; water $<$ 0.1 wt%; nitrobenzene $<$ 5 ppm	40 °C	1.0 bar
Nitrobenzene feed	$\geq$ 99.5 wt%, technical grade	Ambient	1.0 bar
Hydrogen make-up	99.9 mol%, balance N_2	Ambient	5 bar
Treated wastewater	Aniline $<$ 1 ppm before discharge	40 °C	1.0 bar

The 5 ppm nitrobenzene limit in the product is what really controls the design here — nitrobenzene is toxic and it colours the product besides. This limit is what will drive the aniline column design in Report 3.

3 Design Basis and Material Balance

3.1 Basis

Plant capacity	100,000 t/y of aniline
Operating hours	8000 h/y
Production rate	12.5 t/h
Nitrobenzene conversion per pass, X	99 %
Selectivity to aniline, S_{AN}	95 %
Selectivity to cyclohexylamine	1.5 %
Selectivity to benzene (and ammonia)	3.5 %
Hydrogen to nitrobenzene ratio at reactor inlet	10:1 molar
Recycle gas purity (assumed)	95 mol% H_2, balance N_2

Definition of selectivity used throughout this report. All three selectivities are defined on a *moles of nitrobenzene reacted* basis, that is

$$S_i = \frac{\text{moles of nitrobenzene converted into product } i}{\text{total moles of nitrobenzene reacted}}$$

and by definition they sum to 100 %: $95 + 1.5 + 3.5 = 100$. Since every molecule of nitrobenzene carries exactly one benzene ring and one nitrogen atom, and since both side products are formed from aniline itself, this definition coincides with a carbon basis and with a product-mole basis for this particular reaction network. The three definitions are therefore numerically identical here, but the reacted nitrobenzene basis is the one we have used and it should be quoted as such.

3.2 Equations used

Let F be the nitrobenzene entering the reactor, X the conversion per pass and S the selectivity. Then:

$$\text{Aniline produced} = F\,X\,S_{\mathrm{AN}} \tag{4}$$

$$\text{Nitrobenzene reacted} = F\,X \tag{5}$$

$$\text{Unreacted nitrobenzene (recycled)} = F\,(1 - X) \tag{6}$$

$$\text{Water produced} = 2\,F\,X \tag{7}$$

$$\text{Hydrogen consumed} = 3\,(\text{aniline}) + 6\,(\text{cyclohexylamine}) + 4\,(\text{benzene}) \tag{8}$$

Because the unreacted nitrobenzene goes back to the reactor inlet, the fresh make-up equals what's actually consumed by reaction — not the total flow entering the reactor. We've kept that distinction clear in the table below.

3.3 Overall material balance

Stream / quantity	kmol/h	kg/h
Aniline product	134.22	12 500
Nitrobenzene to reactor inlet	142.71	17 571
Nitrobenzene reacted	141.28	17 395
Unreacted nitrobenzene (recycle)	1.43	176
Fresh nitrobenzene make-up	**141.28**	**17 395**
Cyclohexylamine formed	2.12	210
Benzene formed	4.94	386
Ammonia formed	4.94	84
Water formed	282.56	5 091
Hydrogen consumed	435.15	877
Hydrogen at reactor inlet (10:1)	1427.1	2 877
Hydrogen recycled	991.9	2 000

Carbon closes exactly here: $134.22 + 2.12 + 4.94 = 141.28$ kmol/h, matching the nitrobenzene reacted.

3.4 Specific consumption figures

Nitrobenzene per tonne of aniline	1.392 t
Hydrogen consumed per tonne of aniline	70.2 kg
Hydrogen purchased per tonne of aniline	71.6 kg
Water produced per tonne of aniline	0.407 t
Benzene formed per tonne of aniline	30.9 kg
Ammonia formed per tonne of aniline	6.7 kg
Cyclohexylamine formed per tonne of aniline	16.8 kg

3.5 Hydrogen loop — make-up, recycle and purge

The hydrogen figures in the table above describe what happens across the reactor only. Since a purge is unavoidable, the make-up requirement is not the same as the consumption, and the closure of the loop is worked out here explicitly.

Nitrogen enters at 0.1 mol% in the make-up and it does not react. At steady state it has to leave at the same rate, and the purge is the only exit available. Taking the recycle gas purity as 95 mol% H_2, the two balances are:

$$\text{Nitrogen:} \quad 0.001\,M = y_{N_2}\,P \tag{9}$$

$$\text{Hydrogen:} \quad 0.999\,M = 435.15 + y_{H_2}\,P \tag{10}$$

where M is the fresh make-up and P the purge, both in kmol/h. Solving the two together gives $P = 0.02\,M$ and hence:

Quantity	kmol/h	kg/h
Fresh H_2 make-up, 99.9 mol%	444.0	895
Hydrogen consumed by reaction	435.2	877
Hydrogen lost in the purge	8.4	17
Gas leaving the flash separator	1044.2	—
of which nitrogen	52.2	—
Purge stream	8.9	—
Recycle gas returned to the reactor	1035.3	—
Check: H_2 at reactor inlet	1427.1	2 877

The loop therefore closes: $1035.3 \times 0.95 + 444.0 \times 0.999 = 1427.1$ kmol/h, which is exactly the 10:1 ratio on the 142.71 kmol/h of nitrobenzene entering the reactor. The purge is only about 2 % of the make-up and it carries away roughly 1.9 % of the hydrogen purchased.

Purge loss. On a purchased basis the hydrogen requirement is 71.6 kg per tonne of aniline against 70.2 kg/t of pure reaction consumption, so about 2 % of the hydrogen bought leaves with the purge without ever reacting. In the cost analysis of Section 5 we have used the consumption figure; using the purchase figure instead would add about Rs. 259 per tonne to the raw material cost. This shall be carried properly once the recycle loop is converged in Aspen.

The assumed recycle purity of 95 mol% is itself a design variable. A tighter purge raises the hydrogen recovery but allows nitrogen to build up, which lowers the hydrogen partial pressure at the reactor inlet and hurts the rate. The optimum is to be found by a sensitivity study in Report 4.

3.6 Reaction heat duty

The heat release has been computed from the actual reaction extents rather than quoted as a round figure. All the nitrobenzene reacted passes first through the main reaction, and the two side products are then formed from aniline, so the extents are:

Reaction	**Extent (kmol/h)**	ΔH_r **(kJ/mol)**	**Duty (kW)**
$C_6H_5NO_2 + {}_3H_2 \longrightarrow C_6H_5NH_2 + {}_2H_2O$	141.28	-443	17 385
$C_6H_5NH_2 + {}_3H_2 \longrightarrow C_6H_{11}NH_2$	2.12	≈ -193	113
$C_6H_5NH_2 + H_2 \longrightarrow C_6H_6 + NH_3$	4.94	≈ -51	69
Total heat of reaction			**17 568**

That is **17.6 MW**, and not 16.5 MW as would be obtained by taking the aniline production rate of 134.22 kmol/h in place of the nitrobenzene reacted. The distinction matters because every mole of nitrobenzene converted releases the main reaction heat, including the 7.06 kmol/h which subsequently goes on to form side products. The side reactions themselves contribute only about 0.18 MW, which is 1 % of the total, but the correct main-reaction extent is 5 % higher than the aniline rate and that is not negligible when sizing the steam generation.

The heats of the two side reactions have been estimated from standard gas-phase enthalpies of formation and are indicative; they shall be replaced by Aspen values once the reactor is set up with real kinetics.

4 Block Diagram

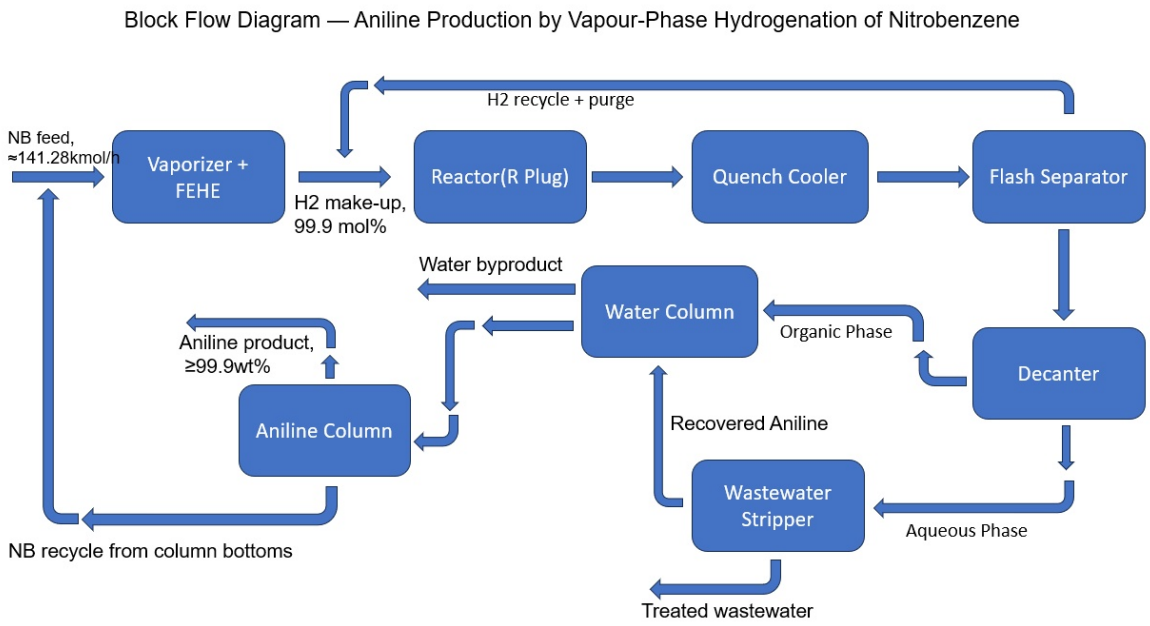


Figure 1: Block flow diagram of the aniline plant, with the three recycle streams labelled.

The reason for the sequence is this: nitrobenzene is vapourised and superheated against the reactor effluent, mixed with make-up and recycled hydrogen, and then passed over the catalyst bed. The effluent is cooled in two stages and the excess hydrogen is flashed off, compressed and returned, with a purge to prevent the buildup of nitrogen and the by-products from light. The liquid which has been condensed divides into two liquid phases in the decanter, which is possible only because aniline and water are partially miscible — this is precisely the property dealt with in Section 6 on thermodynamics. The organic phase is dried in the water column and purified in the aniline column, the bottom portion of which sends nitrobenzene back to the feed. The aniline which is dissolved is removed by means of the aqueous phase before the effluent is treated.

5 Input–Output Cost Analysis and Preliminary Economic Feasibility

5.1 Purpose and approach

Input–output analysis is the crudest economic screen there is, and also the first one worth doing. In its strictest form you only look at the raw materials against the main product. The logic is blunt: if the product's value can't even cover raw material cost, no clever design downstream is going to rescue the project, so it should be dropped right here.

Raw material cost dominates this process, so we didn't stop at the crude version — non-feed operating costs and a limited credit for recoverable by-products and heat were added in too. We've laid the result out as a *margin improvement pathway*, so each commercial lever's contribution is visible on its own rather than the whole result hanging on one convenient assumption.

5.2 Material consumption basis

Nitrobenzene, with recycle

$$\text{Fresh NB factor} = \frac{M_{\text{NB}}}{M_{\text{AN}}} \cdot \frac{1}{S_{\text{AN}}} = \frac{123.11}{93.13} \cdot \frac{1}{0.95} = 1.392 \text{ t/t} \tag{11}$$

Notice conversion doesn't show up in this factor. That's because the flowsheet recycles the unconverted nitrobenzene from the aniline column bottoms straight back to the reactor inlet. So what

the plant actually buys fresh is set by how much reacts, which comes down to selectivity alone. The 99 % conversion figure decides reactor size, recycle flow, energy demand — all of that — but it doesn't mean the leftover 1 % needs buying again. Dividing by conversion *and* selectivity would count the recycled material as if it were lost, and overstate what we actually need to purchase.

Hydrogen

We corrected the hydrogen factor on the same 95 % selectivity basis, folding in the hydrogen consumed by the two side pathways too — 6 mol per mol of cyclohexylamine and 4 mol per mol of benzene. That works out to **0.0702 t/t**. Sticking with the simple main-reaction figure of 0.065 t/t would have understated hydrogen consumption by roughly 8 %.

5.3 Commercial price and procurement basis

Every price below is tied to a specific engineering or procurement strategy — none of them were picked backward from a margin we wanted to show.

Item	Rs./t	Basis / rationale	Reference
Aniline	155,000	Sustained realisation target, equivalent to 1631 USD/t at Rs. 95 per USD. The Indian quoted band was 1500–1600 USD/MT, so this sits marginally above the top of the observed range and requires product quality and market access to be maintained.	Ref. 8(a); FX rate as on 6 August 2026
Nitrobenzene	85,500	Bulk / term procurement target, 900 USD/t, about 6 % below a spot reference of Rs. 91,200/t (960 USD/t). A plant consuming 139,000 t/y is a large and predictable buyer, which gives real negotiating leverage.	Ref. 8(b) for the spot reference
Hydrogen	185,000	On-site SMR *production* cost of 1.95 USD/kg, not a merchant purchase price. Published Indian grey hydrogen production cost is 1.92–2.43 USD/kg at delivered gas of 9–12 USD/MMBtu; we are at the lower end because the site already has natural gas for the gas turbine and the reformer heat can be recovered.	Ref. 8(d)
Benzene	90,250	Used only for the back-integration screening figure of Rs. 9,500/t of aniline; 0.95 USD/kg, with a stoichiometric requirement of 0.634 t benzene per t of nitrobenzene.	Ref. 8(c)
Electricity	embedded	Supplied from the captive gas turbine; included within the operating cost basis.	—

Procurement strategy. We'd want at least two qualified nitrobenzene suppliers on the books, and domestic supply needs to be checked against import parity on a *delivered* basis — freight, insurance, port handling, duties, any trade-remedy measures included — not just FOB price. A benzene-linked formula contract, if one can be arranged, won't raise the average margin by itself, but it cuts exposure to sudden spot spikes and makes financing easier to secure.

5.4 Raw material cost and gross contribution

Item	Rs./t aniline	Rs. crore/y
Aniline revenue	155,000	1550.0
Nitrobenzene, 1.392 × 85,500	118,972	1189.7
Hydrogen, 0.0702 × 185,000	12,987	129.9
Total raw materials	131,959	1319.6
Gross contribution	**23,041**	**230.4**
Gross margin	**14.9 %**	

5.5 Comment on the result

The gross contribution of Rs. 23,041 per tonne, that is 14.9 % of the sales revenue, is enough to clear the input–output screen and the project therefore goes forward.

The table also shows where the money goes. Nitrobenzene alone takes about 77 % of the sales revenue, so procurement is the single most important commercial function in this plant. Since the consumption factor multiplies any price movement, every Rs. 1,000/t rise in the nitrobenzene price adds Rs. 1,392/t to the production cost; the corresponding figure for hydrogen is Rs. 702/t for every Rs. 10,000/t.

5.6 Limitations

- Utilities, labour, maintenance and capital charges are not included, so this is a raw material screen only and not a profitability estimate.
- The Rs. 185,000/t hydrogen cost assumes an on-site reformer and needs an SMR mass and energy balance to confirm it.
- Equipment costs and working capital are not estimated, so ROCE, NPV, IRR and payback are taken up in Report 5.

The remaining weaknesses are collected in Section 9.

6 Thermodynamic Model Selection and Validation

6.1 Why an activity coefficient model is required

The mixture handled in this plant is strongly non-ideal. Aniline and water are only partially miscible whereas aniline and nitrobenzene are completely miscible, and a cubic equation of state cannot represent liquid phase splitting of such a polar–non-polar system. An activity coefficient model is therefore needed for the liquid phase, with an equation of state for the vapour.

NRTL-RK has been selected. NRTL handles the liquid phase activity coefficients and Redlich–Kwong the vapour. NRTL is preferred over Wilson because the Wilson equation cannot predict liquid–liquid splitting at all, and over UNIQUAC because sufficient binary data is available to fit NRTL directly for this system.

Of the phase equilibria relevant here, VLE arises in the columns, LLE in the decanter and VLLE at the heterogeneous azeotrope where the two liquid phases boil together. SLE does not arise, as nothing crystallises in the operating range.

Hydrogen and nitrogen have been declared as **Henry components**. They are supercritical at the operating conditions and so have no pure component vapour pressure; their solubility must be described by Henry's law and not Raoult's law. If this is not done the dissolved hydrogen in the liquid streams is computed wrongly and the overall mass balance does not close.

6.2 The problem with the databank parameters

Before touching anything else, we first checked the binary interaction parameters Aspen Plus pulls in automatically.

Component i	Component j	Source	Temp. Units	AIJ	AJI	BIJ	BJI	CIJ	DIJ	EIJ	EJI	FIJ	FJI	TLOWER	TUPPER
ANILINE	NITRO-01	APV140 VLE-RK	C	0	0	-217.928	287.592	0.3	0	0	0	0	0	83	99
ANILINE	WATER	APV140 VLE-RK	C	-0.775	2.2118	465.474	364.09	0.3	0	0	0	0	0	99	168
ANILINE	BENZE-01	APV140 VLE-RK	C	1.9429	-2.8485	-717.708	1302.27	0.3	0	0	0	0	0	25	180.5
ANILINE	CYCLO-01	APV140 VLE-RK	C	3.0918	-2.1102	-910.317	617.869	0.3	0	0	0	0	0	70.89	181.45
NITRO-01	WATER	APV140 LLE-AS...	C	-5.1549	5.8547	2270.62	229.497	0.2	0	0	0	0	0	0	106.6
NITRO-01	BENZE-01	APV140 VLE-RK	C	0.3005	1.446	40.2665	-653.816	0.3	0	0	0	0	0	70	211
WATER	BENZE-01	APV140 LLE-AS...	C	140.087	45.1905	-5954.31	591.368	0.2	0	-20.0254	-7.5629	0	0	0.8	77
BENZE-01	CYCLO-01	APV140 VLE-RK	C	-0.8986	0.3579	676.613	-388.303	0.3	0	0	0	0	0	60.62	133.77

Figure 2: NRTL binary interaction parameters as retrieved from the Aspen Plus databanks, before regression.

Two things stood out here, and together they're the reason we went ahead with a regression of our own.

1. The ANILINE–WATER row comes from **APV140 VLE-RK** — a *vapour–liquid* data bank — even though this pair is really a liquid–liquid splitting pair. Tellingly, Aspen itself pulled the NITRO-01–WATER and WATER–BENZE-01 pairs from **LLE-ASPEN** with $C_{ij} = 0.2$, which is the correct setting for liquid–liquid systems. So the software treats similar pairs differently, and not in the right direction for the one we care most about.
2. The stated validity range for ANILINE–WATER is $T_{\mathrm{LOWER}} = 99\,^{\circ}\mathrm{C}$ to $T_{\mathrm{UPPER}} = 168\,^{\circ}\mathrm{C}$. Our decanter runs at around $40\,^{\circ}\mathrm{C}$ — nearly 60 degrees below where this correlation is actually valid. Aspen will happily extrapolate and hand you a number, but there's no experimental data backing it up down there.

Given that the whole decanter design, and by extension the entire separation sequence, rides on how well this one pair is represented, we couldn't just accept the default parameters as they stood.

6.3 Experimental data used

We pulled the data through the NIST TDE facility built into Aspen Plus. Aniline + water happens to be one of the recommended reference systems for liquid–liquid equilibrium studies — several independent workers have reported consistent numbers for it, so the data quality here is genuinely good.

TDE handed back 19 LLE sets, 23 VLE sets, one azeotropic set and some diffusion coefficient data. We didn't use all of it — overlapping sets from different authors mostly just add scatter, not information. Here's what we picked:

Data set	Type	Purpose
BLLE001, 002, 005, 019	Binary LLE	Used for the regression
AZEO001	Azeotropic data	**Kept aside for validation, not used in the fit**
BVLE004, 023, 025	Isobaric VLE	Retained for a later check

We dropped the diffusion coefficient data — it's a transport property, and has nothing to do with phase equilibrium anyway.

6.4 Regression procedure and the over-parameterisation problem

We set the regression up with the maximum likelihood objective function and the Britt–Luecke algorithm, using Deming initialisation. The non-randomness parameter C_{ij} was fixed at 0.2 — the standard value for liquid–liquid systems — rather than regressing it alongside the others.

Our first attempt regressed all four parameters, a_{ij}, a_{ji}, b_{ij} and b_{ji}. Here's what came out.

Parameter	Comp. i	Comp. j	Value	Std. deviation
NRTL/1	ANILINE	WATER	−4.579	0.487
NRTL/1	WATER	ANILINE	4.863	0.636
NRTL/2	ANILINE	WATER	1419.39	160.79
NRTL/2	WATER	ANILINE	20.9987	**208.37**

Look at that last parameter: a standard deviation of 208 against a value of just 21. The data simply isn't pinning this parameter down at all — it's floating free, and Aspen flagged a warning that one parameter had hit its bound. Four parameters is one too many for LLE data of this kind.

So we reran it with b_{ji} fixed at zero, fitting only three parameters this time.

Parameter	**Comp. i**	**Comp. j**	**Value**	**Std. deviation**
NRTL/1	ANILINE	WATER	−4.62238	**0.2216**
NRTL/1	WATER	ANILINE	4.92646	**0.0577**
NRTL/2	ANILINE	WATER	1433.47	**74.48**

The parameter values barely moved, but every standard deviation dropped by somewhere between two and tenfold. That settles it — the three-parameter fit is the one we're keeping.

Component i	Component j	Source	Temp. Units	AIJ	AJI	BIJ	BJI	CIJ	DIJ	EIJ	EJI	FIJ	FJI	TLOWER	TUPPER
ANILINE	NITRO-01	APV140 VLE-RK	C	0	0	-217.928	287.592	0.3	0	0	0	0	0	83	99
ANILINE	WATER	R-DR-1	C	-4.62238	4.92646	1433.47	0	0.2	0	0	0	0	0	99	168
ANILINE	BENZE-01	APV140 VLE-RK	C	1.9429	-2.8485	-717.708	1302.27	0.3	0	0	0	0	0	25	180.5
ANILINE	CYCLO-01	APV140 VLE-RK	C	3.0918	-2.1102	-910.317	617.869	0.3	0	0	0	0	0	70.89	181.45
NITRO-01	WATER	APV140 LLE-AS...	C	-5.1549	5.8547	2270.62	229.497	0.2	0	0	0	0	0	0	106.6
NITRO-01	BENZE-01	APV140 VLE-RK	C	0.3005	1.446	40.2665	-653.816	0.3	0	0	0	0	0	70	211
WATER	BENZE-01	APV140 LLE-AS...	C	140.087	45.1905	-5954.31	591.368	0.2	0	-20.0254	-7.5629	0	0	0.8	77
BENZE-01	CYCLO-01	APV140 VLE-RK	C	-0.8986	0.3579	676.613	-388.303	0.3	0	0	0	0	0	60.62	133.77

Figure 3: NRTL parameters after regression. The ANILINE–WATER row now shows the source as `R-DR-1`, that is our own regression case, with C_{ij} corrected to 0.2.

6.5 Quality of the fit

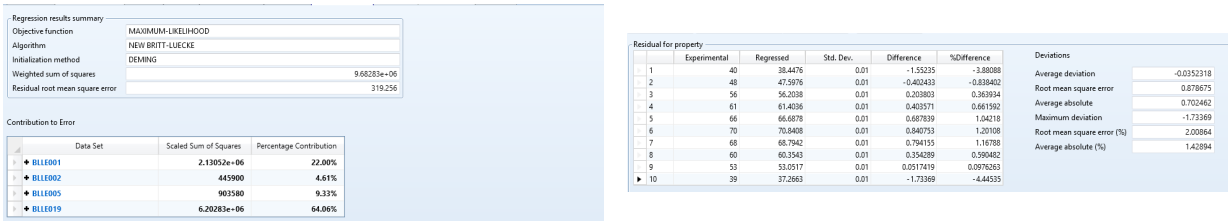


Figure 4: Left: regression summary and the contribution of each data set to the total error. Right: residuals in temperature for BLLE019, the set nearest the decanter operating range.

The regression used the maximum likelihood objective function with the Britt–Luecke algorithm and Deming initialisation, which is the Aspen default for this kind of work. The weighted sum of squares came to 9.68×10^6 with a residual root mean square error of 319.3.

The error is not spread evenly across the four data sets. BLLE019 alone contributes 64 % of the total while BLLE002 contributes only 4.6 %, BLLE001 22 % and BLLE005 9.3 %. This tells us the four sets are not of equal quality, and it is worth knowing which of them the fit is really being pulled by.

Taking the residuals in temperature, BLLE019 gives a root mean square error of 0.879 °C and an average absolute deviation of 0.703 °C, that is about 1.4 % in relative terms. BLLE002 does better at 0.559 °C and 0.554 °C, roughly 1.0 %. We have quoted BLLE019 in particular because it covers 39–70 °C, which is the range the decanter will actually operate in. Both figures are well within what we would accept for design purposes.

6.6 Effect on the phase behaviour

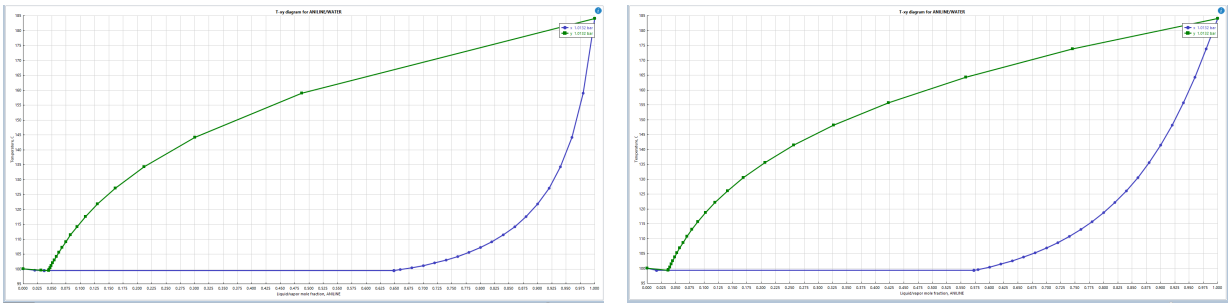


Figure 5: T–xy diagram for aniline–water, before regression (left) and after (right). The flat portion is the two-liquid region.

That flat horizontal stretch in both diagrams is the signature of a **heterogeneous azeotrope**. Two liquid phases coexist across this composition range, so the mixture boils at a fixed temperature and gives off a vapour of fixed composition. This is exactly why the decanter works in our flowsheet — and exactly why a plain distillation column, on its own, can't separate aniline from water.

Regression narrowed the miscibility gap on both sides. Before, it ran from about $x_{\text{aniline}} = 0.05$ to 0.65; afterwards it's tighter, from about 0.037 to 0.575.

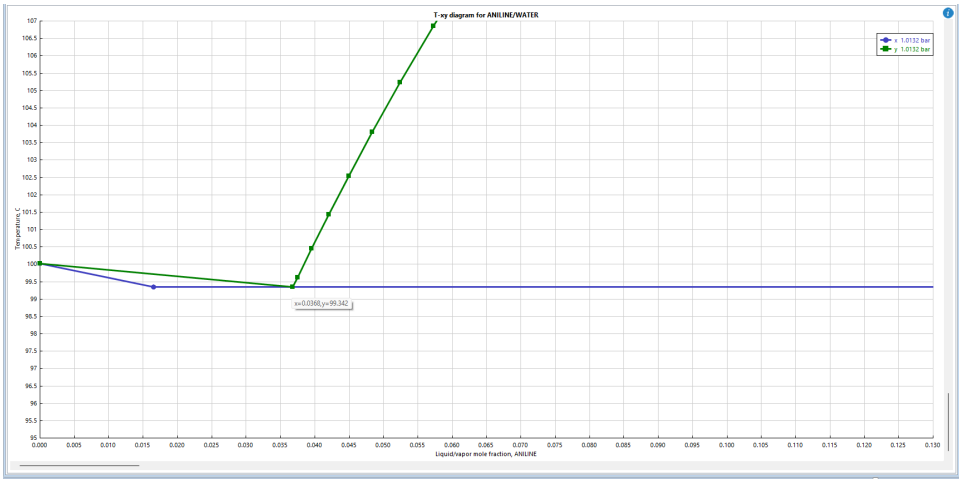


Figure 6: Magnified view near the azeotropic point after regression. The calculated azeotrope is at $x_{\text{aniline}} = 0.0368$ and $T = 99.34\,^{\circ}\text{C}$.

6.7 Validation against independent data

We deliberately held the azeotropic data set AZEO001 out of the regression so it could serve as an independent check afterwards. Here's how the comparison came out.

Quantity	Experimental	Model	Deviation
Azeotropic temperature	372.12 K (98.97 °C)	99.34 °C	0.37 K
Aniline mole fraction	0.04	0.0368	0.0032

This counts as a genuinely *independent* check, not just a self-consistency test — the regression never saw this azeotropic data. We fitted the model purely on liquid–liquid solubility data, and it still reproduced the mixture's boiling behaviour to within 0.37 K. That said, the data set's stated experimental uncertainty is 0.1 K, which puts us roughly four standard deviations off. The agreement is good, but it's not within experimental error, and we'd rather say that plainly than oversell it as a perfect match.

6.8 Ternary behaviour

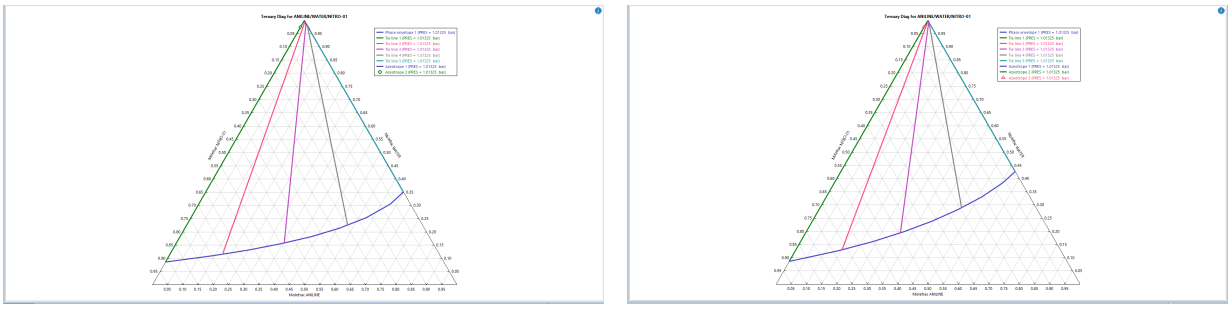


Figure 7: Ternary diagram for aniline/water/nitrobenzene, before regression (left) and after (right).

That wide two-liquid envelope tells us the decanter should give a clean split between an organic-rich phase and an aqueous phase across a broad range of feed compositions — good news for plant operability. Worth flagging too: a third azeotrope has shown up after the regression that wasn't there with the databank parameters. We'll need to pin that down before tackling the distillation sequencing in Report 3.

7 Objectives Which Could Not Be Accomplished, With Reasons

- **We haven't regressed the aniline–nitrobenzene pair yet.** The databank values here are thin too — A_{ij} and A_{ji} are both zero, only the B terms are present, and the validity range is a narrow 83–99 °C. We prioritised aniline–water first since the decanter depends on it directly; this pair is next on the list.

8 Any Other Challenges

- The regression run throws errors under "messages issued during Report Generation". These come from the analysis calculations after the fit, at points where the model predicts a single liquid phase — not from the regression itself, so the parameters and their statistics are still valid.
- We have not yet settled on a single citable source for the LHHW kinetic parameters, so the reactor remains an `RStoic` block until Report 4.

9 Self-Assessment

The points on which we consider this report weakest are as follows.

1. The reactor performance is an assumption and not a result. The conversion and the selectivity split are specified by us on an `RStoic` block, and the material balance, heat duty and cost analysis all inherit that. *Kinetic model in Report 4.*
2. The gross contribution rests on three prices. The aniline realisation of Rs. 155,000/t sits marginally above the top of the observed band, and the nitrobenzene price assumes a term contract has actually been secured. *To be revisited with delivered quotations.*
3. Only the aniline–water pair has been regressed. Aniline–nitrobenzene is still on databank values valid over 83–99 °C, which is narrower than the aniline column will operate over, and the third azeotrope which appeared in the ternary diagram after regression has not yet been identified. *Both before the distillation sequencing of Report 3.*

10 Number of Hours Spent During This Period

Approximately **3 hours per member**, that is about **18 person-hours** for the team in this reporting period. This covers the literature survey, the hand material balance, the Aspen Plus laboratory session for the data retrieval and regression, and the preparation of this report.

11 Contributions From Individual Members

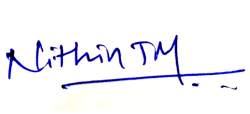
- **Nithin D H and Sajag Masane — process identification, specifications and block diagram.** Literature survey of the alternative synthesis routes and the comparison table, selection of the catalyst system and the reactor operating window, the reaction network including the two side reactions, the feed and product specifications, the hand material balance, and the block flow diagram with the three recycles.
- **Aditya Nitin Patil and Shubham Singh — input–output cost analysis and market analysis.** Market size and end-use survey, price and procurement basis for aniline, nitrobenzene, hydrogen and benzene, the consumption factors and the raw material cost build-up.
- **Nithin T M and Shivam Dongare — thermodynamic tuning, regression and plots.** Property method selection and the Henry component treatment, examination of the databank parameters and their validity limits, retrieval and screening of the NIST TDE data sets, the regression including the diagnosis of the over-parameterised four-parameter fit, the validation against the independent azeotropic data, and all the binary and ternary plots before and after regression.

All the six members have reviewed the complete report and each of us is able to explain the sections owned by the others.

12 References

1. T. Kahl et al., "Aniline", *Ullmann's Encyclopedia of Industrial Chemistry*, DOI 10.1002/14356007.a02_303.
2. B. Amon, H. Redlingshöfer, E. Klemm, E. Dieterich, G. Emig, *Chem. Eng. Process.* 38 (1999) 395–404, DOI 10.1016/S0255-2701(99)00037-9.
3. L. Petrov, K. Kumbilieva, N. Kirkov, kinetic model of nitrobenzene hydrogenation over industrial copper catalyst ($E_a = 35 \pm 1$ kJ/mol).
4. C. Sá Couto, L. M. Madeira, C. P. Nunes, P. Araújo, *Chem. Eng. Technol.* 38 (2015) 1625–1636, DOI 10.1002/ceat.201400468.
5. G. Maria, D.-N. Stefan, *J. Loss Prev. Process Ind.* 23 (2010) 112–126.
6. Aspen Technology, *Aspen Physical Property System — Physical Property Methods*; NIST ThermoData Engine as implemented in Aspen Plus V14.
7. IUPAC–NIST Solubility Data Series (aniline + water as a recommended LLE reference system).
8. Price references, accessed August 2026: (a) aniline, India Q4 2025, 1500–1600 USD/MT, price-watch.ai; (b) nitrobenzene, India Q1 2026, 0.96 USD/kg, Expert Market Research; (c) benzene, India July 2026, 0.95 USD/kg, IMARC Group; (d) grey hydrogen production cost, India, 1.92–2.43 USD/kg, FIPI Hydrogen Market Report. Exchange rate USD 1 = Rs. 95 as on 6 August 2026.

Signatures of members



Nithin T M



Aditya Nitin Patil



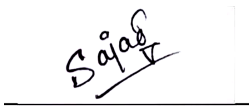
Nithin D H



Shubham Singh



Shivam Dongare



Sajag Masane

Bi-weekly Report 2

Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

CHE453 Process Design Capstone Project

Group No: 14

Date: 23 August 2026

Report 1 precedes this page and is unchanged.

Everything which follows constitutes Report 2.

Bi-weekly Report

Report Number: 02

Group No: 14

Date: 23 August 2026

Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

Team members

Name	Roll No.	Name	Roll No.
Nithin T M	230709	Shubham Singh	230999
Aditya Nitin Patil	230074	Shivam Dongare	230971
Nithin D H	230708	Sajag Masane	230896

Objectives for this report

1. Property analysis and identification of the operating conditions.
2. Process synthesis and development of the process flow diagram.
3. Flowsheet development in Aspen Plus.
4. Material and energy balance using Aspen Plus.

1 Property Analysis and Operating Conditions

1.1 Property model

The property method is **NRTL-RK**, selected and validated in Report 1 Section 6. HYDRO-01 and NITRO-02 are declared as Henry components under the list HC-1, which is attached to every block, since both are supercritical at the flash and decanter conditions. The ANILINE–WATER pair uses our own three-parameter regression R-DR-1 from Report 1 Section 6.4, namely $A_{ij} = -4.62238$, $A_{ji} = 4.92646$, $B_{ij} = 1433.47$, $B_{ji} = 0$ and $C_{ij} = 0.2$. That fit has been made on liquid–liquid data covering 39–70 °C, which brackets the 40 °C at which DEC and FLASH operate, so the decanter is computed inside the range of the data.

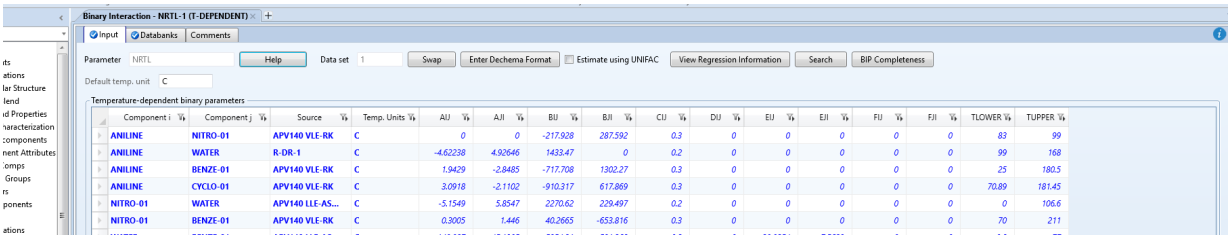


Figure 1: Binary interaction parameters as loaded in the simulation. The ANILINE–WATER row carries the source R-DR-1, that is our own regression from Report 1; every other pair remains on the Aspen databank.

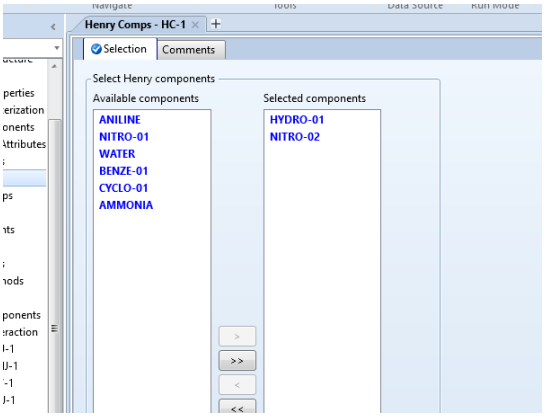


Figure 2: Henry component list HC-1, with HYDRO-01 and NITRO-02 selected.

1.2 Component slate

Aspen ID	Component	Formula	M	T_b (°C)	Role
HYDRO-01	Hydrogen	H_2	2.02	−252.9	Reactant, Henry component, recycled
NITRO-02	Nitrogen	N_2	28.01	−195.8	Inert in the make-up, Henry component
AMMONIA	Ammonia	NH_3	17.03	−33.3	By-product of hydrogenolysis
BENZE-01	Benzene	C_6H_6	78.11	80.1	By-product of hydrogenolysis
WATER	Water	H_2O	18.02	100.0	Main by-product
CYCLO-01	Cyclohexylamine	$C_6H_{13}N$	99.17	134.5	By-product of over-hydrogenation
ANILINE	Aniline	C_6H_7N	93.13	184.1	Product
NITRO-01	Nitrobenzene	$C_6H_5NO_2$	123.11	210.8	Feed, recycled from COL-2 bottoms

The boiling points fix the separation sequence. Aniline and water form a heterogeneous azeotrope at $x_{\text{aniline}} = 0.0368$ and 99.34 °C, so no column can separate the pair and a decanter is used instead. The 26.7 K gap between aniline and nitrobenzene is the split which carries the 5 ppm nitrobenzene limit, and it is therefore placed last, where the feed is dry and the load is smallest.

1.3 Operating conditions

Unit	Condition	Basis
VAP-1	3 bar, VF = 1	The reaction section is held at one pressure so that no compression is required on the organic side. The block delivers saturated vapour at 260.5 ℃.
FEHE	Cold outlet 290 ℃	A 10 K approach to the reactor temperature, recovering heat from the effluent.
REACT	300 ℃, 3 bar	Operating window of the supported copper catalyst. Below about 250 ℃ the rate is inadequate; above about 350 ℃ over-hydrogenation accelerates and the catalyst sinters. Low pressure is preferred as the main reaction loses moles.
COOL-1	120 ℃	Staged, so that this duty is available at a temperature at which steam can be raised.
COOL-2	40 ℃	Cooling-water approach; condenses the organics ahead of the flash.
FLASH	40 ℃, 2.5 bar	0.5 bar below the reaction pressure for line and exchanger losses. The low temperature drives the hydrogen recovery.
SPLIT	Purge fraction 0.0086	Fixed from the hydrogen and nitrogen balances of Section 4.3.
COMP	3.2 bar, $\eta_{is} = 0.75$	Restores the recycle gas above the mixer pressure.
DEC	40 ℃, 1.5 bar, key WATER	As cold as the cooling water permits, since the miscibility gap widens as the temperature falls.
COL-1	1.0 bar top	Atmospheric. The overhead is the water-rich azeotrope near 99 ℃.
COL-2	0.2 bar top	Vacuum. At atmospheric pressure the nitrobenzene bottoms would sit near 211 ℃, at which aniline darkens; at 0.3 bar the reboiler runs at 144 ℃.
STRIP	1.0 bar	Atmospheric stripping of the aqueous phase, reboiler at 101 ℃.

2 Process Synthesis and Process Flow Diagram

2.1 Synthesis decisions

Continuous operation. 100,000 t/y at 8000 h/y is far above the threshold at which batch operation is defensible.

Input–output structure. Nitrobenzene and hydrogen enter; aniline, a water-rich overhead, treated wastewater and a purge gas leave. A purge is required because the make-up hydrogen carries 0.1 mol% nitrogen which does not react.

Recycle structure. Two reactants of very different volatility give two recycles, a gas recycle for the excess hydrogen and a liquid recycle for the unconverted nitrobenzene. Hydrogen is fed in large excess, which suppresses the aniline-consuming side reactions and acts as thermal ballast for the 17.6 MW exotherm. A third recycle, the recovered aniline from the stripper, follows once the separation system is fixed.

Separation system. The vapour–liquid split is taken first and for free in the flash, and the liquid–liquid split for free in the decanter; only the residue is handed to distillation. Within the distillation train the most plentiful component is removed early, water at 282 kmol/h against 133 kmol/h of aniline, and the highest-purity split is left last.

Heat integration. The feed–effluent heat exchanger is placed at this stage because the reactor feed must be raised to 290 ℃ and the effluent cooled from 300 ℃. The remainder of the network is taken up in Report 5.

2.2 Separation sequence

Because the aniline–water azeotrope is heterogeneous, DEC breaks it and therefore comes first, delivering an aniline-rich organic phase and an aqueous phase of water with dissolved aniline. The organic phase is still wet at its saturation limit, so COL-1 takes the water overhead as the azeotrope; it is specified as a three-phase column with WATER as key over stages 2–19, because the overhead condensate itself splits into two liquid phases. The dry bottoms goes to COL-2, where aniline is taken overhead as product and nitrobenzene leaves in the bottoms and returns to the feed. The aqueous phase is stripped in STRIP, the recovered aniline returning to COL-1 and the stripped water leaving as WASTEWTR.

2.3 Process flow diagram

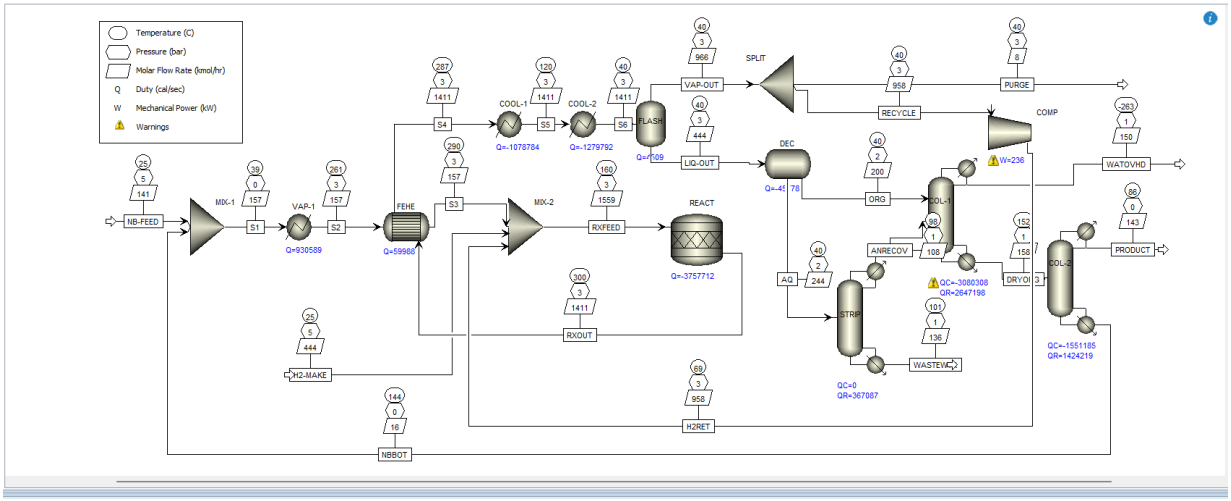


Figure 3: Process flow diagram of the aniline plant as built and converged in Aspen Plus, annotated with temperature (°C), pressure (bar), molar flow (kmol/h), block duty (cal/s) and shaft power (kW).

3 Flowsheet Development in Aspen Plus

3.1 Blocks and specifications

Block	Model	Specification as entered
MIX-1	Mixer	Fresh nitrobenzene with the NBBOT recycle
VAP-1	Heater	3 bar, vapour fraction 1
FEHE	HeatX	Cold outlet 290 °C, minimum approach 1 K
MIX-2	Mixer	Vaporised feed with H2-MAKE and H2RET
REACT	RStoic	300 °C, 3 bar, three reactions, series
COOL-1	Heater	120 °C, $\Delta P = -0.2$ bar
COOL-2	Heater	40 °C, $\Delta P = -0.2$ bar
FLASH	Flash2	40 °C, 2.5 bar
SPLIT	FSplit	Split fraction to PURGE 0.0086
COMP	Compr	Isentropic, 3.2 bar, $\eta_{is} = 0.75$
DEC	Decanter	40 °C, 1.5 bar, key component WATER
COL-1	RadFrac	20 stages, 3 phases, total condenser, kettle reboiler
COL-2	RadFrac	15 stages, 2 phases, total condenser, kettle reboiler
STRIP	RadFrac	8 stages, no condenser, kettle reboiler

3.2 Reaction specification

The three reactions are entered in RStoic with the *series* option enabled, so that each acts on the stream leaving the one before it. Both side reactions consume aniline rather than nitrobenzene, so

unless they follow the main reaction there is no aniline for them to act upon.

No.	Reaction	Conversion, basis	Extent (kmol/h)
1	$C_6H_5NO_2 + 3\,H_2 \longrightarrow C_6H_5NH_2 + 2\,H_2O$	0.99 of NITRO-01	141.113
2	$C_6H_5NH_2 + 3\,H_2 \longrightarrow C_6H_{13}N$	0.01513 of ANILINE	2.367
3	$C_6H_5NH_2 + H_2 \longrightarrow C_6H_6 + NH_3$	0.03530 of ANILINE	5.438

The two fractional conversions have been back-calculated so that, applied in series, they reproduce the 95 / 1.5 / 3.5 selectivity split adopted in Report 1.

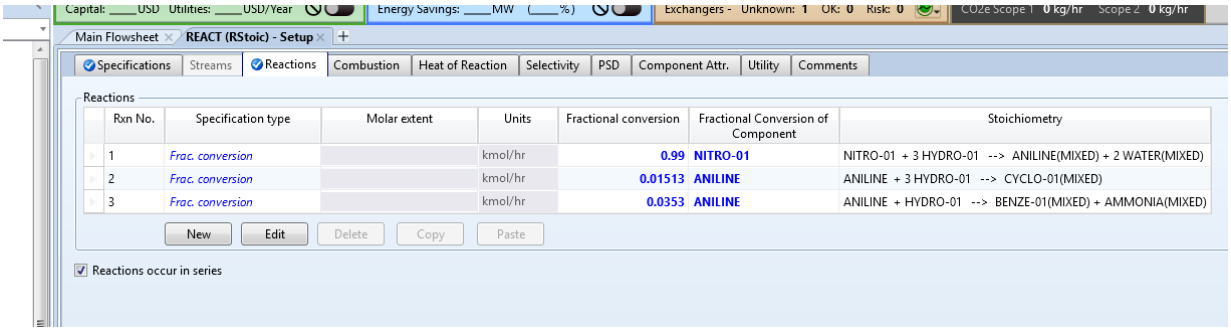


Figure 4: Reaction set entered in the RStoic block, with the three stoichiometries, the fractional conversions and their basis components, and the series option selected.

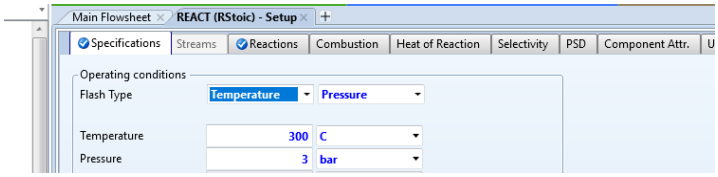


Figure 5: Reactor operating specification, 300 °C and 3 bar.

3.3 Column specifications and results

	COL-1	COL-2	STRIP
Stages	20	15	8
Feed stage	10, above-stage	6, on-stage	1, on-stage
Condenser	Total	Total	None
Reboiler	Kettle	Kettle	Kettle
Convergence	Vapor-Liquid-Liquid	Vapor-Liquid	Standard
Operating specifications	D 150 kmol/h, RR 3	D:F 0.9, RR 2	Boilup ratio 1.0
Top pressure, bar	1.0	0.2	1.0
Reboiler pressure, bar	1.1	0.3	1.05
Distillate, kmol/h	150.00	142.58	108.21
Bottoms, kmol/h	158.42	15.84	136.05
Boilup ratio	5.45	28.67	1.00
Reboiler temperature, °C	151.89	144.15	101.13

3.4 Convergence

Three streams are torn, one for each recycle. The maximum relative errors reported by Aspen Plus are 8.23×10^{-4} on H2RET and 2.66×10^{-5} on S3. The overall mass balance closes to 0.7 kg/h in 18,300 kg/h and the energy balance to 0.003 %, as given in Section 4.

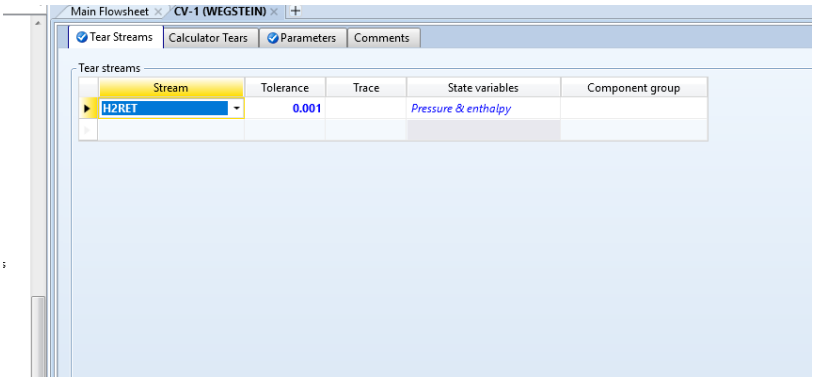
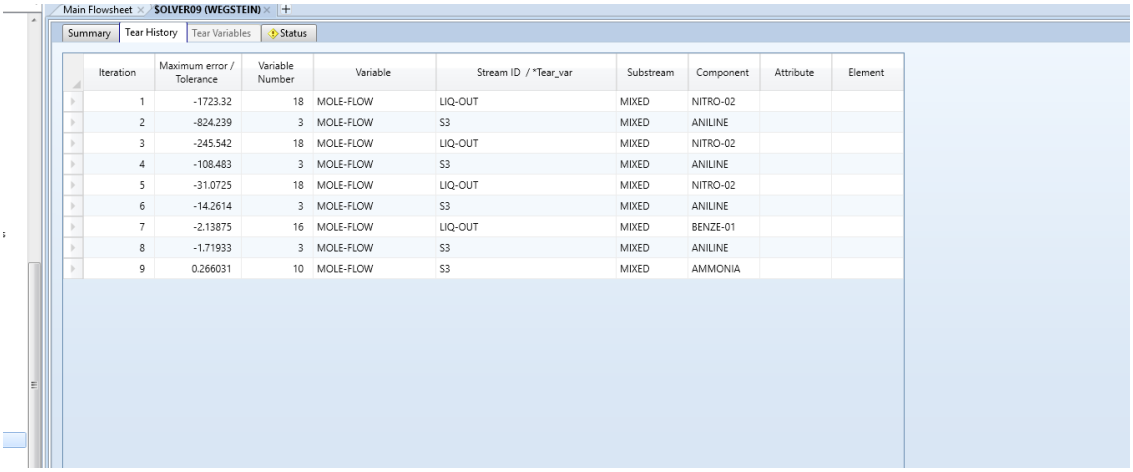


Figure 6: Tear stream definition for the hydrogen loop, converged on pressure and enthalpy to a tolerance of 0.001 by the Wegstein method.



Iteration	Maximum error / Tolerance	Variable Number	Variable	Stream ID / *Tear_var	Substream	Component	Attribute	Element
1	-1723.32	18	MOLE-FLOW	LIQ-OUT	MIXED	NITRO-02		
2	-824.239	3	MOLE-FLOW	S3	MIXED	ANILINE		
3	-245.542	18	MOLE-FLOW	LIQ-OUT	MIXED	NITRO-02		
4	-108.483	3	MOLE-FLOW	S3	MIXED	ANILINE		
5	-31.0725	18	MOLE-FLOW	LIQ-OUT	MIXED	NITRO-02		
6	-14.2614	3	MOLE-FLOW	S3	MIXED	ANILINE		
7	-2.13875	16	MOLE-FLOW	LIQ-OUT	MIXED	BENZE-01		
8	-1.71933	3	MOLE-FLOW	S3	MIXED	ANILINE		
9	0.266031	10	MOLE-FLOW	S3	MIXED	AMMONIA		

Figure 7: Convergence history, showing the maximum error on each torn variable at the final iteration.

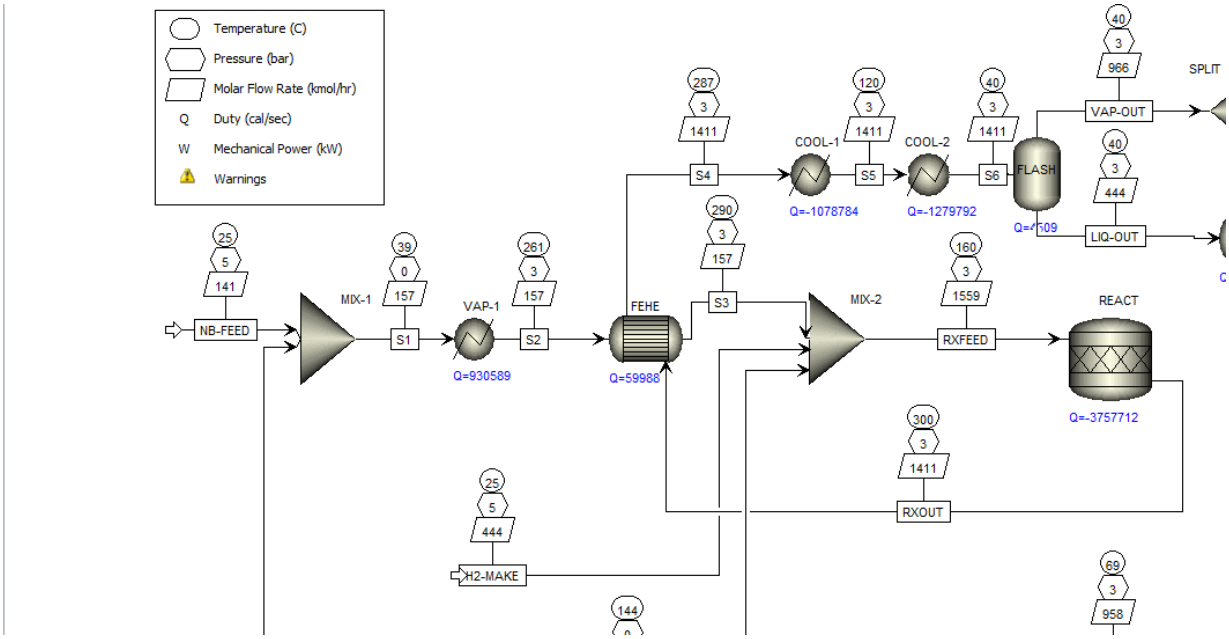


Figure 8: Reaction and hydrogen recycle section of the converged flowsheet.

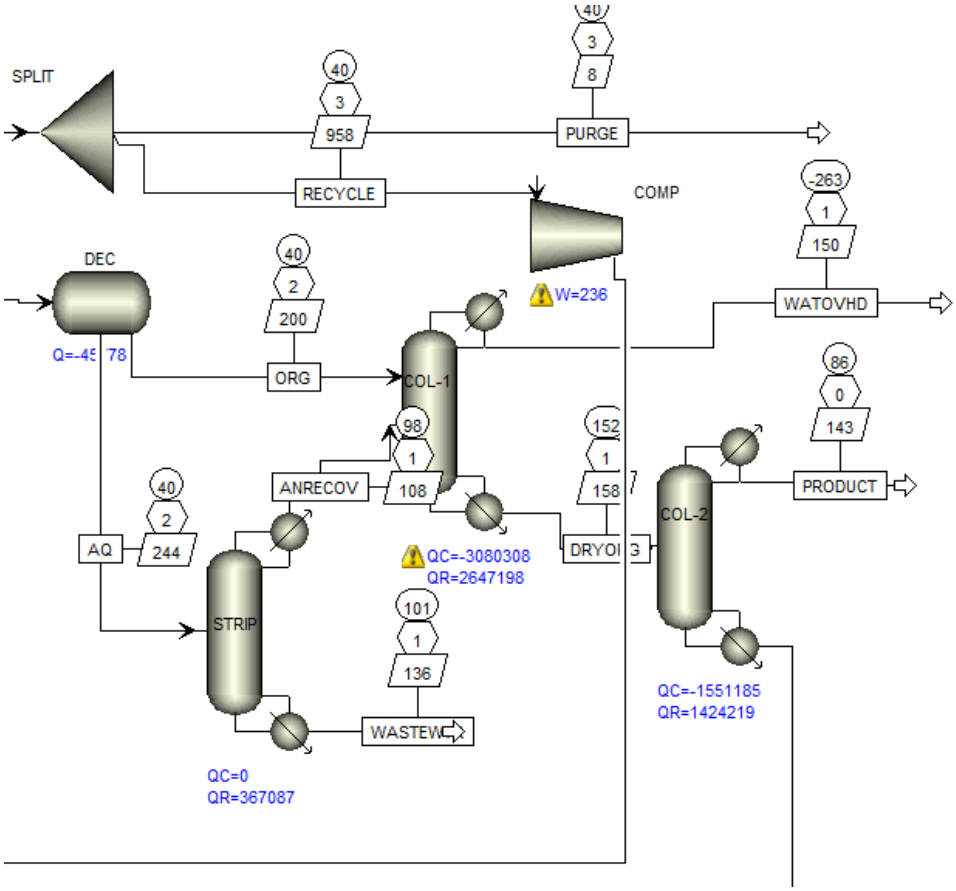


Figure 9: Separation section of the converged flowsheet, with the condenser and reboiler duties of the three columns.

4 Material and Energy Balance

4.1 Overall material balance

In	kmol/h	kg/h	Out	kmol/h	kg/h
NB-FEED	141.28	17,393.2	PRODUCT	142.58	12,346.8
H2-MAKE	444.00	906.6	WATOVHD	150.00	3,372.8
			WASTEWTR	136.05	2,536.8
			PURGE	8.31	42.7
Total		18,299.7	Total		18,299.1

The balance closes to 0.7 kg/h, that is 0.004 %. The converged fresh nitrobenzene feed of 141.28 kmol/h and the hydrogen make-up of 444.0 kmol/h agree exactly with the independent hand balance of Report 1 Sections 3.3 and 3.5.

4.2 Reactor performance

Quantity	Report 1 (hand)	Aspen (converged)
Nitrobenzene to reactor, kmol/h	142.71	142.54
Nitrobenzene reacted, kmol/h	141.28	141.11
Hydrogen at reactor inlet, kmol/h	1427.1	1300.6
Hydrogen-to-nitrobenzene ratio	10.0	9.12
Hydrogen consumed, kmol/h	435.15	435.88
Net aniline produced, kmol/h	134.22	133.31

4.3 Hydrogen loop

Term	kmol/h	Remark
Hydrogen in the make-up	443.56	99.9 mol% in 444.0 kmol/h
Consumed by reaction	435.88	Three reactions
Removed in the purge	7.44	0.86 % split of the flash vapour
Dissolved in the flash liquid	0.13	Computed by Henry's law

The purge fraction follows from the requirement that, at steady state, the hydrogen leaving in the purge equal the make-up less what the reaction consumes and less what dissolves:

$$f_{\text{purge}} = \frac{M - C - D}{V} = \frac{443.56 - 435.88 - 0.13}{864.57} = 0.0087 \tag{1}$$

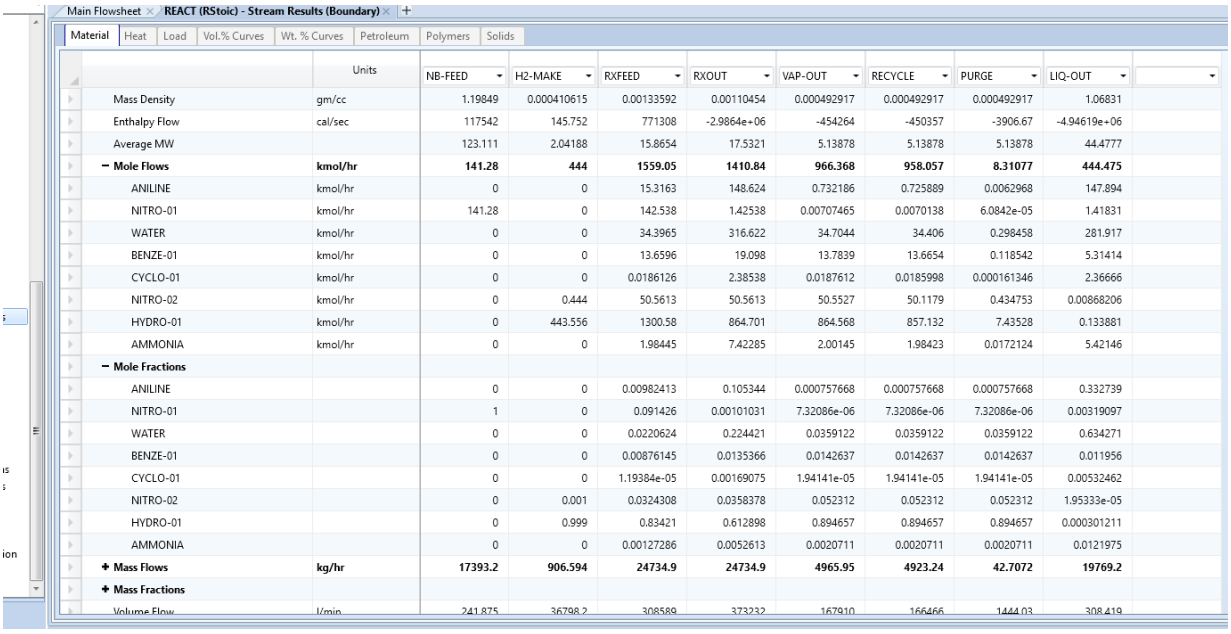
where V is the hydrogen in the flash vapour. The value used in SPLIT is 0.0086. The recycle gas settles at 89.5 mol% hydrogen with 5.2 mol% nitrogen, which is why the reactor ratio holds at 9.12:1 rather than at the 10:1 assumed in Report 1.

4.4 Stream conditions

Stream	T (°C)	P (bar)	Vapour fraction	kmol/h	kg/h
NB-FEED	25.0	5.00	0.000	141.28	17393.2
H2-MAKE	25.0	5.00	1.000	444.00	906.6
S1	38.8	0.30	0.000	157.12	18906.0
S2	260.5	3.00	1.000	157.12	18906.0
S3	290.0	3.00	1.000	157.12	18906.0
RXFEED	159.7	3.00	0.990	1559.05	24734.9
RXOUT	300.0	3.00	1.000	1410.84	24734.9
S4	287.2	3.00	1.000	1410.84	24734.9
S5	120.0	2.80	0.920	1410.84	24734.9
S6	40.0	2.60	0.684	1410.84	24734.9
VAP-OUT	40.0	2.50	1.000	966.37	4965.9
PURGE	40.0	2.50	1.000	8.31	42.7
RECYCLE	40.0	2.50	1.000	958.06	4923.2
H2RET	69.5	3.20	1.000	957.93	4922.3
LIQ-OUT	40.0	2.50	0.000	444.47	19769.2
ORG	40.0	1.50	0.000	200.21	15088.7
AQ	40.0	1.50	0.000	244.27	4680.5
ANRECOV	98.4	1.00	1.000	108.21	2143.7
WATOVHD	—	1.00	0.000	150.00	3372.8
DRYORG	151.9	1.10	0.000	158.42	13859.6
PRODUCT	86.2	0.20	0.000	142.58	12346.8
NBBOT	144.2	0.30	0.000	15.84	1512.9
WASTEWTR	101.1	1.05	0.000	136.05	2536.8

4.5 Component molar flows — reaction section (kmol/h)

	NB-FEED	H2-MAKE	RXFEED	RXOUT	VAP-OUT	RECYCLE	PURGE	LIQ-OUT
Aniline	—	—	15.316	148.624	0.732	0.726	0.006	147.894
Nitrobenzene	141.280	—	142.538	1.425	0.007	0.007	tr	1.418
Water	—	—	34.396	316.622	34.704	34.406	0.298	281.917
Benzene	—	—	13.660	19.098	13.784	13.665	0.119	5.314
Cyclohexylamine	—	—	0.019	2.385	0.019	0.019	tr	2.367
Nitrogen	—	0.444	50.561	50.561	50.553	50.118	0.435	0.009
Hydrogen	—	443.556	1300.578	864.701	864.568	857.132	7.435	0.134
Ammonia	—	—	1.984	7.423	2.001	1.984	0.017	5.421
Total	141.28	444.00	1559.05	1410.84	966.37	958.06	8.31	444.47


Main Flowsheet | REACT (RStoic) - Stream Results (Boundary)

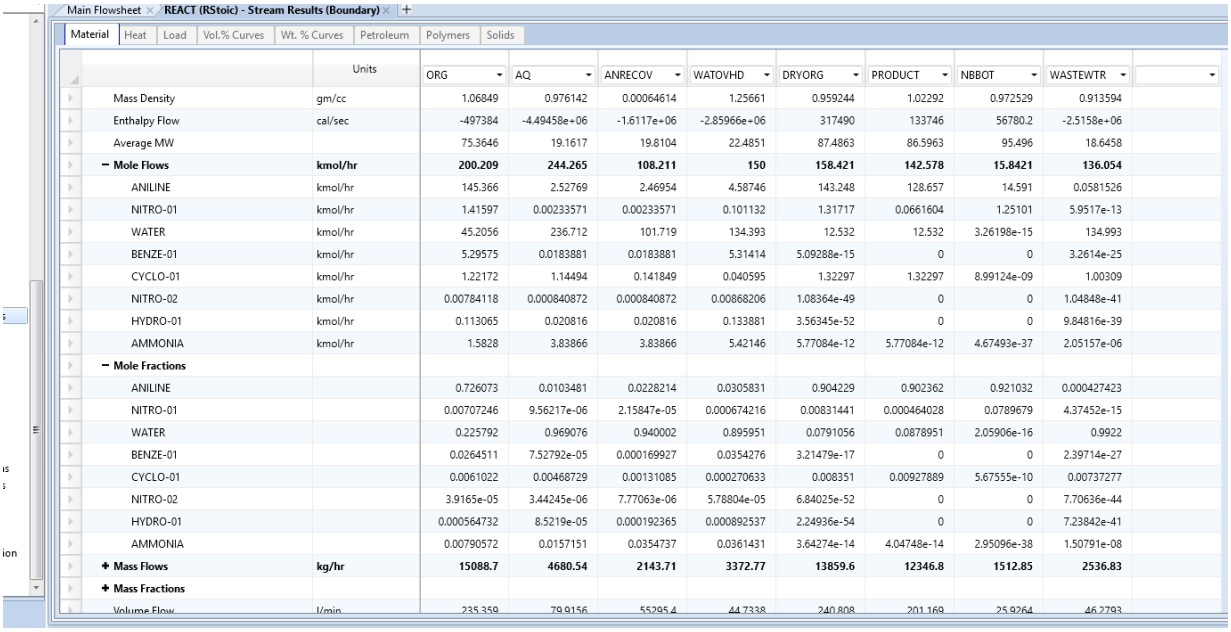
Material | Heat | Load | Vol.% Curves | Wt. % Curves | Petroleum | Polymers | Solids

	Units	NB-FEED	H2-MAKE	RXFEED	RXOUT	VAP-OUT	RECYCLE	PURGE	LIQ-OUT
Mass Density	gm/cc	1.19849	0.000410615	0.00133592	0.00110454	0.000492917	0.000492917	0.000492917	1.06831
Enthalpy Flow	cal/sec	117542	145.752	771308	-2.9864e+06	-454264	-450357	-3906.67	-4.94619e+06
Average MW		123.111	2.04188	15.8654	17.5321	5.13878	5.13878	5.13878	44.4777
− Mole Flows	**kmol/hr**	**141.28**	**444**	**1559.05**	**1410.84**	**966.368**	**958.057**	**8.31077**	**444.475**
ANILINE	kmol/hr	0	0	15.3163	148.624	0.732186	0.725889	0.0062968	147.894
NITRO-01	kmol/hr	141.28	0	142.538	1.42538	0.00707465	0.0070138	6.0842e-05	1.41831
WATER	kmol/hr	0	0	34.3965	316.622	34.7044	34.406	0.298458	281.917
BENZE-01	kmol/hr	0	0	13.6596	19.098	13.7839	13.6654	0.118542	5.31414
CYCLO-01	kmol/hr	0	0	0.0186126	2.38538	0.0187612	0.0185998	0.000161346	2.36666
NITRO-02	kmol/hr	0	0.444	50.5613	50.5613	50.5527	50.1179	0.434753	0.00868206
HYDRO-01	kmol/hr	0	443.556	1300.58	864.701	864.568	857.132	7.43528	0.133881
AMMONIA	kmol/hr	0	0	1.98445	7.42285	2.00145	1.98423	0.0172124	5.42146
− Mole Fractions									
ANILINE		0	0	0.00982413	0.105344	0.000757668	0.000757668	0.000757668	0.332739
NITRO-01		1	0	0.091426	0.00101031	7.32086e-06	7.32086e-06	7.32086e-06	0.00319097
WATER		0	0	0.0220624	0.224421	0.0359122	0.0359122	0.0359122	0.634271
BENZE-01		0	0	0.00876145	0.0135366	0.0142637	0.0142637	0.0142637	0.011956
CYCLO-01		0	0	1.19384e-05	0.00169075	1.94141e-05	1.94141e-05	1.94141e-05	0.00532462
NITRO-02		0	0.001	0.0324308	0.0358378	0.052312	0.052312	0.052312	1.95333e-05
HYDRO-01		0	0.999	0.83421	0.612898	0.894657	0.894657	0.894657	0.000301211
AMMONIA		0	0	0.00127286	0.0052613	0.0020711	0.0020711	0.0020711	0.0121975
+ Mass Flows	**kg/hr**	**17393.2**	**906.594**	**24734.9**	**24734.9**	**4965.95**	**4923.24**	**42.7072**	**19769.2**
+ Mass Fractions									
Volume Flow	l/min	241.875	36798.2	308589	373232	167910	166466	1444.03	308.419

Figure 10: Aspen Plus stream results for the reaction section, as tabulated above.

4.6 Component molar flows — separation section (kmol/h)

	ORG	AQ	ANRECOV	WATOVHD	DRYORG	PRODUCT	NBBOT	WASTEWTR
Aniline	145.366	2.528	2.470	4.587	143.248	128.657	14.591	0.058
Nitrobenzene	1.416	0.002	0.002	0.101	1.317	0.066	1.251	—
Water	45.206	236.712	101.719	134.393	12.532	12.532	—	134.993
Benzene	5.296	0.018	0.018	5.314	—	—	—	—
Cyclohexylamine	1.222	1.145	0.142	0.041	1.323	1.323	tr	1.003
Nitrogen	0.008	0.001	0.001	0.009	—	—	—	—
Hydrogen	0.113	0.021	0.021	0.134	—	—	—	—
Ammonia	1.583	3.839	3.839	5.421	—	—	—	tr
Total	200.21	244.27	108.21	150.00	158.42	142.58	15.84	136.05


Main Flowsheet | REACT (RStoic) - Stream Results (Boundary)

Material | Heat | Load | Vol.% Curves | Wt. % Curves | Petroleum | Polymers | Solids

	Units	ORG	AQ	ANRECOV	WATOVHD	DRYORG	PRODUCT	NBBOT	WASTEWTR
Mass Density	gm/cc	1.06849	0.976142	0.00064614	1.25661	0.959244	1.02292	0.972529	0.913594
Enthalpy Flow	cal/sec	-497384	-4.49458e+06	-1.6117e+06	-2.85966e+06	317490	133746	56780.2	-2.5158e+06
Average MW		75.3646	19.1617	19.8104	22.4851	87.4863	86.5963	95.496	18.6458
− Mole Flows	**kmol/hr**	**200.209**	**244.265**	**108.211**	**150**	**158.421**	**142.578**	**15.8421**	**136.054**
ANILINE	kmol/hr	145.366	2.52769	2.46954	4.58746	143.248	128.657	14.591	0.0581526
NITRO-01	kmol/hr	1.41597	0.00233571	0.00233571	0.101132	1.31717	0.0661604	1.25101	5.9517e-13
WATER	kmol/hr	45.2056	236.712	101.719	134.393	12.532	12.532	3.26198e-15	134.993
BENZE-01	kmol/hr	5.29575	0.0183881	0.0183881	5.31414	5.09288e-15	0	0	3.2614e-25
CYCLO-01	kmol/hr	1.22172	1.14494	0.141849	0.040595	1.32297	1.32297	8.99124e-09	1.00309
NITRO-02	kmol/hr	0.00784118	0.000840872	0.000840872	0.00868206	1.08364e-49	0	0	1.04848e-41
HYDRO-01	kmol/hr	0.113065	0.020816	0.020816	0.133881	3.56345e-52	0	0	9.84816e-39
AMMONIA	kmol/hr	1.5828	3.83866	3.83866	5.42146	5.77084e-12	5.77084e-12	4.67493e-37	2.05157e-06
− Mole Fractions									
ANILINE		0.726073	0.0103481	0.0228214	0.0305831	0.904229	0.902362	0.921032	0.000427423
NITRO-01		0.00707246	9.56217e-06	2.15847e-05	0.000674216	0.00831441	0.000464028	0.0789679	4.37452e-15
WATER		0.225792	0.969076	0.940002	0.895951	0.0791056	0.0878951	2.05906e-16	0.9922
BENZE-01		0.0264511	7.52792e-05	0.000169927	0.0354276	3.21479e-17	0	0	2.39714e-27
CYCLO-01		0.0061022	0.00468729	0.00131085	0.000270633	0.008351	0.00927889	5.67555e-10	0.00737277
NITRO-02		3.9165e-05	3.44245e-06	7.77063e-06	5.78804e-05	6.84025e-52	0	0	7.70636e-44
HYDRO-01		0.000564732	8.5219e-05	0.000192365	0.000892537	2.24936e-54	0	0	7.23842e-41
AMMONIA		0.00790572	0.0157151	0.0354737	0.0361431	3.64274e-14	4.04748e-14	2.95096e-38	1.50791e-08
+ Mass Flows	**kg/hr**	**15088.7**	**4680.54**	**2143.71**	**3372.77**	**13859.6**	**12346.8**	**1512.85**	**2536.83**
+ Mass Fractions									
Volume Flow	l/min	235.359	79.9156	55295.4	44.7338	240.808	201.169	25.9264	46.2793

Figure 11: Aspen Plus stream results for the separation section, as tabulated above.

4.7 Product stream

Component	kmol/h	mol%	wt%
Aniline	128.657	90.24	97.04
Water	12.532	8.79	1.83
Cyclohexylamine	1.323	0.93	1.06
Nitrobenzene	0.066	0.05	0.07
PRODUCT	142.58	100.00	100.00

PRODUCT carries 11,982 kg/h of aniline, that is 11.98 t/h against the design rate of 12.5 t/h. Of the 133.31 kmol/h of aniline produced in the reactor, 128.66 kmol/h reaches the product, giving a recovery of 96.5 % through the separation train.

4.8 Energy balance

Block	Service	cal/s	MW
VAP-1	Vaporiser	+930,589	+3.894
FEHE	Feed–effluent exchanger	59,988	0.251
REACT	Reactor, isothermal at 300 ℃	−3,757,712	−15.722
COOL-1	Effluent cooler to 120 ℃	−1,078,784	−4.514
COOL-2	Effluent cooler to 40 ℃	−1,279,792	−5.354
FLASH	Flash drum	+4,509	+0.019
DEC	Decanter	−45,778	−0.192
COL-1	Condenser	−3,080,308	−12.888
COL-1	Reboiler	+2,647,198	+11.076
COL-2	Condenser	−1,551,185	−6.490
COL-2	Reboiler	+1,424,219	+5.959
STRIP	Reboiler	+367,087	+1.536
COMP	Shaft work	+56,481	+0.236

Taking the FEHE as an internal transfer, the block duties and the compressor work sum to −22.441 MW, against −22.440 MW obtained from the enthalpy flows of the terminal streams, so the energy balance closes to 0.003 %.

The reactor duty of −15.722 MW is the heat of reaction less the sensible heat required to raise the feed from 159.7 to 300 ℃:

$$Q_{\mathrm{REACT}} = -\left(141.113 \times 443 + 2.367 \times 193 + 5.438 \times 51\right) + Q_{\mathrm{sensible}} = -17.569 + 1.847 = -15.722 \ \mathrm{MW} \tag{2}$$

The reaction term of 17.569 MW agrees with the 17.568 MW computed by hand in Report 1 Section 3.6.

5 Results and Discussion

The flowsheet converges with all three recycles closed. The two feed rates reproduce the Report 1 hand balance exactly, the mass balance closes to 0.004 % and the energy balance to 0.003 %, so the simulation and the hand calculation of the previous fortnight are consistent with one another.

The product leaves COL-2 at 97.04 wt% aniline. The residual water at 1.83 wt% and cyclohexylamine at 1.06 wt% are both lighter than aniline and therefore leave overhead with it, so a light-ends draw at the top of COL-2 is the route to the 99.9 wt% specification. Nitrobenzene stands at 660 ppm against a limit of 5 ppm, which calls for a sharper cut in the same column. Both are taken up in the distillation column design of Report 3.

COL-2 returns 14.59 kmol/h of aniline in NBBOT along with 1.25 kmol/h of nitrobenzene, so the reactor feed carries 15.32 kmol/h of aniline before reaction. Since the two side reactions are specified

on aniline, raising the reflux in COL-2 to sharpen this cut recovers that aniline into the product and holds the selectivity at the design value.

COL-1 is specified with a total condenser while its feed carries hydrogen, nitrogen and ammonia. A partial condenser with a vapour draw is the appropriate configuration for a feed containing non-condensables, and this is the first change to be made in Report 3.

ANRECOV returns 101.7 kmol/h of water to COL-1 with the recovered aniline, which is why the COL-1 distillate rate is set at 150 kmol/h and the reboiler duty at 11.08 MW. Routing this stream to DEC instead allows the decanter to remove the same water by phase separation at no energy cost.

Total heating in the flowsheet is 22.48 MW and total cooling 45.16 MW against a reaction exotherm of 17.57 MW, while the FEHE presently recovers 0.251 MW. The heat exchanger network design of Report 5 shall address this.

6 Objectives Which Could Not Be Accomplished, With Reasons

- The aniline–nitrobenzene binary pair has not yet been regressed, the laboratory time in this period having gone to building and converging the flowsheet. The pair bears directly on COL-2, whose reboiler runs at 144 °C while the databank parameters are valid over 83–99 °C, and the same shall be completed before the column design of Report 3.

7 Any Other Challenges

- MIX-1 has been left at a pressure specification of zero, so its outlet takes the lowest inlet pressure, which is the 0.3 bar of the COL-2 bottoms. A pump belongs at that point and shall be added with the equipment list.
- COL-1, with 20 stages and three-phase convergence, is the slowest block in the flowsheet and is re-solved on every recycle iteration. The same shall need attention when the loop is run repeatedly for the sensitivity studies of Report 4.

8 Number of Hours Spent During This Period

Approximately **4 hours per member**, that is about **24 person-hours** for the team in this reporting period. This covers the property and operating-condition analysis, the process synthesis and the flow diagram, the Aspen Plus laboratory sessions in which the flowsheet was built and converged, the extraction of the balances and the preparation of this report.

9 Contributions From Individual Members

- **Nithin T M — flowsheet development in Aspen Plus.** Construction of the entire flowsheet block by block, the property set-up with the Henry component list and the regressed parameters, the reaction specification in RStoic, the configuration of the decanter and all three RadFrac columns, the closing of the three recycle loops, the purge specification and the convergence of the complete case.
- **Sajag Masane — process synthesis and the process flow diagram.** The hierarchy of synthesis decisions, the recycle structure, the argument for taking the phase splits before distillation, the ordering of the separation train against the boiling point data, the placement of the feed–effluent exchanger, and the preparation of the process flow diagram.
- **Aditya Nitin Patil — material and energy balance.** Extraction of the converged stream and block results, the overall and component balances, the hydrogen loop and purge calculation, the energy balance with the reactor duty decomposition, and the closure checks against Report 1.
- **Nithin D H — component slate and operating conditions.** Compilation of the pure component boiling point data and the tabulation of the operating conditions with their bases.

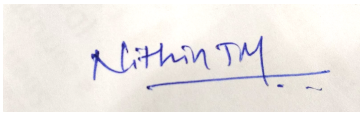
- **Shubham Singh — column specifications.** Tabulation of the converged column specifications and results, and cross-checking of the reported condenser and reboiler duties.
- **Shivam Dongare — documentation.** Assembly of the figures from the Aspen Plus flowsheet, the reference list and the compilation of this report.

All the six members have reviewed the complete report and each of us is able to explain the sections owned by the others.

10 References

1. T. Kahl et al., "Aniline", *Ullmann's Encyclopedia of Industrial Chemistry*, DOI 10.1002/14356007.a02_303.
2. B. Amon, H. Redlingshöfer, E. Klemm, E. Dieterich, G. Emig, *Chem. Eng. Process.* 38 (1999) 395–404, DOI 10.1016/S0255-2701(99)00037-9.
3. J. M. Douglas, *Conceptual Design of Chemical Processes*, McGraw-Hill, 1988.
4. W. D. Seider, J. D. Seader, D. R. Lewin, *Product and Process Design Principles*, 3rd ed., Wiley.
5. W. L. Luyben, *Distillation Design and Control Using Aspen Simulation*, 2nd ed., Wiley.
6. Aspen Technology, *Aspen Plus V14 Unit Operation Models* and *Physical Property Methods*.
7. Own regression case R-DR-1, ANILINE–WATER NRTL parameters, Report 1 Section 6.4.
8. Aspen Plus simulation file `Simulation1_che453.bkp`, with results exported as the stream results and model results summary workbooks, 21 August 2026.

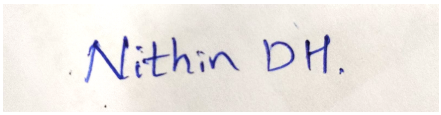
Signatures of members



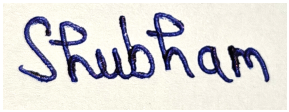
Nithin T M



Aditya Nitin Patil



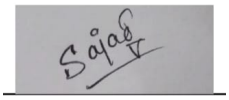
Nithin D H



Shubham Singh



Shivam Dongare



Sajag Masane

Bi-weekly Report 3

Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

CHE453 Process Design Capstone Project

Group No: 14

Date: 6 September 2026

Reports 1 and 2 precede this page and are unchanged.
Everything which follows constitutes Report 3.

Bi-weekly Report

Report Number: 03
Group No: 14
Date: 6 September 2026

Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

Team members

Name	Roll No.	Name	Roll No.
Nithin T M	230709	Shubham Singh	230999
Aditya Nitin Patil	230074	Shivam Dongare	230971
Nithin D H	230708	Sajag Masane	230896

Objectives for this report

1. Distillation column design and sequencing.
2. Gas-turbine based captive power plant.
3. Cooling water loop.

1 Basis

The flowsheet delivered in Report 2 was converged and its balances closed, but the product was not on specification and a few defects were noted at that time. Those items are listed below. The present report takes up all of them, and whatever is still open is given in Section 7.

Table 1: Position at the close of Report 2

Item	Report 2 status	Committed in
Aniline purity, 99.9 wt% required	97.04 wt%	Report 2, Section 5
Water in product, below 0.1 wt%	1.83 wt%	Report 2, Section 5
Nitrobenzene in product, below 5 ppm	660 ppm	Report 1, Section 2
Aniline in wastewater, below 1 ppm	2135 ppm	Report 1, Section 2
Product at 40 °C, 1.0 bar	86.2 °C, 0.2 bar	Report 2, Section 5
COL-1 total condenser on a feed carrying non-condensables	Condenser at −263.15 °C	Report 2, Section 5
ANRECOV returning water to COL-1	101.7 kmol/h returned	Report 2, Section 5
Aniline returned to the reactor in NBBOT	14.59 kmol/h	Report 2, Section 5
MIX-1 pressure specification of zero	Pump absent	Report 2, Section 7

Two Aspen Plus V14 files have been used. The process flowsheet `che453_R3_12.apw` keeps NRTL-RK, with HYDRO-01 and NITRO-02 as Henry components and the ANILINE–WATER pair on

our regression case R-DR-1 from Report 1. The captive power plant is a separate file on Peng–Robinson, which suits a gas-phase combustion system with no liquid phase. The two files are not linked; the purge is taken as a fixed feed to the power plant.

1.1 Key results

Aniline recovery through the separation train is 98.09 per cent, against 96.5 per cent in Report 2. The product leaves at 40 °C and 1.0 bar with 99.39 wt per cent aniline and 0.04 wt per cent water, against 97.04 and 1.83 wt per cent in Report 2. The selectivities have come back to the design values of Report 1. Total heating is 20.40 MW and total cooling 41.89 MW. The cooling water loop circulates 1578 t/h with 43.6 t/h of make-up. The captive gas turbine gives 709.1 kW net against a plant demand of 708.5 kW.

2 Distillation Column Design and Sequencing

2.1 The sequence and the reason for it

The sequence is fixed by the aniline–water heterogeneous azeotrope at $x_{\text{aniline}} = 0.0368$ and 99.34 °C, which we obtained from our own regression in Report 1. No single column can separate this pair, because across the two-liquid region the mixture boils at one temperature and gives a vapour of one composition. A decanter breaks the azeotrope by using the miscibility gap directly, so the decanter has to come before the distillation train. This is not a choice, it follows from the phase behaviour.

The remaining choices follow the usual heuristics, which here do not conflict.

- **Free splits first.** The vapour–liquid split is obtained at no separation cost in FLASH and the liquid–liquid split at no separation cost in DEC. Only the residue is handed to distillation.
- **Most plentiful component first.** The organic phase carries 47.9 kmol/h of water against 134.7 kmol/h of aniline, and the water is removed first in COL-1.
- **Hardest split last.** Aniline and nitrobenzene differ by 26.7 K in boiling point and this split carries the 5 ppm nitrobenzene limit. Placed last, its feed is dry and its throughput the smallest of the three, so the column required is the smallest.

The aqueous phase is stripped separately, its treatment being a discharge requirement rather than a product requirement.

2.2 COL-1 — water column

Stages required

COL-1 dries rather than fractionates. At the reboiler the vapour pressure of water is of order 5 bar against 0.13 bar for aniline, and water is strongly positive-deviating in aniline, giving a relative volatility of order 100. Fenske for 99.9 per cent recovery of each key:

$$N_{\min} = \frac{\ln\!\left[\left(\frac{x_{LD}}{x_{HD}}\right)\left(\frac{x_{HB}}{x_{LB}}\right)\right]}{\ln \alpha} = \frac{\ln\!\left[(0.999/0.001)^2\right]}{\ln 100} = \frac{13.81}{4.605} \approx 3$$

Three theoretical stages at total reflux. The twenty stages of Report 2 were therefore not doing separation work; they were present because the misrouted ANRECOV stream was returning

water to the column from which water was being removed, so the same duty was being performed twice. Twelve stages have been adopted, four times the minimum, which leaves margin for stage efficiency.

Condenser

A total condenser has to give a fully liquid distillate. The COL-1 overhead carries hydrogen and nitrogen, which are supercritical here and have no condensation temperature, so no solution exists and the solver goes to its lower flash bound of 10 K. That is the −263.15 °C of Report 2, and the −12.888 MW with it is not a condenser duty at all.

A Partial-Vapor-Liquid condenser gives a vapour and a liquid distillate together and needs only the ratio between them. The non-condensables leave in VENT. Three values of the distillate vapour fraction were tried.

Table 2: Selection of the distillate vapour fraction on COL-1

Distillate vapour fraction	Condenser T, °C	Condenser duty, MW	Outcome
0.037	55.7	−7.24	Converged; approach to CW only 10.7 K
0.060	71.0	−6.86	Converged
0.100	—	—	Failed to converge on Report 2 feed
0.100	66.9	−2.76	Converged after the reroute; adopted

At 0.10 on the original feed the run stopped with SOLVER01 not converging, because taking too much of the overhead as vapour leaves the top stage short of liquid. After the ANRECOV reroute of Section 2.3 the feed is lighter and 0.10 converges. It has been kept, since it gives the largest approach to the cooling water.

Drying specification and the steam level

Water leaving in the COL-1 bottoms follows the aniline into the COL-2 distillate and appears directly in the product, so the bottoms water content controls the product water specification. A design specification has been set on the water mole fraction in DRYORG at 0.002, manipulating the distillate rate over 50 to 75 kmol/h. It converged at 60.19 kmol/h with $x_{H_2O} = 0.00200$.

Design specifications have been preferred to sensitivity analyses throughout, for the reason in Section 9.

The consequence of drying properly is that the reboiler rises from 151.89 °C to 185.95 °C, this being the bubble point of essentially dry aniline at 1.099 bar. Saturated steam at 10 bar is available at 184 °C and cannot drive it. COL-1 has therefore been placed on 16 bar steam at 201 °C. This is unavoidable if the water specification is to be met, and the consumption at that level is given in Section 3.8.

Aniline lost overhead: why reflux cannot recover it

A second design specification was attempted on aniline mole recovery to the bottoms at 0.995, manipulating the reflux ratio over 2 to 8. It failed with

```
DESIGN SPEC IS NOT SATISFIED BECAUSE ONE OR MORE
MANIPULATED VARIABLE IS AT ITS BOUND
```

The aniline overhead is not there because of poor rectification. It is there because the overhead composition sits at the heterogeneous azeotrope. WATOVHD carries $x_{\text{aniline}} = 0.0464$ against an azeotropic composition of 0.0368, so the column is already up against the azeotrope and no reflux ratio can get past it. The specification was deleted.

The column was then put back on Vapor-Liquid-Liquid with WATER as the key component, so that the two liquid phases formed on condensation are represented. Report 2 declared three phases over stages 2–19; here it is kept to stage 1 only, since that is where the split happens, and this also keeps the solver load down. The condenser rose from 55.7 to 66.9 °C and both duties fell slightly. The aniline lost overhead did not change much, being fixed by the azeotrope; it is given in Section 3.3.

Table 3: COL-1, Report 2 against Report 3

	Report 2	Report 3
Number of stages	20	12
Feed stages	ORG 10, ANRECOV 2	ORG 4
Condenser	Total	Partial-Vapor-Liquid
Valid phases	VLL, stages 2–19	VLL, stage 1
Distillate vapour fraction	—	0.10
Operating specifications	D 150 kmol/h, RR 3	$x_{\text{H}_2\text{O},B} = 0.002$, RR 3
Top / reboiler pressure, bar	1.0 / 1.1	1.0 / 1.099
Distillate, kmol/h	150.00	60.19
Bottoms, kmol/h	158.42	134.29
Boilup ratio	5.45	2.25
Condenser temperature, °C	−263.15	66.91
Condenser duty, MW	−12.888	−2.763
Reboiler temperature, °C	151.89	185.95
Reboiler duty, MW	+11.076	+3.896
Water in DRYORG, mol %	9.0	0.20
Steam level	10 bar	16 bar

2.3 Rerouting of the recovered aniline

ANRECOV was returned to COL-1 in Report 2 and carried water into the column. It has been routed to DEC instead, where the decanter performs the same phase separation at no energy cost.

ANRECOV leaves the stripper as vapour at 97.5 °C and 1.0 bar whereas DEC operates at 40 °C and 1.5 bar, so a condenser ARCOND has been introduced at 40 °C and 1.5 bar. It is specified liquid-only. A pump placed downstream of a two-phase specification failed with

```
NET POSITIVE SUCTION HEAD IS LESS THAN ZERO
FLASH CALCULATIONS FAILED TO CONVERGE IN 30 ITERATIONS
```

because the stream carries hydrogen, nitrogen and ammonia and still has a small vapour fraction at 40 °C. Setting the block liquid-only removes this. A pump of below 0.2 kW is needed here in the real plant and is listed as an equipment item though not shown as a block.

The first attempt at the rerouted case also failed inside COL-1 with

```
THE FOLLOWING STAGES DRIED UP (VAPOR OR LIQUID FLOW
APPROACHES 0):  8 9 11
COMPONENT BALANCE EQUATION FAILED TO CONVERGE
BLOCK COL-1 IS NOT IN MASS BALANCE, RELATIVE DIFFERENCE = 0.229
```

The problem was the operating specification, not the piping. The column was set on a fixed bottoms rate of 158 kmol/h while its feed had dropped from 308 to about 205 kmol/h, so during the recycle iterations the feed went below the bottoms rate, the distillate turned negative and the stages ran dry. Specifying the distillate rate instead, and letting the bottoms follow by difference, fixed it.

The rerouting is not free. The COL-1 reboiler falls by 7.18 MW, but a condensing duty of 7.659 MW appears at ARCOND. So the total duty is about the same. What changes is the grade: a duty which earlier needed 16 bar steam is now taken away by cooling water, which is the cheapest utility on the plant. The other effect is that the circulation around the decanter and the stripper goes up a lot, the stripper feed rising from 244 to 851 kmol/h, which is taken up in Section 2.5.

2.4 COL-2 — aniline column

The feed changed entirely with the COL-1 redesign, the water in DRYORG falling from 9.0 to 0.20 mol per cent, so the column was re-specified rather than re-tuned.

The defect lay in the operating specification. A distillate-to-feed ratio of 0.9 forces the bottoms to be one tenth of the feed irrespective of how much nitrobenzene is present. The feed carries 1.32 kmol/h of nitrobenzene, so a bottoms of 14.6 kmol/h is oversized elevenfold and must carry aniline with it. Replacing the ratio by a bottoms rate of 2 kmol/h reduced the aniline returned to the reactor from 13.39 to 0.831 kmol/h.

This matters beyond the separation train. Both side reactions are specified on aniline, so aniline returned in NBBOT re-enters the reactor and is partly destroyed. Removing it restored the selectivity to the design values, as shown in Section 3.2.

Two further changes were tested and both rejected on the evidence.

Table 4: COL-2 parameters tested and rejected

Change	Gain	Cost	Decision
Reflux ratio 2 → 4	NB 713 → 295 ppm	Reboiler duty +74 %	Rejected
Bottoms 2 → 3 kmol/h	NB 1466 → 1230 ppm	Aniline recovery 99.4 → 98.6 %	Rejected

In the first case the purity scarcely moved despite the duty rise, because at that point cyclohexylamine and water rather than nitrobenzene were the binding impurities; the reflux was being spent on the wrong separation. In the second the nitrobenzene fell only 16 per cent while a further 0.97 kmol/h of aniline was pushed into the recycle. Both were reverted.

The ANILINE–NITRO-01 pair is presently on the Aspen databank, whose stated temperature range is 83 to 99 °C. Regression of this pair against NIST TDE data, on the same lines as the ANILINE–WATER regression of Report 1, is planned in the next period, and the nitrobenzene figures above shall be confirmed once that is in place.

Table 5: COL-2, Report 2 against Report 3

	Report 2	Report 3
Number of stages, feed stage	15, 6 on-stage	15, 6 on-stage
Operating specifications	$D{:}F$ 0.9, RR 2	Bottoms 2 kmol/h, RR 2
Top / reboiler pressure, bar	0.2 / 0.3	0.2 / 0.3
Distillate, kmol/h	142.58	132.29
Bottoms, kmol/h	15.84	2.00
Aniline in bottoms, kmol/h	14.591	0.831
Bottoms composition	92 mol % aniline	58 mol % nitrobenzene
Condenser temperature, °C	—	129.10
Condenser duty, MW	−6.490	−5.322
Reboiler temperature, °C	144.15	153.15
Reboiler duty, MW	+5.959	+4.852

Why the light-ends draw proposed in Report 2 does not work

Report 2 proposed a light-ends draw as the route to 99.9 wt per cent, both water and cyclohexylamine being lighter than aniline. Since COL-1 now removes the water, cyclohexylamine is the only remaining light end of consequence, and the natural place to take it is COL-1, where an intermediate boiler between water at 100.0 °C and aniline at 184.1 °C should concentrate. Two draws were tested.

Table 6: Cyclohexylamine side draw trials on COL-1

Location	Rate, kmol/h	CHA in draw, mol %	Draw composition
Stage 8	2.0	0.21	67.5 mol % water, 32.0 mol % aniline
Stage 6	1.0	0.12	Water and aniline

Both draws took out water and aniline and left the cyclohexylamine behind. The stage 8 draw carried 0.64 kmol/h of aniline out for only 0.004 kmol/h of cyclohexylamine, and cyclohexylamine in DRYORG did not fall. The composition profile shows no intermediate peak anywhere in COL-1. Both draws were deleted.

The reason is that boiling point alone is a poor guide here. Cyclohexylamine and aniline are both amines of similar polarity and similar molar mass, so even with a 49.6 K difference in boiling point their relative volatility in the aniline-rich liquid is close to one. It is the same kind of effect as the aniline–water azeotrope: what the vapour pressures suggest is misleading, and the activity coefficients decide the outcome.

Two routes remain. One is a separate light-ends column on DRYORG, which means a fourth column. The better one is to attack it at source, since cyclohexylamine comes from over-hydrogenation of aniline at 1.5 per cent selectivity, and a more selective catalyst or lower hydrogen partial pressure would cut it down and improve the yield together. This links to the kinetic model due in Report 4.

2.5 STRIP — wastewater stripper

The aqueous phase rose from 244 to 851 kmol/h with the ANRECOV rerouting, the water now leaving through the stripper rather than the COL-1 overhead. At the boilup ratio of unity carried from Report 2 the residual aniline rose to 2.6 wt per cent, and the boilup ratio was studied as a series of discrete converged cases.

Table 7: Effect of the stripper boilup ratio

Boilup ratio	Reboiler duty MW	Overhead kmol/h	Aniline in WASTEWTR kmol/h	Aniline ppm mass
1.0, as Report 2	2.62	187	1.237	26,200
3.0, adopted	7.97	615	0.0073	154
5.0	13.06	1025	0.0027	57

From unity to three the residual falls by a factor of 177. From three to five it falls by a factor of 2.7 only, at the cost of a further 5.09 MW of reboiler duty and a further 410 kmol/h of overhead returning to the decanter and thence to COL-1. Neither case meets the 1 ppm discharge limit and both therefore require the same polishing step, so the case of lower duty has been adopted. The saving is 5.09 MW of 10 bar steam, which is about 13 per cent of the total plant heating duty.

The residual of 154 ppm is a reduction by a factor of 13.9 on the 2135 ppm of Report 2 but remains two orders of magnitude above the limit. Equilibrium stripping cannot reach 1 ppm for this system at any practicable boilup, and a polishing step, either activated carbon adsorption or biological treatment, shall be required. This is a limitation of the unit operation, not of the model.

The wastewater also carries 1.637 kmol/h of cyclohexylamine, that is 3.7 wt per cent, which is a larger discharge problem than the aniline and is recorded in Section 8.

2.6 Hydraulic completion and product conditioning

Table 8: Blocks added

Block	Model	Specification	Result
ARCOND	Heater	40 °C, 1.5 bar, liquid only	−7.659 MW
P-1	Pump	0.2 to 1.0 bar, $\eta = 0.75$, driver 0.95	0.392 kW, head 8.34 m
PRODCOOL	Heater	40 °C, 1.0 bar	−0.664 MW
P-2	Pump	0.3 to 3.0 bar, $\eta = 0.75$, driver 0.95	0.023 kW, head 27.03 m

P-2 closes the defect of Report 2 Section 7, MIX-1 having adopted the 0.3 bar of the COL-2 bottoms with VAP-1 thereafter repressurising to 3 bar without a pump. The MIX-1 pressure specification has been set to 3.0 bar accordingly. P-1 and PRODCOOL bring the product from 129.10 °C and 0.2 bar to 40 °C and 1.0 bar as specified.

One point for detailed design: the pump discharge computes at 129.13 °C against a condenser temperature of 129.10 °C, so the pump takes suction at the bubble point and the available NPSH is nominal. Either a few degrees of subcooling shall be taken before the pump or adequate static head provided at the reflux drum.

3 Material and Energy Balance

3.1 Overall material balance

Table 9: Overall material balance

In	kmol/h	kg/h	Out	kmol/h	kg/h
NB-FEED	141.28	17,393.2	PRODUCT	132.29	12,307.2
H2-MAKE	444.00	906.6	WATOVHD	54.17	1,306.1
			WASTEWTR	235.44	4,374.9
			VENT	6.02	262.6
			PURGE	9.79	49.0
Total		18,299.7	Total		18,299.8

The balance closes to 0.01 kg/h in 18,299.7 kg/h, that is 0.00006 per cent, against 0.7 kg/h in Report 2. The fresh nitrobenzene of 141.28 kmol/h and hydrogen make-up of 444.0 kmol/h are unchanged from Report 2 and agree with the hand balance of Report 1.

3.2 Reactor performance and the hydrogen loop

Table 10: Reactor performance against Report 1 and Report 2

Quantity	Report 1, hand	Report 2	Report 3
Nitrobenzene to reactor, kmol/h	142.71	142.54	142.46
Nitrobenzene reacted, kmol/h	141.28	141.11	141.03
Conversion per pass, %	99.0	99.0	99.0
Hydrogen at reactor inlet, kmol/h	1427.1	1300.6	1461.8
Hydrogen-to-nitrobenzene ratio	10.00	9.12	10.26
Hydrogen consumed, kmol/h	435.15	435.88	434.54
Net aniline produced, kmol/h	134.22	133.31	133.91

The selectivities have returned to the design values of Report 1, which the reduction of the aniline recycle in Section 2.4 was intended to achieve:

$$S_{\mathrm{AN}} = \frac{133.911}{141.034} = 94.95\ \%, \qquad S_{\mathrm{CHA}} = \frac{2.160}{141.034} = 1.53\ \%, \qquad S_{\mathrm{BZ}} = \frac{4.963}{141.034} = 3.52\ \%$$

against the design split of 95 / 1.5 / 3.5 per cent and against 94.47 per cent achieved in Report 2. The hydrogen-to-nitrobenzene ratio of 10.26 is also closer to the Report 1 assumption of 10 than the 9.12 obtained in Report 2, the recycle gas now settling at 90.2 mol per cent hydrogen and 4.5 mol per cent nitrogen.

The purge fraction remains as derived in Report 1, the hydrogen leaving in the purge equalling the make-up less what reacts and less what dissolves:

$$f_{\mathrm{purge}} = \frac{M - C - D}{V} = \frac{443.556 - 434.543 - 0.128}{1027.105} = 0.00865$$

against the 0.0086 entered in SPLIT.

3.3 Aniline accounting

Table 11: Disposition of the aniline produced in the reactor, 133.911 kmol/h

Destination	kmol/h	kg/h	% of production
PRODUCT	131.351	12,232.5	98.09
WATOVHD	2.515	234.2	1.88
VENT	0.031	2.8	0.02
WASTEWTR	0.007	0.7	0.01
PURGE	0.008	0.7	0.01

Recovery through the separation train is 98.09 per cent against 96.5 per cent in Report 2. The dominant remaining loss is WATOVHD at 234 kg/h, discussed in Section 8; the 0.831 kmol/h in NBBOT is not a loss, being returned to the reactor.

3.4 Component molar flows, reaction section (kmol/h)

	NB-FEED	H2-MAKE	RXFEED	RXOUT	VAP-OUT	RECYCLE	PURGE	LIQ-OUT
Aniline	—	—	1.709	135.620	0.885	0.877	0.008	134.735
Nitrobenzene	141.280	—	142.458	1.425	0.010	0.010	tr	1.415
Water	—	—	40.053	322.121	40.407	40.059	0.347	281.714
Benzene	—	—	17.211	22.174	17.358	17.208	0.149	4.816
Cyclohexylamine	—	—	0.020	2.180	0.020	0.020	tr	2.160
Nitrogen	—	0.444	50.822	50.822	50.815	50.378	0.437	0.007
Hydrogen	—	443.556	1461.776	1027.233	1027.105	1018.272	8.833	0.128
Ammonia	—	—	2.072	7.035	2.091	2.073	0.018	4.944
Total	141.280	444.000	1716.121	1568.609	1138.690	1128.897	9.793	429.919

3.5 Component molar flows, separation section (kmol/h)

	ORG	AQ	ANRECOV	VENT	WATOVHD	DRYORG	PRODUCT	NBBOT	WASTEWTR
Aniline	134.728	15.308	15.301	0.031	2.515	132.183	131.351	0.831	0.007
Nitrobenzene	1.415	0.024	0.024	0.001	0.099	1.315	0.147	1.169	—
Water	47.919	793.864	560.068	1.537	46.113	0.269	0.269	—	233.795
Benzene	4.816	0.154	0.154	2.586	2.230	tr	tr	—	—
Cyclohexylamine	0.523	2.037	0.401	tr	tr	0.523	0.523	—	1.637
Nitrogen	0.007	0.004	0.004	0.007	—	—	—	—	—
Hydrogen	0.128	0.112	0.112	0.128	—	—	—	—	—
Ammonia	4.944	39.422	39.422	1.729	3.215	—	—	—	—
Total	194.480	850.925	615.486	6.019	54.172	134.289	132.289	2.000	235.439

One point from these tables matters for sizing. Benzene is made at 4.963 kmol/h per pass but 17.21 kmol/h circulates in the gas loop, and ammonia is made at 4.963 kmol/h but 39.42 kmol/h circulates in the aqueous loop. Both close at steady state, but the circulating amounts are three to eight times the production rates, and it is these that fix the equipment sizes.

3.6 Stream conditions

Stream	T, °C	P, bar	Vapour fraction	kmol/h	kg/h
NB-FEED	25.00	5.00	0.000	141.28	17,393.2
NBRECYC	153.25	3.00	0.000	2.00	221.3
S1	27.12	3.00	0.000	143.28	17,614.4
S2	263.05	3.00	1.000	143.28	17,614.4
S3	290.00	3.00	1.000	143.28	17,614.4
H2-MAKE	25.00	5.00	1.000	444.00	906.6
H2RET	69.41	3.20	1.000	1128.84	5,650.1
RXFEED	153.33	3.00	0.987	1716.12	24,171.1
RXOUT	300.00	3.00	1.000	1568.61	24,171.1
S4	289.66	3.00	1.000	1568.61	24,171.1
S5	120.00	2.80	0.945	1568.61	24,171.1
S6	40.00	2.60	0.725	1568.61	24,171.1
VAP-OUT	40.00	2.50	1.000	1138.69	5,699.1
PURGE	40.00	2.50	1.000	9.79	49.0
RECYCLE	40.00	2.50	1.000	1128.90	5,650.1
LIQ-OUT	40.00	2.50	0.000	429.92	18,472.0
ANRECOV	97.50	1.00	1.000	615.49	12,241.2
ARLIQ	40.00	1.50	0.000	615.49	12,241.2
ORG	40.00	1.50	0.000	194.48	14,097.2
AQ	40.00	1.50	0.000	850.92	16,616.1
VENT	66.91	1.00	1.000	6.02	262.6
WATOVHD	66.91	1.00	0.000	54.17	1,306.1
DRYORG	185.95	1.099	0.000	134.29	12,528.5
WASTEWTR	101.15	1.05	0.000	235.44	4,374.9
PRODHOT	129.10	0.20	0.000	132.29	12,307.2
PRODWARM	129.13	1.00	0.000	132.29	12,307.2
PRODUCT	40.00	1.00	0.000	132.29	12,307.2
NBBOT	153.15	0.30	0.000	2.00	221.3

3.7 Product stream

The contained aniline is 12,232.5 kg/h, that is 12.23 t/h against a design rate of 12.5 t/h and 97,860 t/y against a design basis of 100,000 t/y, or 97.9 per cent of capacity. Report 2 delivered 11.98 t/h.

The water content of 0.04 wt per cent is within the 0.1 wt per cent limit, having been closed upstream in COL-1 rather than in the product column, and the delivery conditions of 40 °C and 1.0 bar follow from the conditioning of Section 2.6. The residual 0.61 wt per cent of impurity is chiefly cyclohexylamine, for the reason established in Section 2.4. The nitrobenzene content of 1466 ppm is higher than the 660 ppm of Report 2, this being the accepted consequence of reducing the bottoms cut: the trade is 12.6 kmol/h of aniline recovered against 0.08 kmol/h of additional nitrobenzene admitted.

Table 12: PRODUCT, 132.29 kmol/h and 12,307.2 kg/h at 40 °C and 1.0 bar

Component	kmol/h	mol %	kg/h	wt %	Report 2, wt %
Aniline	131.351	99.29	12,232.5	99.39	97.04
Cyclohexylamine	0.523	0.40	51.8	0.42	1.06
Nitrobenzene	0.147	0.11	18.0	0.15	0.07
Water	0.269	0.20	4.8	0.04	1.83
Total	132.29	100.00	12,307.2	100.00	

3.8 Energy balance

Table 13: Block duties and assigned utilities

Block	Service	cal/s	MW	Utility
VAP-1	Feed vaporiser	+874,407	+3.659	HPSTEAM
FEHE	Feed–effluent exchanger	50,324	0.211	Internal
REACT	Reactor, isothermal	−3,669,783	−15.354	Unassigned, steam raising
COOL-1	Effluent cooler to 120 °C	−1,046,828	−4.380	Unassigned, steam raising
COOL-2	Effluent cooler to 40 °C	−1,331,805	−5.572	CW
FLASH	Flash drum	+5,188	+0.022	Negligible
DEC	Decanter	−42,163	−0.176	Chilled water or air cooler
ARCOND	Stripper overhead condenser	−1,830,523	−7.659	CW
COL-1	Condenser	−660,333	−2.763	CW
COL-1	Reboiler	+931,068	+3.896	MPSTEAM
COL-2	Condenser	−1,271,875	−5.322	CW
COL-2	Reboiler	+1,159,755	+4.852	LPSTEAM
STRIP	Reboiler	+1,905,416	+7.972	LPSTEAM
PRODCOOL	Product cooler	−158,760	−0.664	CW
COMP	Shaft work	—	+0.278	Electrical
P-1, P-2	Shaft work	—	+0.0004	Electrical

Taking FEHE as an internal transfer, the block duties and shaft work sum to −21.211 MW against −21.210 MW obtained from the enthalpy flows of the terminal streams, so the energy balance closes to 0.005 per cent. Total heating is 20.40 MW and total cooling 41.89 MW, against 22.48 and 45.16 MW in Report 2.

The steam level for each heated service is fixed by the temperature it has to drive and not by the size of its duty. VAP-1 vaporises the feed at 263.05 °C, so it needs 70 bar steam at 285.8 °C, which gives a 22.8 K approach; 50 bar steam at 263.9 °C would leave less than 1 K and is of no use. COL-1 reboils at 185.95 °C and cannot be served by 10 bar steam at 184 °C, so it takes the 16 bar level at 201 °C. COL-2 at 153.15 °C and STRIP at 101.15 °C are both comfortably served by 10 bar steam, and STRIP therefore takes the cheapest level in the plant even though it carries the largest duty of the four.

Three points follow. The stripper reboiler at 7.972 MW is now the largest single heating load, exceeding the COL-1 reboiler twofold; the ANRECOV rerouting moved the burden rather than eliminating it, and moved it from a 16 bar to a 10 bar steam service, which is the gain. The reaction exotherm of 15.354 MW at 300 °C against a heating requirement of 20.40 MW leaves

Table 14: Utility consumption, from Aspen Plus

Utility	Block	Service	Usage, kg/h
Steam, 70 bar	VAP-1	Feed vaporiser	8,753
Steam, 16 bar	COL-1	Reboiler	7,261
Steam, 10 bar	STRIP	Reboiler	14,265
Steam, 10 bar	COL-2	Reboiler	8,683
Steam total			38,962
Cooling water	ARCOND	Condenser	552,133
Cooling water	COOL-2	Effluent cooler	401,711
Cooling water	COL-2	Condenser	383,631
Cooling water	COL-1	Condenser	199,179
Cooling water	PRODCOOL	Product cooler	47,886
Cooling water total			1,584,540

the plant short by approximately 5 MW even under perfect integration. And heat integration remains nominal: FEHE recovers only 0.211 MW because it heats the nitrobenzene vapour alone at 143 kmol/h while 1573 kmol/h of hydrogen joins downstream at MIX-2, so 92 per cent of the reactor feed by mole bypasses the exchanger and RXFEED reaches only 153 °C. This is the subject of Report 5.

4 Cooling Water Loop

4.1 Assignment of services

A service is placed on cooling water where its heat is rejected too low to be recovered usefully. Three are excluded, for two different reasons.

REACT rejects 15.354 MW isothermally at 300 °C and COOL-1 rejects 4.380 MW between 287 and 120 °C. Both are in the range where steam can be raised. Report 1 gave recovery of the exotherm as steam as one of the reasons for choosing the vapour-phase route, and sending that heat to cooling water would go against it.

Taking the COOL-1 duty as fully recoverable, a waste-heat boiler there would raise about 7.8 t/h of 10 bar steam, which is 20 per cent of the plant steam demand given in Section 3.8, and would also cut the cooling water circulation by 315 t/h and the make-up by 8.7 t/h. This is a screening figure only. The stream is cooled down to 120 °C while the steam is raised at 184 °C, so the lower part of the duty cannot be recovered at that level, and the recoverable fraction has to be fixed by a temperature-profile calculation on the exchanger. That is taken up with the pinch analysis of Report 5. On the strength of the screening figure COOL-1 has been kept off the cooling water header; the alternative case, with COOL-1 on cooling water, needs 1893 t/h of circulation and 52.3 t/h of make-up and is noted for comparison only.

DEC is excluded because the cooling water cannot reach it. The block operates isothermally at 40 °C while the water returns at 42 °C, and Aspen Plus rejected the assignment with

```
TEMPERATURES CROSS WITH UTILITY CW IN DECANTER BLOCK DEC
BLOCK TEMPERATURES 3.131497D+02 K AND 3.131500D+02 K
UTILITY TEMPERATURES 3.181500D+02 K AND 3.031500D+02 K
```

This was a correct diagnosis of a real design problem and not a modelling error. The duty is

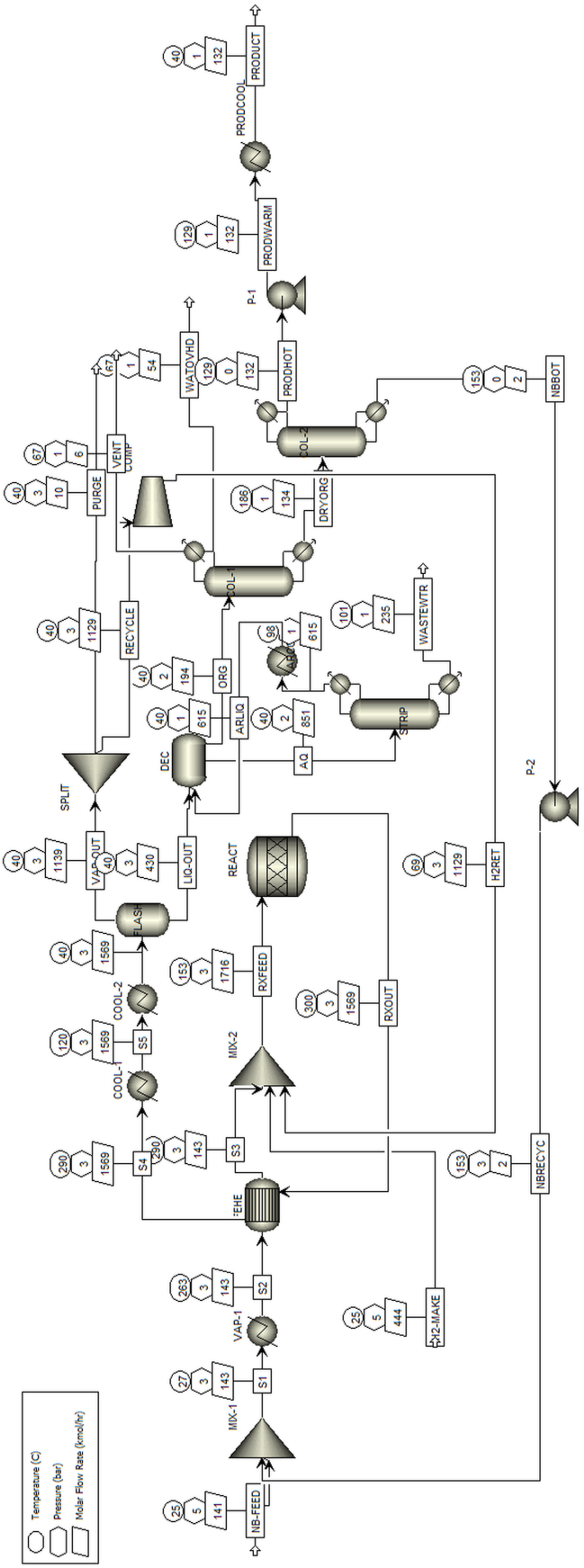


Figure 1: Process flow diagram as converged, file `che453_R3_12.apw`, annotated with temperature (°C), pressure (bar) and molar flow (kmol/h). ARCOND, P-1, P-2, PRODCOOL and the stream VENT are new in this report.

0.176 MW, 0.8 per cent of the load, and shall be handled by a trim exchanger on chilled water or an air-cooled unit.

The same consideration applies with greater force to COOL-2, which also operates at 40 °C but carries 5.572 MW. Supply and return were first set at 30 and 45 °C, a 15 K range, which places the return above the process temperature and is infeasible. They were revised to 30 and 42 °C, a 12 K range, which retains COOL-2 on the main header at a 2 K approach. The narrower range raises the circulation, and the trade is discussed in Section 4.4.

4.2 Circulation rate and verification

With water supplied at 30 °C and returned at 42 °C and a mean specific heat of 4.178 kJ/kg·K,

$$\dot{m} = \frac{Q}{c_p\,\Delta T} = \frac{21.980 \times 10^{6}}{4178 \times 12} = 438.4 \text{ kg/s} = 1578 \text{ t/h}$$

The loop was designed by hand and then computed independently in Aspen Plus by defining cooling water as a utility and assigning it to each block, so that the two may be compared.

Table 15: Cooling water services, hand calculation against Aspen Plus

Service	Hot side	Duty, MW	Hand, t/h	Aspen, t/h	Deviation, %
ARCOND	97.5 to 40 °C	7.659	550.0	552.1	0.38
COOL-2	120 to 40 °C	5.572	400.1	401.7	0.40
COL-2 condenser	129.1 °C condensing	5.322	382.2	383.6	0.37
COL-1 condenser	66.9 °C condensing	2.763	198.4	199.2	0.40
PRODCOOL	129 to 40 °C	0.664	47.7	47.9	0.42
Total		21.980	1578.3	1584.5	0.39

The two agree to 0.39 per cent overall and better than 0.5 per cent on every service. The residual arises from the mean specific heat, Aspen Plus evaluating it from the water property model across the actual range rather than at a single mean temperature.

4.3 Tower water balance

The tower balance is established by hand, no facility for it existing in Aspen Plus. With a mean latent heat of 2418 kJ/kg, four cycles of concentration and drift at 0.02 per cent of circulation:

$$E = \frac{Q}{\lambda} = \frac{21.980 \times 10^{6}}{2.418 \times 10^{6}} = 9.09 \text{ kg/s} = 32.7 \text{ t/h} \tag{1}$$

$$D = 0.0002 \times 1578 = 0.32 \text{ t/h} \tag{2}$$

$$B = \frac{E}{N-1} - D = \frac{32.7}{3} - 0.32 = 10.6 \text{ t/h} \tag{3}$$

$$M = E + D + B = \frac{NE}{N-1} = 43.6 \text{ t/h} \tag{4}$$

Equation (3) follows from the solids balance. If N is the ratio of dissolved solids in the circulating water to that in the make-up, then $M x_M = (B + D) N x_M$; eliminating M between this and the water balance $M = E + D + B$ gives the result.

As a check, the evaporation of 32.7 t/h is 2.07 per cent of circulation, against the usual rule of about one per cent per 5.6 K of range, which gives 2.14 per cent for a 12 K range. The closed form $M = NE/(N-1) = 43.6$ t/h also matches the make-up got by adding the three losses.

The figure is on the safe side, since part of the tower duty leaves as sensible heat to the air rather than as latent heat. Make-up at 43.6 t/h is 2.8 per cent of circulation, or 349,000 t/y over 8000 hours.

4.4 Equipment and design conditions

Pump power, with $Q = \dot{m}/\rho = 438.4/993 = 0.4415$ m^3/s, $H = 35$ m and $\rho = 993$ kg/m^3:

$$P_{\mathrm{hyd}} = \rho g H Q = 993 \times 9.81 \times 35 \times 0.4415 = 150.5 \text{ kW}, \qquad P_{\mathrm{elec}} = \frac{150.5}{0.75 \times 0.95} = 211 \text{ kW}$$

Fan power, at $L/G = 1.2$ giving an air rate of $438.4/1.2 = 365.3$ kg/s, that is 317.7 m^3/s at 1.15 kg/m^3, against 180 Pa static:

$$P_{\mathrm{elec}} = \frac{317.7 \times 180}{0.60 \times 0.95} = 100.3 \text{ kW}$$

Table 16: Cooling water loop equipment

Item	Basis	Result
Circulation	21.980 MW, 12 K range	1578 t/h, 1589 m^3/h
Circulation pump	35 m head, $\eta = 0.75$, motor 0.95	211 kW electrical
Induced-draft fans	$L/G = 1.2$, 180 Pa, $\eta = 0.60$	100 kW electrical
Tower cells	400 m^3/h per cell	4 duty, 1 standby
Basin	10 minutes holdup	265 m^3
Evaporation / make-up	4 cycles of concentration	32.7 / 43.6 t/h
Total parasitic power		311 kW

The 35 m head comprises approximately 10 m of tower lift, 15 m across the exchanger network and 10 m of piping and valve losses, and shall be firmed up once the piping is routed.

Two points are noted for later. A supply of 30 °C needs a design wet-bulb of about 25 °C at a 5 K approach, whereas Kanpur is 27 to 29 °C before the monsoon, so 33 to 34 °C is the realistic figure for detailed design. Also the 12 K range taken here to keep COOL-2 feasible is narrow; a 15 K range would bring the circulation down to 1263 t/h and the pump to 169 kW. Both are to be settled in Report 5 along with the 40 °C services.

5 Gas Turbine Captive Power Plant

5.1 Plant electrical demand

The plant is to be sized on the electrical load it has to carry, so that load is established first.

Table 17: Plant electrical demand

Consumer	kW	Source
COMP, recycle gas compressor	278.4	Aspen Plus block result
Cooling water circulation pump	211.3	Section 4.4
Cooling tower fans	100.3	Section 4.4
P-1 and P-2	0.4	Aspen Plus block results
Lighting, instrumentation and miscellaneous, 20 %	118.1	Allowance
Total	708.5	

5.2 Basis and sizing

The purge is the only combustible waste stream in the plant. Its lower heating value is 0.610 MW on the Report 2 flow, of which hydrogen contributes 82 per cent, so it cannot support a power plant alone and a natural gas make-up is provided.

The plant is sized so that the net electrical output matches the plant electrical demand of Section 5.1. A simple open Brayton cycle with exhaust heat recovery has been adopted, this being appropriate for a unit of this size on a chemical plant.

Air is supplied at 290 kmol/h, corresponding to 200 per cent excess over stoichiometric, selected to hold the turbine inlet temperature near 1150 °C, the limit consistent with uncooled blading in a machine of this class. The value obtained is 1153.85 °C.

A first case was built at 1207 kmol/h of air and 39.9 kmol/h of natural gas, which gave 2.966 MW net, four times the plant demand. The flame temperature depends on the fuel-to-air ratio and the efficiency on the pressure ratio and turbine inlet temperature, not on size, so the case was scaled down at the same ratio to 290 and 7.5 kmol/h. The flame temperature moved by only 0.5 K and nothing else needed changing.

5.3 Configuration and results

Table 18: Gas turbine blocks and results

Block	Model	Specification	Result
AIRCOMP	Compr	Isentropic, 1 to 12 bar, $\eta_{is} = 0.85$, $\eta_m = 0.98$	+864.9 kW, 377.3 °C
FUELCOMP	Compr	Isentropic, 2.5 to 12 bar, $\eta_{is} = 0.85$	+13.4 kW, 209.5 °C
COMBUST	RGibbs	12 bar, adiabatic, full equilibrium	1153.85 °C
TURBINE	Compr, turbine	Isentropic, 12 to 1 bar, $\eta_{is} = 0.88$	−1617.0 kW, 593.2 °C
HRSG	Heater	150 °C, 1 bar	−1.179 MW

The combustor is an RGibbs block minimising Gibbs free energy at constant pressure with zero heat duty, so the adiabatic flame temperature is an output and not an assumption. The stack temperature of 150 °C is kept well above the water dew point of the flue gas, so as to leave a margin against condensation and corrosion in the stack. No sulphur species are present in this fuel, so an acid dew point does not arise here.

Table 19: Fuel input

Fuel	kmol/h	LHV, MJ/kmol	Thermal, MW	Share, %
Purge, hydrogen	7.435	241.8	0.499	21.9
Purge, benzene	0.119	3135.6	0.104	4.5
Purge, aniline	0.006	3290	0.005	0.2
Purge, ammonia	0.017	316.8	0.001	0.1
Purge subtotal	8.31		0.610	26.7
Natural gas, as methane	7.5	802.6	1.672	73.3
Total			2.282	100.0

5.4 Fuel and performance

$$\text{Net shaft power} = 1617.0 - 864.9 - 13.4 = 738.7 \text{ kW} \tag{5}$$

$$\text{Net electrical power} = 738.7 \times 0.96 = 709.1 \text{ kW} \tag{6}$$

$$\eta_{\text{thermal}} = 709.1/2282 = 31.1\ \% \tag{7}$$

$$\eta_{\text{CHP}} = (0.709 + 1.179)/2.282 = 82.7\ \% \tag{8}$$

The net electrical output of 709.1 kW matches the plant demand of 708.5 kW of Section 5.1 to 0.08 per cent, which is what was required. The simple-cycle efficiency of 31.1 per cent is consistent with commercial machines of this class, which achieve 30 to 33 per cent, so the model has not returned an optimistic figure.

The purge now gives 26.7 per cent of the fuel input, against 6.4 per cent in the oversized case, so the waste hydrogen is a real part of the fuel and not a token.

The HRSG recovers 1.179 MW, raising approximately 1.8 t/h of 10 bar steam from feedwater at 105 °C. Against a plant steam demand of 39.0 t/h this is 4.6 per cent, so the recovered heat is a side benefit and not a solution to the steam shortfall noted in Section 3.8.

Matching the demand exactly leaves no margin for start-up or for the miscellaneous allowance being low. Normal practice is to size 10 to 15 per cent above demand and keep a grid connection for standby. The matched case is given here as required, and the margin is left to detailed design.

5.5 Combustion products and balance checks

The flue gas leaves at 302.16 kmol/h and 8529.6 kg/h, comprising 76.0 mol per cent nitrogen, 13.6 per cent oxygen, 7.7 per cent water and 2.7 per cent carbon dioxide, which is a normal lean gas-turbine exhaust. Three checks were made.

Carbon. Methane, benzene and aniline supply $7.5 + 0.119 \times 6 + 0.006 \times 6 = 8.250$ kmol/h of carbon, and the flue gas carries 8.25 kmol/h of carbon dioxide.

Oxygen. $290 \times 0.21 = 60.9$ kmol/h is supplied and 41.23 kmol/h remains, so 19.67 kmol/h is consumed. The stoichiometric requirement, taking 2 mol of oxygen per mol of methane, 0.5 per mol of hydrogen, 7.5 per mol of benzene, 7.75 per mol of aniline and 0.75 per mol of ammonia, is

$$7.5 \times 2 + \frac{7.435}{2} + 0.119 \times 7.5 + 0.006 \times 7.75 + 0.017 \times 0.75 = 19.669 \text{ kmol/h}$$

The residual oxygen corresponds to 200 per cent excess air as specified.

Completeness. Residual methane is of order 10^{-29} kmol/h, hydrogen 10^{-5} and ammonia 10^{-12}, confirming that the excess air is adequate and no unburnt fuel leaves the combustor.

The carbon dioxide emission is 363 kg/h, that is 2900 t/y, or 0.030 t per tonne of aniline.

Carbon monoxide and nitrogen oxides were not put in the component list, so the model cannot show them. At 200 per cent excess air the carbon monoxide would be negligible. The ammonia in the purge is fuel-bound nitrogen and would give some NOx, which is small here but is noted for the environmental section of a later report.

FUELCOMP reports `FEED STREAM IS BELOW DEW POINT`, the purge at 40 °C and 2.5 bar being just inside the two-phase region. A suction knockout drum belongs on the fuel compressor in the real plant.

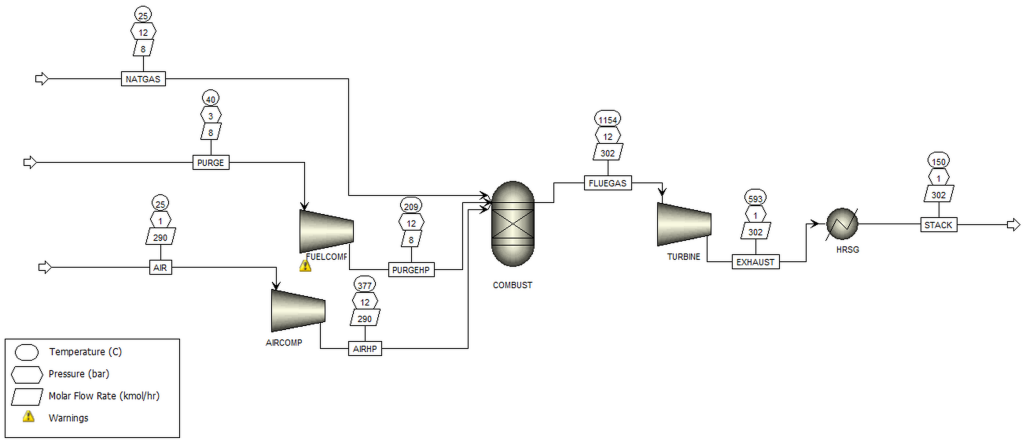


Figure 2: Gas turbine captive power plant as converged, annotated with temperature (°C), pressure (bar) and molar flow (kmol/h). Molar flows are shown rounded, so NATGAS at 7.5 kmol/h and PURGE at 8.31 kmol/h both appear as 8.

6 Objectives Which Could Not Be Accomplished, With Reasons

- **Aniline purity.** 99.39 wt % reached against 97.04 wt % in Report 2, the target being 99.9 wt %. The impurity holding it back is cyclohexylamine at 0.42 wt %, which cannot be taken out in either column for the reason given in Section 2.4. A light-ends column, or a lower over-hydrogenation selectivity in the reactor, is needed.
- **Nitrobenzene in the product.** 1466 ppm against a target of 5 ppm. A sharper cut needs more stages in COL-2 or a third column, and the binary parameters should be regressed first.
- **Aniline in the wastewater.** 154 ppm against 2135 ppm in Report 2, the target being 1 ppm. Section 2.5 shows that stripping alone cannot reach 1 ppm at any workable boilup, so a polishing step is needed.
- **Aniline in WATOVHD.** The stream carries 2.515 kmol/h of aniline, 234 kg/h. Its composition lies inside the miscibility gap, so sending it to the stripper or to a second decanter would recover most of it. To be taken up in Report 4.
- **Cyclohexylamine in the wastewater.** WASTEWTR carries 1.637 kmol/h, that is 3.7 wt %, to be handled along with the wastewater polishing step.
- **Heat integration.** A pinch analysis of the full network, including steam raising from REACT

and COOL-1, is left to Report 5 as stated in Report 2.

7 Any Other Challenges

- **The partial condenser had crashed the engine earlier.** The cause was a redundant specification. A Partial-Vapor-Liquid condenser already gives two distillate streams and needs only the ratio between them, so an extra liquid side draw over-specifies the column. The engine reported `PRODUCTS NOT COMPLETE` and then `SUM OF PRODUCT FLOW SPECIFICATIONS EXCEEDS FEED FLOW` before stopping. Deleting the side draw fixed it.
- **Design specifications were used in place of sensitivity analyses.** A sensitivity on the stripper boilup tried earlier gave `ERROR WHILE CHECKING INPUT SPECIFICATIONS FOR VARIED VARIABLE` and then a hundred `FORTRAN DIVIDE BY ZERO` errors which stopped the run. The stripper sits inside the recycle loop, so each sensitivity point re-tears all three recycles and a step landing on zero internal flow cascades. The boilup study of Section 2.5 was therefore done as separate converged runs.
- **The Strongly Non-Ideal Liquid algorithm could not be used**, Aspen Plus locking the algorithm to Standard whenever design specifications are active on a RadFrac block. Standard was adequate once three-phase behaviour was kept to the condenser stage.
- **Parameters tested and abandoned because another variable overshot.** Recorded in the sections concerned and collected here: the COL-1 distillate vapour fraction of 0.10 on the Report 2 feed, which failed to converge; COL-2 reflux ratio 4, which reduced nitrobenzene but raised the reboiler duty 74 per cent for no purity gain; COL-2 bottoms 3 kmol/h, which reduced nitrobenzene 16 per cent while pushing 0.97 kmol/h more aniline into the recycle; stripper boilup 5, which reduced residual aniline 2.7-fold for 5.09 MW of additional duty and 410 kmol/h of additional recycle; and the two cyclohexylamine side draws, which removed water and aniline instead of the impurity.
- **Work planned for the next period.** The ANILINE–NITRO-01 pair is to be regressed against NIST TDE data, on the same lines as the ANILINE–WATER regression of Report 1, and an Azeotrope Search is to be run over the aniline/water/nitrobenzene ternary for the third azeotrope noted in Report 1 Section 6.8.
- **Purge basis for the power plant.** The power plant is built on the Report 2 purge of 8.31 kmol/h. The converged flowsheet here gives 9.79 kmol/h, so the purge fuel value is about 0.72 MW instead of 0.610 MW. At the same turbine inlet temperature the natural gas simply comes down by the same amount, so the net power changes by well under one per cent and the sizing of Section 5 holds. The power plant file is to be run again on the new purge next period.
- **VENT needs disposal.** The new vapour distillate from COL-1 is 6.02 kmol/h and 262.6 kg/h at 43 mol per cent benzene and 29 per cent ammonia, so a flare or thermal oxidiser has to be added to the equipment list.
- **The work was done as thirteen successive files**, `che453_R3_00` through `che453_R3_12`, one change at a time, since every change to the liquid train alters the feed to the next unit and all three recycles have to re-converge.

8 Number of Hours Spent During This Period

Approximately **6 hours per member**, that is about **36 person-hours** for the team.

9 Contributions From Individual Members

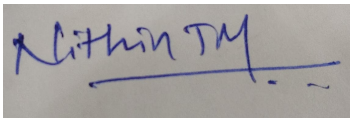
- **Nithin T M and Aditya Nitin Patil** — COL-1 and COL-2 redesign, the stripper boilup study, the pumps and product cooler, convergence of the flowsheet, and the material and energy balances.
- **Shubham Singh and Shivam Dongare** — the captive power plant flowsheet, its sizing against the plant electrical load, and the combustion balance checks.
- **Nithin D H and Sajag Masane** — the cooling water loop and its hand calculations, the utility definitions in Aspen Plus, the figures and the compilation of this report.

All six members have reviewed the complete report.

10 References

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14. Group 14, CHE453 Bi-weekly Report 2, 23 August 2026.
15. Aspen Plus simulation files `che453_R3_12.apw` and the separate captive power plant file, with the model summary, stream summary and utility results exported as workbooks, 5 September 2026.

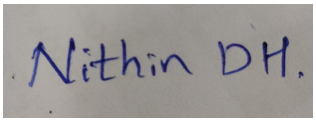
Signatures of members



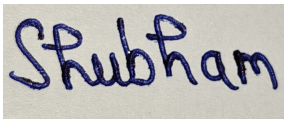
Nithin T M



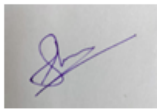
Aditya Nitin Patil



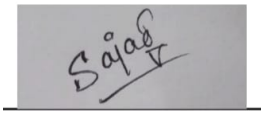
Nithin D H



Shubham Singh



Shivam Dongare



Sajag Masane

Bi-weekly Report 4

Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

CHE453 Process Design Capstone Project

Group No: 14

Date: 27 September 2026

Reports 1 to 3 precede this page and are unchanged.

Everything which follows constitutes Report 4.

Bi-weekly Report

Report Number: 04
Group No: 14
Date: 27 September 2026

Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

Team members

Name	Roll No.	Name	Roll No.
Nithin T M	230709	Shubham Singh	230999
Aditya Nitin Patil	230074	Shivam Dongare	230971
Nithin D H	230708	Sajag Masane	230896

Objectives for this report

1. Other separators design.
2. Reactor design and sequencing with detailed kinetics.
3. One heat exchanger detailed design.
4. Simulation of overall process with recycle loops.

1 Basis

The flowsheet delivered in Report 3 was converged and its balances closed, but the reactor in it was a stoichiometric block and several items were left open at that time. Those items are listed below. The present report takes them up, and whatever is still open is given in Section 7.

Table 1: Position at the close of Report 3

Item	Report 3 status	Taken up in
Reactor model	Stoichiometric	Section 2
Vessels other than the columns	Not sized	Section 3
Heat exchangers	Shortcut duty only	Section 4
Aniline–nitrobenzene binary parameters	Databank, 83–99 °C	Section 5.1
Nitrobenzene in product, below 5 ppm	1466 ppm	Section 5.3
Aniline purity, 99.9 wt% required	99.39 wt%	Section 5.5
Aniline in wastewater, below 1 ppm	154 ppm	Section 3.5
Aniline in WATOVHD	2.515 kmol/h	Section 3.3

All simulation work is in Aspen Plus V14 with NRTL-RK as the property method. Hydrogen and nitrogen are Henry components under the HC-1 set. The process flowsheet is `che453_R4_`

`FINAL.apw`. The binary regression is a separate file, `che453_R4_C_regress.apw`, and the exchanger is designed in Aspen Exchanger Design and Rating in the file `B3.EDR`. The gas turbine remains a separate file on Peng–Robinson and is not linked to the process flowsheet; the purge is taken to it as a fixed feed. No costing is given in this report. Costing is Report 5.

1.1 Key results

The reactor is now a kinetic plug-flow model. It gives 99.49 per cent conversion per pass against the 99.0 per cent that was set by hand in Report 3, with selectivities of 94.06, 1.76 and 4.19 per cent to aniline, cyclohexylamine and benzene. All four recycle loops are closed and converged together, and the overall mass balance closes to 7 parts per million on 18,300 kg/h.

The product leaves at 40 °C and 1.0 bar with 99.45 wt per cent aniline against 99.39 in Report 3, and 313 ppm nitrobenzene against 1466 ppm. That reduction came from regressing the aniline–nitrobenzene pair and then adding ten stages to COL-2, which is what Report 3 said was needed. It cost no extra reboiler duty. The 99.9 wt per cent and 5 ppm specifications of Report 1 are not yet reached; the position on each is in Section 7.

Total heating is 22.24 MW and the cooling water loop circulates 1691 t/h. The reactor removes a further 15.93 MW. On the thermal study, 25 mm tubes hold the bed peak at 390.7 °C against 640.4 °C for 52.5 mm tubes. The aniline recovery condenser B3 needs 132.87 m^2 and the design delivers 147.03 m^2, an over-surface of 10.7 per cent.

2 Reactor Design with Detailed Kinetics

2.1 Reaction scheme

Three reactions are taken. The main reaction makes aniline. The other two are the side reactions seen on copper catalyst, which are over-hydrogenation to cyclohexylamine and hydrodenitrogenation to benzene and ammonia.

$$\mathrm{C_6H_5NO_2 + 3\,H_2 \longrightarrow C_6H_5NH_2 + 2\,H_2O} \tag{1}$$

$$\mathrm{C_6H_5NH_2 + 3\,H_2 \longrightarrow C_6H_{11}NH_2} \tag{2}$$

$$\mathrm{C_6H_5NH_2 + H_2 \longrightarrow C_6H_6 + NH_3} \tag{3}$$

2.2 Rate expressions

Power-law kinetics are used in the vapour phase. Concentrations are molar, in kmol/m^3, and the rate is on reactor volume, in kmol/(m^3·s). Aspen writes the rate constant in the reference-temperature form

$$k(T) = k_0 \left(\frac{T}{T_0}\right)^{n} \exp\!\left[-\frac{E}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right] \tag{4}$$

with $T_0 = 573.15$ K and $n = 0$ for all three reactions. The parameters are in Table 2. Because the three rate expressions have different overall orders, the units of k_0 differ between them and are given with each value.

Table 2: Power-law kinetic parameters. Vapour phase, concentration in kmol/m^3, rate in kmol/(m^3·s), $T_0 = 573.15$ K, $n = 0$.

Rxn	Reaction	k_0	Units of k_0	E (kJ/kmol)	Exponents
1	Nitrobenzene → aniline	42	$(\mathrm{m^3/kmol})^{0.5}\,\mathrm{s^{-1}}$	35,000	NB 1, H_2 0.5
2	Aniline → cyclohexylamine	0.88	$\mathrm{m^3/(kmol{\cdot}s)}$	60,000	Aniline 1, H_2 1
3	Aniline → benzene + ammonia	0.41	$(\mathrm{m^3/kmol})^{0.5}\,\mathrm{s^{-1}}$	75,000	Aniline 1, H_2 0.5

The half-order in hydrogen for reactions 1 and 3 represents dissociative adsorption of hydrogen on the copper surface. Reaction 2 is taken first order in hydrogen because cyclohexylamine formation needs three hydrogen molecules on neighbouring sites and is not limited by dissociation.

The activation energy of the main reaction is taken from Petrov, Kumbilieva and Kirkov for an industrial copper catalyst, which gives $E_a = 35 \pm 1$ kJ/mol [3]. The two side reactions are not independently regressed. Their activation energies are set above that of the main reaction, at 60 and 75 kJ/mol, because both side products are reported to grow with temperature on this catalyst [2, 4], which requires E_2 and E_3 to exceed E_1. Their pre-exponential factors are then fitted so that the bed reproduces the conversion and the selectivity assumed in Report 3. The three rate constants are therefore a working set for this flowsheet rather than a measured kinetic model, and an independent regression against rate data is the way to firm them up.

2.3 Bed geometry and pressure drop

The reactor is a multitubular fixed bed with the catalyst in the tubes and the coolant on the shell. The geometry is in Table 3.

Table 3: Reactor bed geometry.

Item	Value	Basis
Number of tubes	2200	
Tube inside diameter	52.5 mm	See Section 2.4
Tube length	2.0 m	
Total tube cross-section	4.76 m^2	
Packed bed volume	9.53 m^3	Tube volume, 2200 tubes × 2.0 m
Bed voidage	0.40	
Catalyst solid volume	5.72 m^3	0.60 of the bed volume
Catalyst mass	11,430 kg	Bulk density 1200 kg/m^3 on bed volume
Catalyst	Cu on silica	
Particle diameter	6 mm	
Gas hourly space velocity	1808 h^{-1}	Inlet actual volumetric flow on bed volume
Superficial velocity	1.01 m/s inlet, 1.26 m/s outlet	On the 4.76 m^2 cross-section
Pressure drop	0.131 bar	Ergun, see below

The 9.53 m^3 is the packed bed volume, that is the volume enclosed by the tubes. At a voidage of

0.40 the catalyst solids occupy 5.72 m^3 of it. The bulk density of 1200 kg/m^3 is defined on bed volume, so the charge is 11,430 kg of catalyst.

Pressure drop goes as L/d_p^2 at a fixed superficial velocity, so the bed was set short and the particles coarse. The 0.131 bar in Table 3 is the Ergun result taken from the non-isothermal 52.5 mm case of Section 2.4, where the gas expands through the hot spot and the velocity rises. It is the larger of the two Ergun values available and is 4.4 per cent of the operating pressure.

In the converged flowsheet this 0.131 bar is imposed as a fixed drop rather than recomputed by Ergun at every recycle iteration, which keeps the integration stable during tearing. The converged answer is the same.

2.4 Thermal behaviour and choice of tube diameter

The main reaction is strongly exothermic. The reactions release 18.75 MW over the bed, and 8.04 MW of that, which is 43 per cent, comes off in the first 0.2 m. So the axial temperature profile is what decides the tube diameter.

Copper sinters by bulk diffusion above its Tammann temperature, which is half the melting point in kelvin. For copper this is 679 K, that is 406 °C. A maximum allowable bed temperature of **400 °C** is therefore taken for the Cu/silica charge, below which loss of copper surface area is slow enough for a normal catalyst life.

Two non-isothermal cases were run on the same packed bed volume and the same feed, with only the tube diameter changed. Holding 9.53 m^3 at 2.0 m length fixes the total cross-section at 4.76 m^2, so the 25 mm case needs 9700 tubes in place of 2200. The results are in Table 4.

Table 4: Effect of tube diameter on the axial temperature profile. Same packed bed volume, same bed length, same feed.

Item	52.5 mm tubes	25 mm tubes
Number of tubes	2200	9700
Inlet temperature	138.0 °C	138.0 °C
Peak temperature	640.4 °C	390.7 °C
Position of peak	0.341 m	0.293 m
Outlet temperature	240.4 °C	242.5 °C
Pressure drop	0.130 bar	0.279 bar
Maximum allowable, 400 °C	Exceeded	Satisfied

Heat removal per unit volume goes as the surface-to-volume ratio of the tube, which is $4/d$, so the 25 mm tube removes heat twice as fast for the same driving force. The peak falls by 250 °C and comes in at 390.7 °C, which is under the 400 °C limit. The 52.5 mm case is well above it. **25 mm tubes are therefore taken for the reactor.** The cost is 0.15 bar of extra pressure drop. The composition results are unchanged by the choice, because the nitrobenzene is almost fully converted inside the first 0.5 m in both cases.

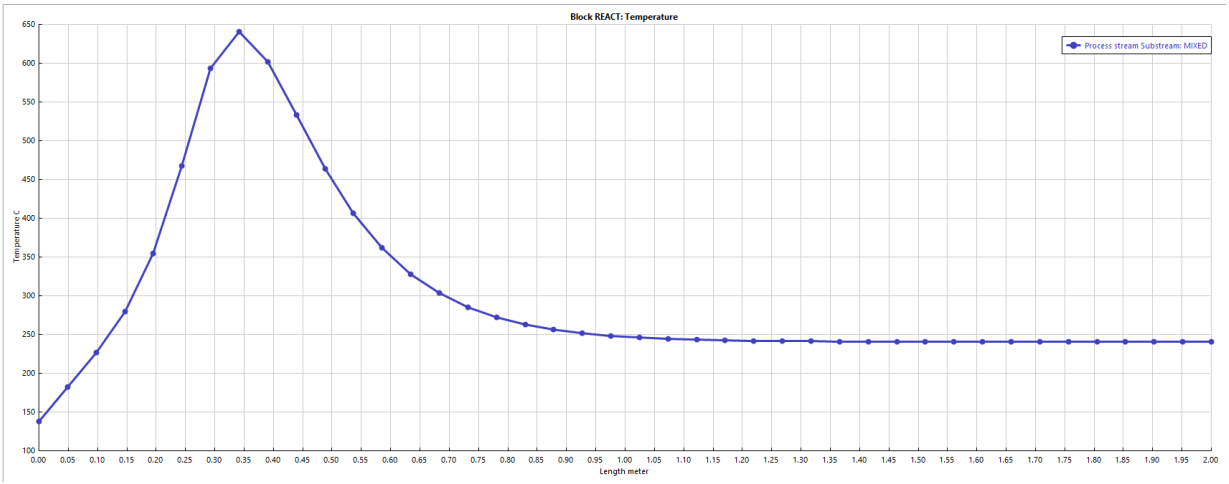


Figure 1: Axial temperature profile in the catalyst bed with 52.5 mm tubes. The peak is 640.4 °C at 0.341 m.

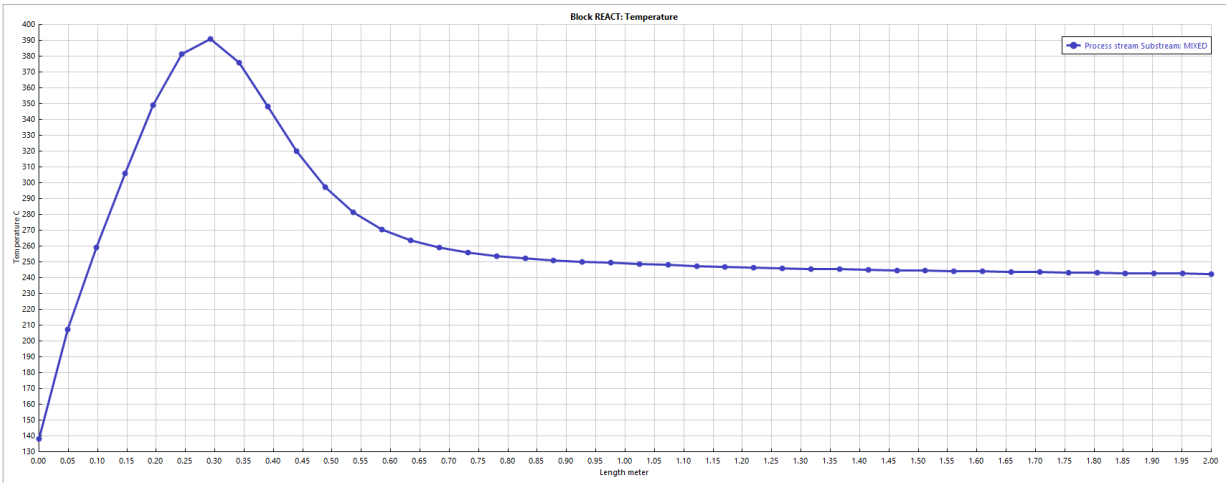


Figure 2: Axial temperature profile in the catalyst bed with 25 mm tubes. The peak is 390.7 °C at 0.293 m, below the 400 °C limit.

2.5 Reactor in the converged flowsheet

In the flowsheet the reactor is run as a constant-temperature RPlug at 300 °C. This is the limiting case of a tightly cooled bed and it returns the net duty the coolant has to take away. It does not resolve the axial profile, which is why the non-isothermal cases of Section 2.3 are run separately to fix the tube diameter. The two are consistent in that the 25 mm bed holds its peak within 91 °C of the 300 °C isothermal level over the first half metre and then settles close to it.

Table 5 gives the reactor performance in the converged flowsheet.

Table 5: Reactor performance in the converged flowsheet.

Item	Value
Nitrobenzene into reactor (RXFEED)	141.888 kmol/h
Nitrobenzene out (RXOUT)	0.725 kmol/h
Conversion per pass	99.49 %
Selectivity to aniline	94.06 %
Selectivity to cyclohexylamine	1.76 %
Selectivity to benzene	4.19 %
Hydrogen consumed	436.85 kmol/h
Inlet temperature	160.0 °C
Outlet temperature	300.0 °C
Heat of reaction released	18.75 MW
Feed heating, 160 to 300 °C	2.82 MW
Net duty removed by the coolant	15.93 MW
Residence time	0.63 s

The block duty and the reaction heat are not the same number. The reactions release 18.75 MW, of which 2.82 MW is absorbed in bringing the feed from 160 °C to the 300 °C bed temperature, so the coolant removes 15.93 MW. It is the 15.93 MW that appears against REACT in the flowsheet.

The hydrogen consumption checks against stoichiometry. Three moles per mole of nitrobenzene converted gives 423.5 kmol/h, three more per mole of cyclohexylamine gives 7.4 kmol/h, and one per mole of benzene gives 5.9 kmol/h. The total is 436.8 kmol/h, which matches the simulation.

In Report 3 the stoichiometric reactor was set at 99.0 per cent conversion with selectivities of 94.95, 1.53 and 3.52 per cent. The kinetic reactor gives 99.49, 94.06, 1.76 and 4.19 per cent on the same bed, so the rest of the flowsheet built on the Report 3 numbers still works.

3 Design of the Other Separators

The distillation columns were done in Report 3. This section covers the vessels. All of them are sized on the converged flows of the final run and the checks are repeated in Section 3.6.

3.1 Flash drum

The reactor effluent is cooled and flashed to take the hydrogen-rich gas off the top for recycle. The drum is a vertical knockout drum with a demister.

The diameter comes from the Souders–Brown equation,

$$u_{\max} = K \sqrt{\frac{\rho_L - \rho_V}{\rho_V}} \tag{5}$$

with $K = 0.1067$ m/s, which is 0.35 ft/s, the standard value for a vertical drum with a demister. The length comes from a five-minute liquid holdup plus 1.5 m of vapour space above the high liquid level, which covers the clearance to the demister, the demister pad itself and the disengaging height above it.

Table 6: Flash drum sizing.

Item	Value	Note
Vapour flow	2.522 m^3/s	VAP-OUT at 40 °C, 2.5 bar
Vapour density	0.531 kg/m^3	
Liquid flow	17.43 m^3/h	LIQ-OUT
Liquid density	1064.9 kg/m^3	
Maximum vapour velocity $u_{\max}$	4.78 m/s	Souders–Brown
Minimum diameter	0.820 m	
Selected diameter	0.92 m	
Actual vapour velocity	3.79 m/s	79 % of $u_{\max}$
Selected length, tangent to tangent	3.8 m	$L/D = 4.1$
Liquid holdup time	5.27 min	

3.2 Decanter

The condensed reactor effluent separates into an organic phase rich in aniline and an aqueous phase. A horizontal decanter with $L/D = 5$ is used.

The dispersed phase was identified with the Selker–Sleicher criterion as given by Towler and Sinnott [6],

$$\psi = \frac{Q_L}{Q_H}\left(\frac{\rho_L \mu_H}{\rho_H \mu_L}\right)^{0.3} \tag{6}$$

which gives $\psi = 1.65$. This falls in the band where either phase can be the dispersed one, so the vessel was sized on the slower of the two settling velocities, which is the aqueous droplet rising through the organic phase.

Settling is by Stokes law on a 150 μm design droplet,

$$u = \frac{d^2 g\,\Delta\rho}{18\,\mu_c} \tag{7}$$

Table 7: Decanter sizing.

Item	Value	Note
Organic flow	13.29 m^3/h	1067.0 kg/m^3
Aqueous flow	17.33 m^3/h	965.8 kg/m^3
Density difference	101.2 kg/m^3	
Design droplet	150 μm	Standard decanter basis
Design settling velocity	8.12×10^{-4} m/s	Slower of the two phases
Diameter	1.25 m	
Length	6.25 m	$L/D = 5$
Residence time	15.0 min	
Droplet settling time	12.8 min	Less than residence time
Interface velocity	4.7×10^{-4} m/s	Below settling velocity

The settling time is below the residence time and the continuous-phase velocity along the interface is below the settling velocity, so both of the usual decanter checks are satisfied.

3.3 Second decanter on the column overhead

The overhead from the first column, stream WATOVHD, carries 2.61 kmol/h of aniline along with water and benzene. A second decanter on this stream recovers 76.4 per cent of that aniline, which is 1.99 kmol/h or 185 kg/h, into an organic phase.

It is reported open loop, the recovered organic not being fed back. WATOVHD is the only exit benzene has from the process, benzene being made by reaction 3 and consumed nowhere. Keeping the stream open therefore reports the aniline available for recovery while leaving that exit intact. A benzene column is needed before the loop can be closed.

3.4 Vent thermal oxidiser

The vapour distillate from the first column, stream VENT, carries benzene, ammonia and hydrogen. It goes to a thermal oxidiser.

Table 8: Thermal oxidiser sizing.

Item	Value	Note
VENT flow	6.39 kmol/h	277.9 kg/h
Benzene in VENT	2.74 kmol/h	Sets most of the duty
Ammonia in VENT	2.02 kmol/h	
Heat release	2.60 MW	Lower heating value basis
Chamber temperature	870 °C	For $\geq$ 98 % destruction
Combustion and quench air	337 kmol/h	
Chamber diameter	1.6 m	
Chamber length	4.8 m	$L/D = 3$
Residence time	1.07 s	At least 1 s required

The heat release is enough for the chamber to run without support fuel.

3.5 Carbon adsorber on the wastewater

The stripper bottoms still carry a little aniline. A granular activated carbon bed is put on this stream as a polishing step, this being the route Report 3 identified when it showed that stripping alone cannot reach the discharge limit at any workable boilup.

Table 9: Carbon adsorber sizing.

Item	Value	Note
Wastewater flow	4375.8 kg/h	4.41 m^3/h at 40 °C
Aniline into the bed	216 ppm by weight	0.945 kg/h, converged run
Aniline to remove	22.7 kg/day	
Carbon loading	0.2 kg/kg	Aniline on granular carbon
Carbon consumption	113 kg/day	
Cycle length to breakthrough	30 days	Design basis
Carbon per bed	3400 kg	7.6 m^3 at 450 kg/m^3
Bed diameter	1.2 m	
Bed height	6.7 m	
Empty bed contact time	104 min	
Aniline out of the bed	Below 1 ppm	Until breakthrough, see below

The bed volume is set by the carbon holding capacity over a 30-day cycle. The contact time that follows is well above the 10 to 30 minutes usually needed, so the bed is not mass-transfer limited and the treated water stays below the 1 ppm discharge limit for the whole cycle, breakthrough being the point at which the bed is changed. Two beds in lead–lag are used so that the lag bed holds the discharge while the lead bed is being changed. Confirming the 30-day cycle needs a breakthrough curve from adsorption isotherm data on this water, and that is left to Report 5.

The wastewater also carries 4.28 weight per cent cyclohexylamine, which is far more soluble in water than aniline and is not held on carbon. It is taken up with the wastewater treatment in Report 5.

3.6 Consistency check on the converged flows

All the vessels above were first sized on the Report 3 stream table. The flows changed slightly when the kinetic reactor and the closed recycles were brought in. Every vessel was re-checked on the final converged flows. Table 10 gives the result.

Table 10: Vessel checks repeated on the final converged flows.

Vessel	Criterion	Requirement	Final run
Flash drum	Vapour velocity	≤ 100 % of $u_{\max}$	79 %
Flash drum	Liquid holdup	≥ 5 min	5.27 min
Decanter	Settling vs residence time	Settling shorter	12.8 vs 15.0 min
Oxidiser	Residence time	≥ 1 s	1.07 s
Carbon bed	Cycle length	30 days	30 days

The flash drum gained margin. The vapour rate fell and the vapour density rose in the final run, so the drum now runs at 79 per cent of the flooding velocity instead of 99 per cent. The decanter and the oxidiser moved by less than one per cent and both stay inside their criteria on the original dimensions. No vessel had to be resized.

4 Detailed Design of One Heat Exchanger

The exchanger chosen for detailed design is the condenser on the aniline recovery column, block B3 in the flowsheet, which is the ARCOND service. It condenses the ANRECOV vapour against cooling water. It was chosen because it has the largest duty of any water-cooled service in the plant.

The design was done in Aspen Exchanger Design and Rating in rigorous shell and tube mode, in design (sizing) mode, with the process conditions taken from the converged flowsheet.

4.1 Process conditions

Table 11: Process conditions for exchanger B3.

Item	Hot side (shell)	Cold side (tube)
Stream	ANRECOV → ARLIQ	Cooling water
Mass flow	3.400 kg/s	152.78 kg/s
Inlet temperature	97.5 °C	30.0 °C
Outlet temperature	40.0 °C	42.1 °C
Inlet vapour fraction	1.000	0
Outlet vapour fraction	7.2×10^{-5}	0
Inlet pressure	1.0 bar	4.0 bar
Allowable pressure drop	0.1 bar	0.5 bar
Calculated pressure drop	0.055 bar	0.458 bar
Fouling resistance	0.0002 m^2K/W	0.0002 m^2K/W

The hot fluid is put on the shell side because it is condensing. A condensing stream on the shell side has a large free volume to disengage in and a low pressure drop, and the cooling water goes in the tubes where it can be cleaned mechanically.

Fouling resistance of 0.0002 m^2K/W is applied on both sides, which is the standard value for clean organics and for treated cooling water. This is what sets the area, as Section 4.3 shows.

4.2 Geometry

Table 12: Exchanger B3 geometry.

Item	Value
TEMA type	AES
Shell inside diameter	812.8 mm
Tube outside diameter	19 mm
Tube pitch	25 mm, 30° triangular
Tube length	3.658 m
Number of tubes	721
Number of tube passes	2
Shells in series / parallel	1 / 1
Baffle type	Single segmental, horizontal cut
Number of baffles	4
Baffle spacing	590.6 mm
Material	Carbon steel

TEMA AES is chosen because the removable channel cover on the A head lets the tubes be cleaned without breaking the piping, and the floating head on the S rear lets the bundle move when the tube wall runs 50 °C colder than the shell wall. Carbon steel is acceptable because dry aniline is not corrosive to steel and the water side is treated cooling water.

The baffle cut is horizontal because the shell side is condensing. A horizontal cut makes the vapour flow up and down across the bundle and lets the condensate drain along the bottom of the shell instead of collecting in the baffle windows.

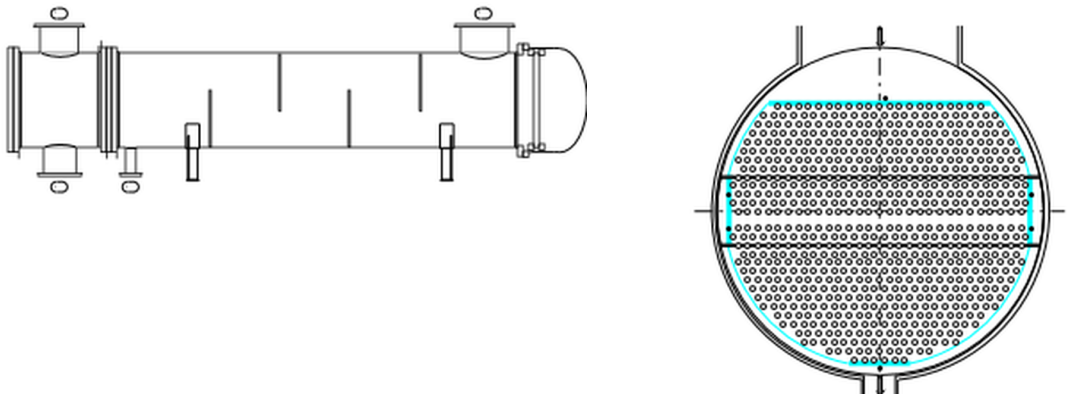


Figure 3: Exchanger B3 from Aspen EDR. Left, setting plan showing the A head, the four single segmental baffles and the S floating head. Right, tube layout of the 721 tubes on 25 mm 30° triangular pitch, with the two tube passes divided by the pass partition.

4.3 Thermal results

Table 13: Thermal results for exchanger B3.

Item	Value
Heat duty	7663.7 kW
End-point LMTD, counter-current	26.52 °C
Zone-weighted mean temperature difference (EDR)	48.23 °C
Ratio of the two, reported by EDR as the LMTD correction factor	1.819
UA	158.9 kW/K
Overall coefficient, clean	2.638 kW/m^{2}K
Overall coefficient, dirty	1.196 kW/m^{2}K
Required area	132.87 m^2
Actual area	147.03 m^2
Over-surface	10.7 %

The figure of 1.819 needs a word, because a conventional LMTD correction factor for a 1–2 shell is always below one. It is not that quantity here. The end-point LMTD computed from the four terminal temperatures is

$$\Delta T_{\mathrm{lm}} = \frac{(97.5 - 42.1) - (40 - 30)}{\ln[(97.5 - 42.1)/(40 - 30)]} = 26.52\ ^{\circ}\mathrm{C} \tag{8}$$

The shell side desuperheats, condenses and then subcools, so EDR divides the exchanger into zones and computes a duty-weighted mean temperature difference across them, which comes to 48.23 °C. EDR reports the ratio 48.23/26.52 = 1.819 in the field labelled LMTD correction factor. It is therefore a zone-weighted mean temperature difference ratio and not the conventional F factor, and it is above one because most of the duty is released in the condensing zone where the local temperature difference is much larger than the end-point value suggests. The area follows from the zone-weighted value: $UA = 7663.7/48.23 = 158.9$ kW/K and $A = 158.9/1.196 = 132.9$ m^2, which is the required area in the table.

The over-surface of 10.7 per cent is in the normal range. The dirty coefficient is 45 per cent of the clean one, which shows that fouling, and not the film coefficients, controls the area.

The duty of 7663.7 kW agrees with the flowsheet value of 7626 kW for the ARCOND service to within 0.5 per cent, and the cooling water requirement of 550,008 kg/h agrees with the flowsheet value of 549,472 kg/h to within 0.1 per cent.

5 Simulation of the Overall Process with Recycle Loops

5.1 Regressed binary interaction pair

The aniline–nitrobenzene pair decides the separation in the second column, which is the column that sets the final product purity. The Aspen databank parameters for this pair are fitted over 83 to 99 °C only. The second column runs from 129 to 145 °C, so the databank values are being used outside the range they were fitted in. Report 3 recorded that this pair should be regressed, in Section 10 of that report. This section does that.

Four binary vapour–liquid data sets for aniline and nitrobenzene were taken from the NIST ThermoData Engine inside Aspen Plus and regressed together by maximum likelihood with the Britt–Luecke algorithm and Deming initialisation. Regression case DR-1 in the file `che453_R4_C_regress.apw`. Only the temperature-dependent terms were fitted and the non-randomness parameter was held at 0.3.

Table 14: Binary data sets used in the regression.

Data set	Pressure	Temperature range
BVLE007	0.00667 bar	56–70 °C
BVLE008	0.01333 bar	69–83 °C
BVLE009	0.00416 bar	43–98 °C
BVLE010	Atmospheric	179–210 °C

Taken together the four sets span 43 to 210 °C. The second column runs from 129 to 145 °C, which is inside that span, and BVLE010 brackets it directly. The databank parameters cover only 83 to 99 °C, so they have to be extrapolated to reach the column.

Table 15: NRTL parameters for the aniline–nitrobenzene pair.

Parameter	Databank	Regressed (DR-1)
a_{ij}	0	0
a_{ji}	0	0
b_{ij} (K)	−217.928	488.44
b_{ji} (K)	+287.592	−310.108
c_{ij}	0.3	0.3
Valid range	83–99 °C	−273 to 727 °C

Both b terms changed sign. The regressed pair puts aniline and nitrobenzene closer in volatility than the databank pair did, so the split between them is harder than the Report 3 model showed. This changed the design of the second column, as Section 5.3 describes.



Figure 4: Regressed NRTL fit for the aniline–nitrobenzene pair, data set BVLE010, which is the set closest to the operating range of the second column. Points are experimental, lines are the fitted model.

5.2 Recycle structure and convergence

The flowsheet has four recycle loops. Aspen tears them at the streams listed in Table 16 and converges them with Wegstein. The hydrogen loop is on its own convergence block CV-1 because it carries a design specification on the purge rate. The other three are solved together on $OLVER01.

Table 16: Tear streams and convergence status in the final run.

Tear stream	What it carries	Convergence block	Status
H2RET	Recycle gas to the reactor	CV-1	Converged
S3	Nitrobenzene recycle	$OLVER01	Converged
LIQ-OUT	Flash liquid to the decanter	$OLVER01	Converged
ARLIQ	Aniline recovery liquid	$OLVER01	Converged

All parameter studies were run as separate converged cases rather than with a Sensitivity block, which re-tears every loop at each point when placed on a variable inside a recycle. This is the same practice as Report 3 Section 7.

5.3 Second column: effect of the design variables

With the regressed pair in place, the second column was studied in four converged runs. One variable was changed per run. Table 17 gives the result.

Table 17: Second column: effect of reflux ratio, bottoms rate and stage count on the product.

Run	Reflux ratio	Bottoms (kmol/h)	Stages	NB in product	Aniline purity
R4_02	2	2	15	1896 ppm	99.29 wt%
R4_03	3	2	15	1469 ppm	99.33 wt%
R4_04	3	3	15	1274 ppm	99.36 wt%
R4_05	3	3	25	**313 ppm**	**99.45 wt%**

Reflux ratio 2 to 3 took the nitrobenzene from 1896 to 1469 ppm for 36 per cent more reboiler duty. Widening the bottoms draw from 2 to 3 kmol/h took it to 1274 ppm at no extra duty. Report 3 found the same on 15 stages and set both variables aside.

The bottoms composition shows why. At 2 kmol/h the bottoms is already 24 mole per cent nitrobenzene on a 15-stage column, so more reflux only raises the vapour traffic without sharpening the split. The boilup ratio of 249 in run R4_03 is the same evidence.

Ten more stages took the nitrobenzene to 313 ppm and the aniline to 99.45 weight per cent with no rise in reboiler duty, because the taller column reaches the separation at the same vapour rate. Stage count is the variable that controls this split. The 25-stage column with reflux ratio 3 and a 3 kmol/h bottoms draw is carried forward. The trend also gives the route to the 5 ppm specification, which is further stages, and that is taken up in Section 7.

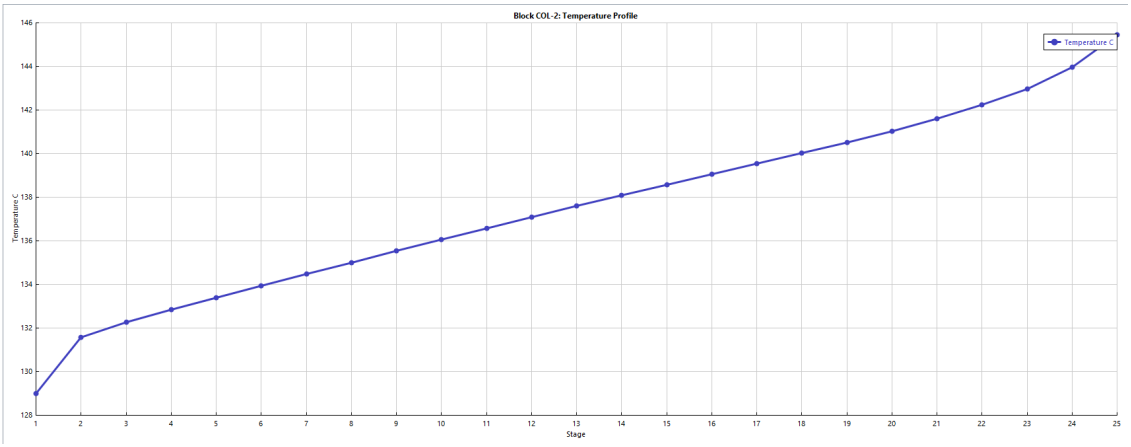


Figure 5: Temperature profile of the second column in the final run.

5.4 Overall mass balance

Table 18: Overall mass balance of the converged flowsheet.

In	kmol/h	kg/h	Out	kmol/h	kg/h
NB-FEED	141.28	17,393.16	PRODUCT	131.02	12,185.80
H2-MAKE	444.00	906.59	WASTEWTR	234.35	4,375.76
			WATOVHD	57.48	1,418.94
			VENT	6.39	277.88
			PURGE	7.49	41.49
Total		**18,299.75**	**Total**		**18,299.87**

The balance closes to 0.12 kg/h on 18,300 kg/h, which is 7 parts per million.

5.5 Product

Table 19: Product stream composition.

Component	kmol/h	Mole fraction	Weight
Aniline	130.123	0.99317	99.45 wt%
Cyclohexylamine	0.596	0.00455	0.48 wt%
Water	0.268	0.00205	0.04 wt%
Nitrobenzene	0.031	0.00024	313 ppm
Total	**131.017**		**12,185.8 kg/h**

The PRODUCT stream leaves at 12,185.8 kg/h, which at 8200 operating hours per year is 99,920 t/y of product. This is the product stream rate. The aniline it contains is 12,118 kg/h, that is 99,370 t/y, the balance being the cyclohexylamine and water of Table 19. Closing that gap is the same work as reaching the 99.9 wt per cent specification.

The product leaves at 40 °C and 1.0 bar, and the water content of 0.04 weight per cent is inside the 0.1 weight per cent set in Report 1. Nitrobenzene has come down from 1466 ppm to 313 ppm, a factor of 4.7. The remaining 0.48 weight per cent is cyclohexylamine. The purity and nitrobenzene specifications are both taken up in Section 7.

Table 20 compares the product against the Report 3 flowsheet.

Table 20: Comparison with Report 3.

Item	Report 3	Report 4 final
Reactor model	Stoichiometric	Kinetic plug flow
Conversion per pass	99.0 % (set)	99.49 % (calculated)
Aniline purity	99.39 wt%	99.45 wt%
Nitrobenzene in product	1466 ppm	313 ppm
Product rate	12,320 kg/h	12,186 kg/h
Second column stages	15	25

5.6 Utilities

Table 21: Utility summary of the converged flowsheet. Totals are formed from unrounded values, so they may differ by 0.01 from the sum of the rounded entries.

Utility	Service	Duty (MW)	Usage (kg/h)
High pressure steam	VAP-1	3.68	8,792
Medium pressure steam	COL-1 reboiler	4.06	7,564
Low pressure steam	STRIP reboiler	7.94	14,199
Low pressure steam	COL-2 reboiler	6.56	11,723
	Total heating	**22.24**	**42,279**
Cooling water	ARCOND	7.63	549,472
Cooling water	COOL-2	5.24	377,467
Cooling water	COL-2 condenser	7.02	506,019
Cooling water	COL-1 condenser	2.93	211,156
Cooling water	PRODCOOL	0.66	47,381
	Total cooling water	**23.48**	**1,691,496**

The reactor removes a further 15.93 MW, which is not in this table because no utility is assigned to it in the simulation. In practice this heat would raise steam on the shell side of the reactor. That is taken up in Report 5.

Total heating is 22.24 MW against 20.40 MW in Report 3. The difference of 1.84 MW is low pressure steam into the second column, and that is what took the nitrobenzene in the product from 1466 ppm down to 313 ppm.

6 Gas Turbine on the Purge Stream

The purge from the hydrogen recycle loop still carries 6.64 kmol/h of hydrogen. This was run through a gas turbine in a separate file, with natural gas as supplementary fuel and a design specification holding the turbine inlet temperature.

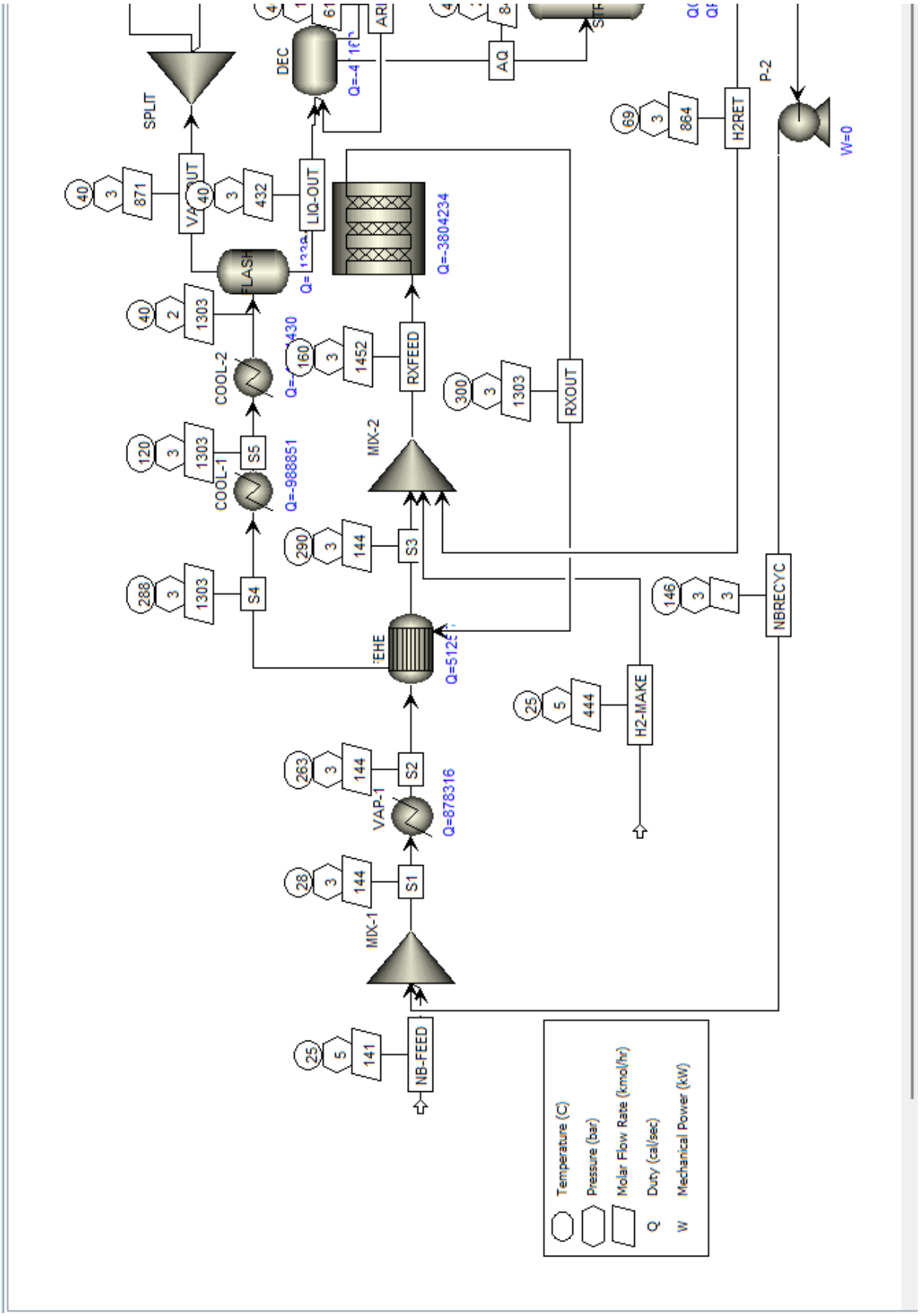


Figure 6: Converged flowsheet, feed and reaction section, from the nitrobenzene feed through the reactor, flash drum and decanter. Annotated with temperature (°C), pressure (bar), molar flow (kmol/h) and duty (cal/s). Continued in Figure 7.

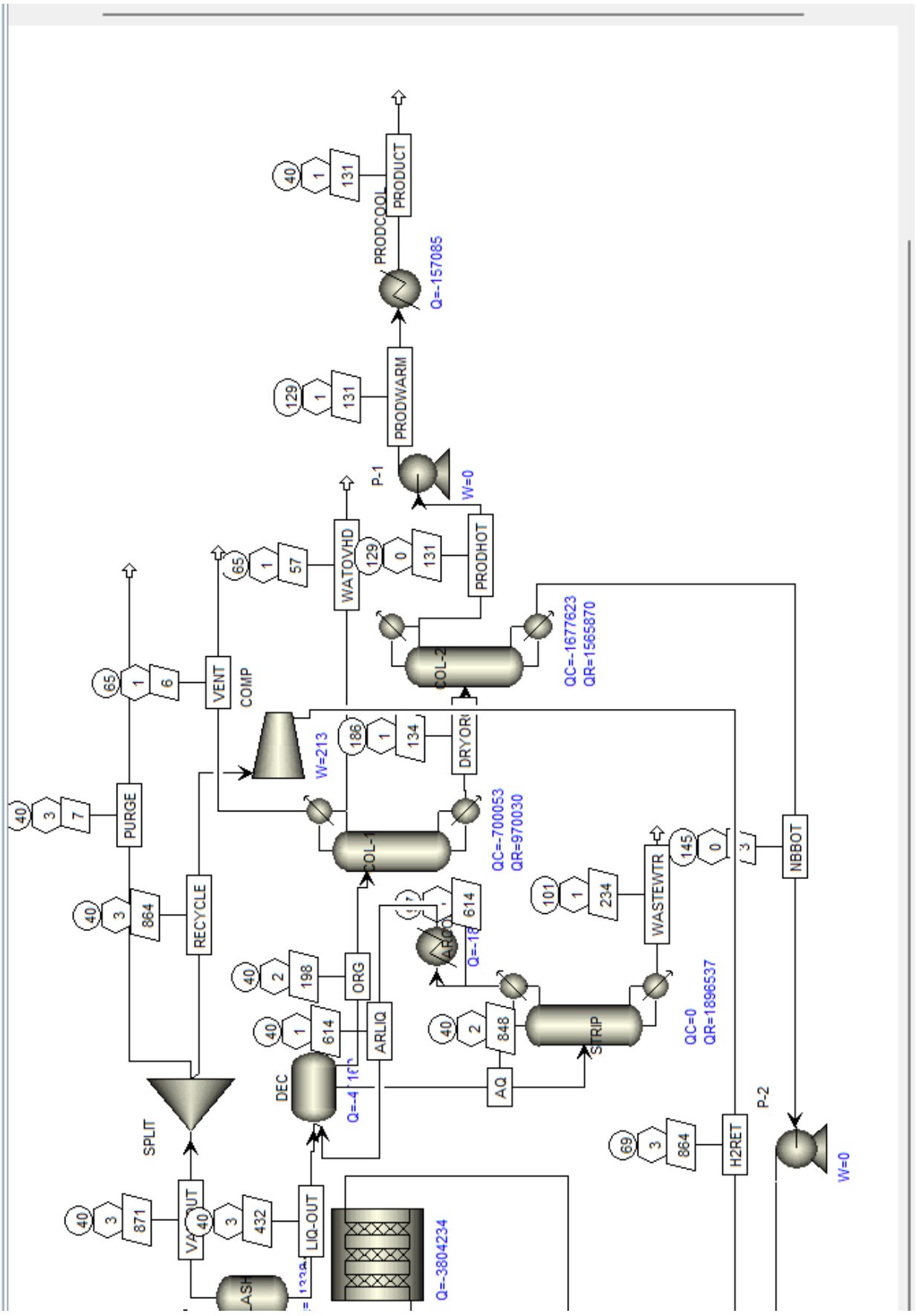


Figure 7: Converged flowsheet, separation section, from the decanter through the stripper, the two columns and the product cooler, with the hydrogen and nitrobenzene recycles returning to Figure 6.

Table 22: Gas turbine on the purge stream. Simulated case, on a purge of 7.37 kmol/h.

Item	Value
Purge to turbine	7.37 kmol/h
Hydrogen in purge	6.52 kmol/h
Supplementary natural gas	7.71 kmol/h
Air	290 kmol/h at 12 bar
Turbine inlet temperature	1153.9 °C
Turbine exhaust temperature	593.3 °C
Air compressor work	864.9 kW
Fuel compressor work	16.3 kW
Turbine work	1657.3 kW
Net shaft power	776.1 kW
Net electrical power, simulated	745.1 kW

The Report 3 baseline was 709.1 kW of net electrical power. The purge rate changed when the recycle loops were closed, and the turbine gains 5.1 per cent on the new purge.

The simulated case of Table 22 was run on a purge of 7.37 kmol/h taken from an intermediate converged run. The final run gives 7.49 kmol/h, which is 1.6 per cent higher. Scaling the output linearly with the fuel received gives an estimate of about 757 kW on the final purge. That 757 kW is an estimate and not a simulated result; the simulated figure is the 745.1 kW in the table. The turbine is not coupled to the main flowsheet and is reported as a separate case.

7 Objectives Which Could Not Be Accomplished, With Reasons

The four objectives of this report are design and simulation tasks and all four are complete. They are separate from the product and discharge specifications set in Report 1, which are carried forward from report to report. The position on each specification is below.

Aniline purity. 99.45 wt % reached against 99.39 wt % in Report 3, the target being 99.9 wt %. This target is not yet met. The impurity holding it back is cyclohexylamine at 0.48 wt %, which cannot be taken out in either column for the reason given in Report 3 Section 2.4. It boils below aniline, so a light-ends draw at the top of COL-2 is the route, and a lower over-hydrogenation selectivity in the reactor would reduce the load on that draw.

Nitrobenzene in the product. 313 ppm against 1466 ppm in Report 3, the target being 5 ppm. This target is not yet met. Section 5.3 shows the split responds to stage count and weakly to everything else, so a taller COL-2 is the route. The four cases in Table 17 give the trend from which the stage count for 5 ppm can be set.

Aniline in the wastewater. The stripper bottoms carry 216 ppm, against 154 ppm in Report 3 on the Report 3 flows, the discharge limit being 1 ppm. The carbon adsorber of Section 3.5 is sized to hold the treated water below 1 ppm for a 30-day cycle. That outlet concentration is a design basis taken from the carbon loading, not a simulated result, and confirming it needs a breakthrough curve from adsorption data. Until that is done the target is treated as pending.

Cyclohexylamine in the wastewater. WASTEWTR carries 4.28 wt %, which carbon does not hold. It is to be handled with the wastewater treatment in Report 5.

Second decanter on WATOVHD. Reported open loop in Section 3.3, because WATOVHD is the only exit benzene has. A benzene column is needed before that loop can be closed.

Reactor heat. 15.93 MW is available from REACT as a steam-raising duty. Its assignment belongs with the heat integration left to Report 5 in Report 2.

Reaction kinetics. The two side-reaction rate constants are fitted to reproduce the Report 3 selectivity rather than regressed against rate data, as Section 2.2 states. An independent regression would firm them up.

8 Any Other Challenges

Reacting phase had to be set to Vapor on all three reactions. On Liquid every rate came out as zero in an all-vapour stream. A Reset was needed after the change, the tear streams otherwise holding the earlier solution.

The reactor profile grid needed 41 points. On the default grid the peak read 353 °C and 589 °C against 391 and 640 °C on the fine grid, the integrator stepping over the hot spot.

Ergun was replaced by a fixed pressure drop inside the recycle loop. The correlation is used to fix the value, as Section 2.3 describes, and the converged answer is unchanged.

Design specifications were used in place of sensitivity analyses, as in Report 3, each parameter case being run as its own converged file. One change was made per run and each run saved as a new numbered file, `che453_R4_00` through `che453_R4_05`.

9 Number of Hours Spent During This Period

Approximately 8 hours per member, which is 48 hours for the group.

10 Contributions From Individual Members

Nithin T M and Aditya Nitin Patil — the kinetic model and its rate parameters, the conversion of the reactor to RPlug, the bed geometry and pressure drop, convergence of the flowsheet with all four recycles, the COL-2 study, and the compilation of this report.

Nithin D H and Shubham Singh — the flash drum, decanter, second decanter, thermal oxidiser and carbon adsorber, the re-check of every vessel on the converged flows, and the gas turbine case on the new purge.

Shivam Dongare and Sajag Masane — the regression of the ANILINE–NITRO-01 pair against NIST TDE data, and the detailed design of exchanger B3 in Aspen EDR with its fouling and over-surface checks.

All six members have reviewed the complete report.

11 References

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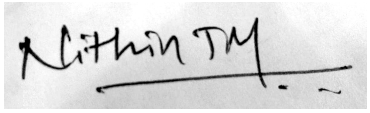
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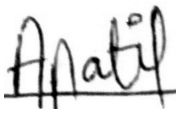
trial copper catalyst, $E_a = 35 \pm 1$ kJ/mol.

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15. Own regression case R-DR-1, ANILINE–WATER NRTL parameters, Report 1 Section 6.4.
16. Group 14, CHE453 Bi-weekly Reports 1, 2 and 3.
17. Aspen Plus simulation file `che453_R4_FINAL.apw`, with the stream summary, model summary, utility results and block profiles exported as workbooks, and the exchanger file `B3.EDR`, 27 September 2026.

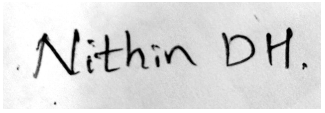
Signatures of members



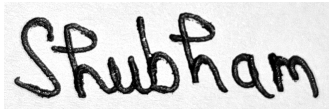
Nithin T M



Aditya Nitin Patil



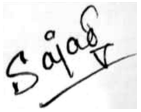
Nithin D H



Shubham Singh



Shivam Dongare



Sajag Masane

Bi-weekly Report 5

Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

CHE453 Process Design Capstone Project

Group No: 14

Date: 25 October 2026

Reports 1 to 4 precede this page and are unchanged.
Everything which follows constitutes Report 5.

Bi-weekly Report

Report Number: 05
Group No: 14
Date: 25 October 2026

Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

Team members

Name	Roll No.	Name	Roll No.
Nithin T M	230709	Shubham Singh	230999
Aditya Nitin Patil	230074	Shivam Dongare	230971
Nithin D H	230708	Sajag Masane	230896

Objectives for this report

1. Heat exchanger network design using pinch technology.
2. Estimate utility costs. Perform detailed techno-economic analysis.

1 Basis

Report 4 closed the flowsheet with all four recycles converged. Heat integration and costing were left for this report. The items open at that time are in Table 1.

Table 1: Position at the close of Report 4.

Item	Report 4 status	Taken up in
Pinch analysis of the full network	Not done	Section 2
Reactor heat, 15.93 MW	No utility assigned	Section 3
COOL-1 duty, 4.14 MW	No utility assigned	Section 3
Utility costs	Not estimated	Section 4
Capital cost	Not estimated	Section 5
NPV, IRR and payback	Not estimated	Section 6
Aniline purity, 99.9 wt% required	99.45 wt%	Section 7
Nitrobenzene in product, below 5 ppm	313 ppm	Section 7
Aniline in wastewater, below 1 ppm	216 ppm into carbon bed	Section 7
Cyclohexylamine in wastewater	4.28 wt%	Sections 6, 7

The flowsheet is same as Report 4. A copy of che453_R4_FINAL is saved as che453_R5_00.apwz and the Energy Analysis of Aspen Plus V14 is run on it. The stream data is then taken into Aspen Energy Analyzer. The targets and the network are in che453_R5_01_Copy1.hch, design

A_Design2. The final Aspen Plus file is che453_R5_01.apwz. All costs are in US dollars at the 2023 annual CEPCI of 797.9, and Rs. 95 per USD is used as in Report 1.

1.1 Key results

On the Aspen Energy Analyzer data the plant is a threshold problem. At $\Delta T_{min} = 10$ K the minimum hot utility is zero and the minimum cold utility is 21.33 MW. The reactor alone gives 15.93 MW at 300°C, which is more than enough for every heater and reboiler in the plant.

Aspen Energy Analyzer gives a network of 17 exchangers which meets this target exactly. In the plant the reactor coolant cannot be as hot as Aspen assumes, because of the bed temperature limit of Report 4. So the final network has the reactor shell raising 40 bar steam, with three new exchangers. Purchased heating falls from 22.24 MW to 1.80 MW. Cooling water falls from 3137 t/h to 1327 t/h, taking REACT and COOL-1 on cooling water in the base case. The 1.80 MW left is the vaporisation of nitrobenzene at 262°C, which is above the 40 bar steam temperature. Utility cost falls from USD 10.43 million per year to USD 0.67 million per year.

The fixed capital is USD 18.2 million (Rs. 173 crore) and the total capital USD 20.9 million (Rs. 199 crore). At Report 1 prices the project gives an NPV of USD 61.3 million at 12 per cent, an IRR of 50.4 per cent, a return on investment of 55 per cent and a payback of 1.6 years. Without the heat integration the NPV is only USD 16.4 million.

2 Pinch Analysis

2.1 Stream data

The stream data was taken by the Energy Analysis of Aspen Plus straight from the converged flowsheet. Every duty was checked against the utility and model workbooks of R4_05 and agrees to the third decimal. Three corrections were needed before the targets could be used.

- REACT was read in as a cold stream from 160.2 to 300°C, heated by fired heat. This is because the reactor outlet is hotter than its inlet. In reality the bed gives off 15.93 MW at 300°C. It was deleted and entered again as a hot stream from 301 to 300°C with the same heat load.
- DEC and FLASH were read as small duties at 40°C, 0.18 MW together. Nothing can use or remove heat at 40°C against 30°C cooling water, so both were removed.
- VAP-1 starts at 28.0°C and not at 144°C. The 144 is the molar flow of S1 in kmol/h. The duty of 3.68 MW is unchanged.

The final data is in Table 2. VAP-1 and the FEHE cold side make one stream, S1 to S3. COOL-1, COOL-2 and the FEHE hot side make one stream, RXOUT to S6.

Table 2: Stream data used for pinch analysis, from Aspen Energy Analyzer.

Stream	Type	T_{in} (°C)	T_{out} (°C)	Duty (MW)
REACTOR (bed, entered again)	Hot	301.0	300.0	15.94
RXOUT to S6 (FEHE, COOL-1, COOL-2)	Hot	300.0	40.0	9.58
ANRECOV to ARLIQ (ARCOND)	Hot	97.2	40.0	7.63
COL-2 condenser	Hot	131.6	129.0	7.02
COL-1 condenser	Hot	101.8	64.7	2.96
PRODWARM to PRODUCT (PRODCOOL)	Hot	129.0	40.0	0.66
S1 to S3 (VAP-1, FEHE)	Cold	28.0	290.0	3.89
COL-1 reboiler	Cold	176.1	185.8	4.07
COL-2 reboiler	Cold	143.9	145.4	6.56
STRIP reboiler	Cold	100.9	101.2	7.94
Total hot / total cold				43.78 / 22.45

2.2 Manual problem table

The targets were also worked out by hand with the problem table method, following the temperature-interval steps of Seider, Seader, Lewin and Widagdo [24]. Hot and cold totals in Table 2 do not match, and they need not, because the plant has a heat surplus. Each stream is taken with constant $CP = Q/(T_{in} - T_{out})$. The reactor is a heat source at about 300°C, so it is put in as a 1 K stream. Its CP of 15.93 MW/K is only a number for the method, not a real heat capacity.

Step 1, shifted temperatures. Hot streams are moved down by $\Delta T_{min}/2 = 5$ K and cold streams up by 5 K, so any match in the same interval has at least 10 K driving force.

Table 3: Shifted temperatures and CP values, $\Delta T_{min} = 10$ K.

Stream	Type	T^*_{in} (°C)	T^*_{out} (°C)	CP (MW/K)
REACTOR	Hot	296.0	295.0	15.9300
RXOUT	Hot	295.0	35.0	0.0368
ARCOND	Hot	92.2	35.0	0.1334
COL-2 cond	Hot	126.6	124.0	2.7000
COL-1 cond	Hot	96.8	59.7	0.0790
PRODCOOL	Hot	124.0	35.0	0.0074
S1–S3	Cold	295.0	33.0	0.0148
COL-1 reb	Cold	190.8	181.1	0.4186
COL-2 reb	Cold	150.4	148.9	4.3733
STRIP reb	Cold	106.2	105.9	26.4667

Step 2, intervals. The shifted temperatures in falling order are 296.0, 295.0, 190.8, 181.1, 150.4, 148.9, 126.6, 124.0, 106.2, 105.9, 96.8, 92.2, 59.7, 35.0 and 33.0°C. This gives 14 intervals.

Step 3, interval balance. For each interval $\Delta H_i = (\sum CP_{hot} - \sum CP_{cold})\,\Delta T^*_i$. A positive value is surplus heat and a negative value is a deficit.

Step 4, cascade. Starting with $Q_H = 0$ at the top, each ΔH_i is added and the heat flowing out of every interval is listed in the last column of Table 4.

Table 4: Problem table and heat cascade with $Q_H = 0$. MW.

#	T^* interval (°C)	Streams present	$\sum CP_H - \sum CP_C$	ΔT^*	ΔH_i	Cascade out
1	296.0–295.0	REACTOR	15.9300	1.0	15.930	15.930
2	295.0–190.8	RXOUT, S1–S3	0.0220	104.2	2.292	18.222
3	190.8–181.1	RXOUT, S1–S3, COL-1 reb	-0.3966	9.7	-3.847	14.376
4	181.1–150.4	RXOUT, S1–S3	0.0220	30.7	0.675	15.051
5	150.4–148.9	RXOUT, S1–S3, COL-2 reb	-4.3513	1.5	-6.527	8.524
6	148.9–126.6	RXOUT, S1–S3	0.0220	22.3	0.491	9.015
7	126.6–124.0	RXOUT, COL-2 cond, S1–S3	2.7220	2.6	7.077	16.092
8	124.0–106.2	RXOUT, PRODCOOL, S1–S3	0.0294	17.8	0.524	16.615
9	106.2–105.9	RXOUT, PRODCOOL, S1–S3, STRIP reb	-26.4373	0.3	-7.931	8.684
10	105.9–96.8	RXOUT, PRODCOOL, S1–S3	0.0294	9.1	0.268	8.952
11	96.8–92.2	RXOUT, COL-1 cond, PRODCOOL, S1–S3	0.1084	4.6	0.499	9.450
12	92.2–59.7	RXOUT, ARCOND, COL-1 cond, PRODCOOL, S1–S3	0.2418	32.5	7.858	17.308
13	59.7–35.0	RXOUT, ARCOND, PRODCOOL, S1–S3	0.1628	24.7	4.021	21.330
14	35.0–33.0	S1–S3	-0.0148	2.0	-0.030	21.300

Step 5, targets. No value in the cascade goes below zero. The smallest heat flow is 8.52 MW, below interval 5 at the COL-2 reboiler. So no hot utility has to be added at the top, and $Q_{H,min} = 0$. The heat leaving the bottom of the cascade is the cold utility, $Q_{C,min} = 21.30$ MW. Aspen gives 21.33 MW. The 0.03 MW gap is because Aspen uses the real enthalpy curve of S1 to S3, with the vaporisation at 262°C, while the hand method takes one straight-line CP. Since the cascade never touches zero, there is no process pinch, which confirms the threshold result.

2.3 Targets

The composite curves are in Figure 1 and the grand composite curve in Figure 2. The hot curve is fully above the cold curve. So there is no process pinch and this is a threshold problem. When MP steam raising is added as a utility, Aspen puts a utility pinch at 153.9°C hot and 143.9°C cold, just above the COL-2 reboiler. The targets are in Table 5.

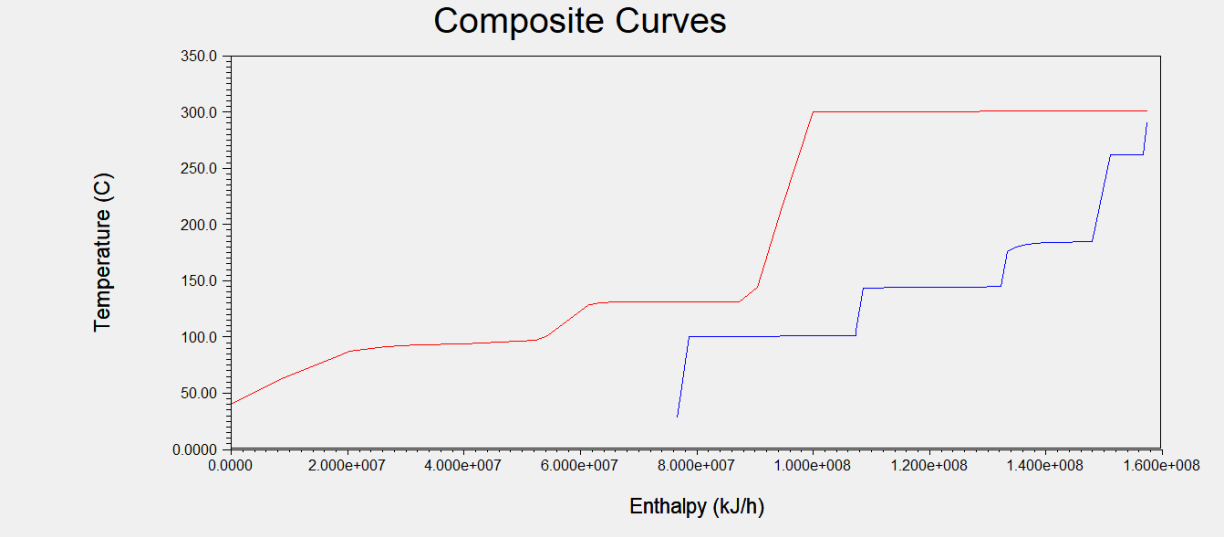


Figure 1: Composite curves at $\Delta T_{min} = 10$ K, from Aspen Energy Analyzer. Red is the hot composite and blue the cold composite.

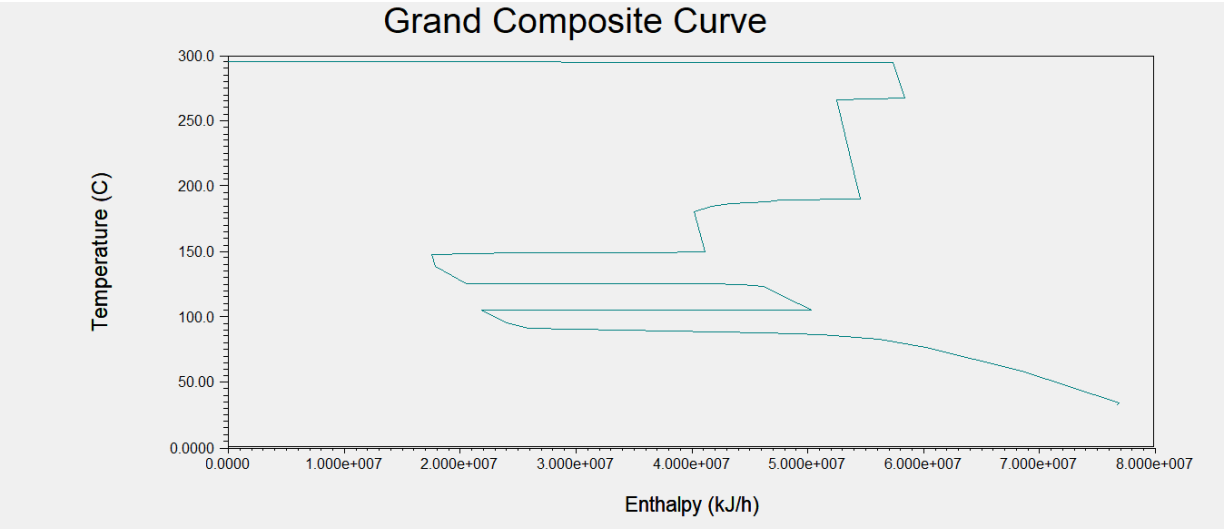


Figure 2: Grand composite curve at $\Delta T_{min} = 10$ K, from Aspen Energy Analyzer.

Table 5: Energy targets at $\Delta T_{min} = 10$ K, Aspen Energy Analyzer.

Item	Value
Minimum hot utility	0 MW
Minimum cold utility	21.33 MW
Current heating in flowsheet	22.24 MW
Number of units, minimum / at maximum energy recovery	11 / 12
Number of shells	18
Area target, counter-current / 1-2 shell and tube	4389 / 4445 m^2
MP steam that can be raised, at 175°C	4.70 MW
Cooling water left, with steam raised	16.63 MW
Utility pinch, hot / cold	153.9 / 143.9°C

The GCC shows a large pocket from 300 down to about 100°C. Inside it the reactor heat runs down into the reboilers. Below 100°C all the heat goes to cooling water.

Table 6 gives the effect of ΔT_{min}, from a problem table on the same data. The hot utility stays near zero up to 30 K. At 40 K it jumps to 1.83 MW, because the reactor at 300°C can no longer vaporise the feed at 262°C. This value comes back in Section 3.

Table 6: Effect of ΔT_{min} on the targets.

ΔT_{min} (K)	Hot utility (MW)	Cold utility (MW)
5	0	21.31
10	0	21.31
20	0.07	21.38
30	0.15	21.46
40	1.83	23.15

3 Heat Exchanger Network Design

3.1 Network from Aspen Energy Analyzer

The network was first made in Aspen Energy Analyzer with Recommend Designs, on the corrected stream data and with HP and MP steam raising as cold utilities. Design A_Design2 is taken. It is in Figure 3 and its numbers are in Table 7.

Table 7: Aspen Energy Analyzer network, design A_Design2.

Item	Value
Heat exchangers	17
Shells	20
Total area	3834 m^2
Hot utility	0 MW
Cooling water	16.63 MW
MP steam raised	4.70 MW
Smallest approach	9.4 K, on ARCOND and PRODCOOL

All ten streams reach their target temperatures, so the design meets the energy target exactly. The reactor heat goes to the S1 feed (preheat, vaporising and superheat), to both reboilers of COL-1 and COL-2, and the balance raises MP steam. RXOUT gives heat to the COL-2 reboiler, to the S1 preheat and to the stripper, and then goes to cooling water. The COL-2 condenser drives the stripper reboiler in three exchangers. ARCOND and PRODCOOL run 0.6 K below the 10 K approach on the cooling water side, which is why Aspen marks them.

3.2 Steam level on the reactor

In A_Design2 the reactor is a source at 300°C, and it vaporises the feed at 262°C. In the plant this heat is carried by the coolant on the reactor shell, so the coolant would have to be above 273°C. Report 4 showed that the 25 mm bed peaks at 390.7°C against a 400°C limit, with its outlet near 240°C. A coolant at 286°C, which is 70 bar steam, pushes the peak above the limit.

So 40 bar steam is taken on the shell, saturated at 250.4°C. The mean driving force against the 300°C bed is then about 50 K. The 9700 tubes of 25 mm give 1524 m^2 inside area, so the overall coefficient needed is

$$U = \frac{15.93 \times 10^6}{1524 \times 50} = 209 \text{ W/m}^2\text{K}, \tag{1}$$

which is a normal value for a packed-bed tube wall.

The 50 K is a rough mean: the bed is about 300°C over most of its length and peaks at 390.7°C near the inlet, against 250.4°C on the shell. The 1524 m^2 is the inside area of the Report 4 tubes, so the check is preliminary. Steam pressure is held by the drum pressure controller, which sets the coolant temperature and so keeps the bed below 400°C.

So the reactor works at an effective ΔT of about 50 K and not 10 K. From Table 6 this costs about 1.8 MW of hot utility, which is the vaporisation of the feed at 262°C.

3.3 Final network

The final network keeps the idea of A_Design2 but carries the reactor heat through the 40 bar steam header. The vaporiser stays on HP steam. Some small matches of A_Design2 are merged, so the unit count comes down from 17 to 14. The exchangers are in Table 8. Areas are from $Q = UA\,\Delta T_{lm}$, with U from the typical ranges in Sinnott and Towler [6]. We tried to put this network into Aspen Energy Analyzer also, as a second design, but it did not close in the time we had. So its balance is checked by hand, in Table 9.

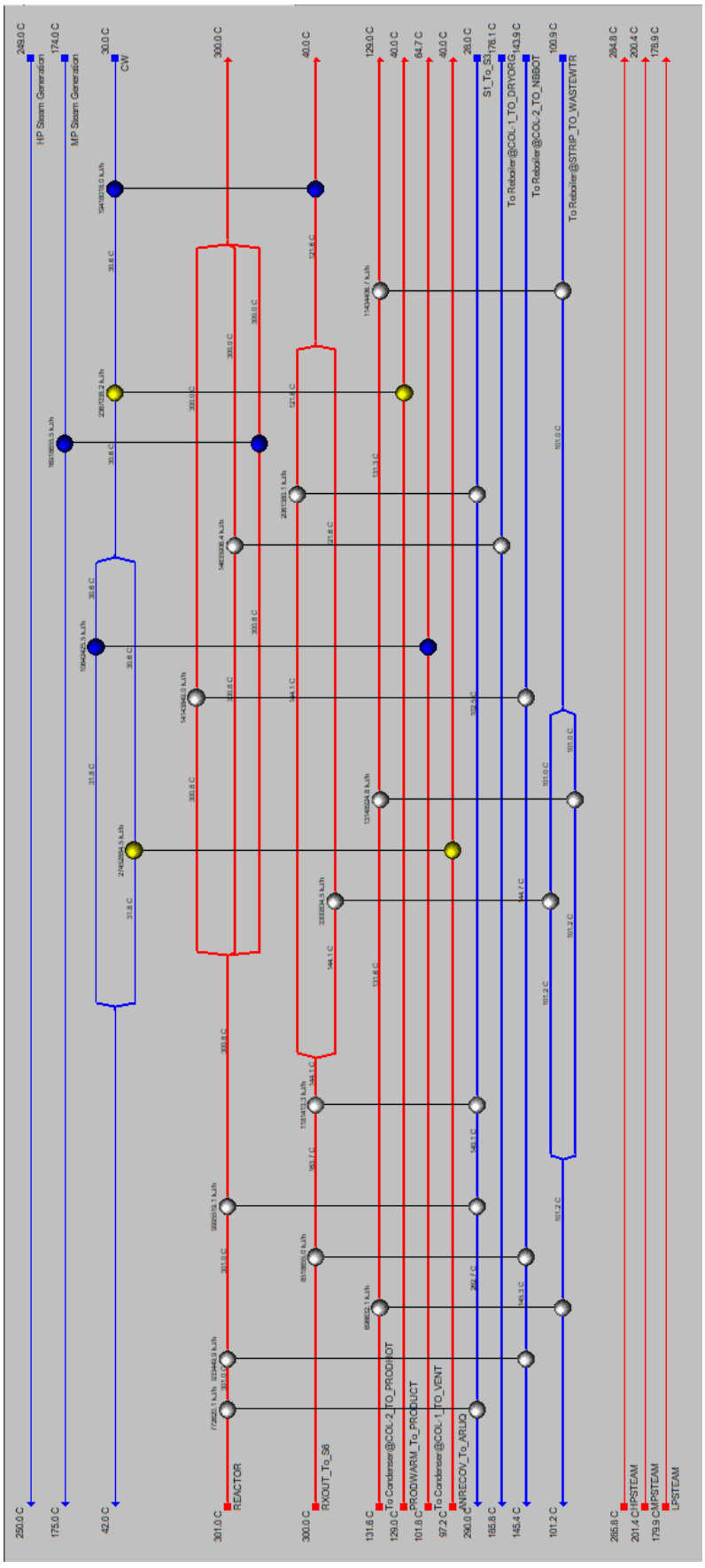


Figure 3: Heat exchanger network A_Design2 from Aspen Energy Analyzer, $\Delta T_{min} = 10$ K. Red lines are hot streams, blue lines are cold streams and utilities. Duties in kJ/h.

Table 8: Final heat exchanger network. New units are marked N.

Unit	Hot side	Cold side	Duty (MW)	U (W/m^{2}K)	Area (m^2)
E-101 (N)	REACT bed, 300°C	BFW to 40 bar steam	15.93	209	1524
E-102A (N)	40 bar steam	S1, 28.0 to 240.4°C	1.88	500	54.8
VAP-1	70 bar steam	S1, 240.4 to 262.7°C	1.80	500/900	87.6
FEHE	RXOUT, 300.0 to 288.2°C	S4, 262.7 to 290.0°C	0.21	100	129.8
E-104 (N)	RXOUT, 288.2 to 200.0°C	BFW to 10 bar steam	2.17	120	345.4
COOL-1	RXOUT, 200.0 to 120.0°C	Cooling water	1.97	150	108.7
COOL-2	RXOUT, 120.0 to 40.0°C	Cooling water	5.24	300	527.5
E-103 (N)	COL-2 overhead, 131.6 to 129.0°C	STRIP bottoms, 100.9 to 101.2°C	7.02	800	300.3
STRIP reb.	10 bar steam	STRIP bottoms	0.92	1500	67.1
COL-1 reb.	16 bar steam, let down from 40 bar	COL-1 bottoms	4.06	1000	202.4
COL-2 reb.	10 bar steam, let down from 40 bar	COL-2 bottoms	6.56	1000	186.0
ARCOND	ANRECOV, 97.2 to 40°C	Cooling water	7.63	EDR	147.0
COL-1 cond.	COL-1 overhead	Cooling water	2.93	600	105.9
PRODCOOL	Product, 129 to 40°C	Cooling water	0.66	350	52.8

E-101 is the reactor shell itself, so it needs only a steam drum and no extra tube area. E-103 replaces the COL-2 condenser. The COL-2 overhead at 0.2 bar condenses on the shell side of the stripper reboiler, with 28 K driving force. The vapour line at 0.2 bar is large, so the stripper is placed next to COL-2. VAP-1 is now only the vaporiser and E-102A preheats its feed. COOL-1 becomes a trim cooler after E-104.

3.4 Steam balance and comparison with target

Table 9: Steam balance of the integrated plant, MW.

40 bar header		10 bar header	
Raised in E-101	15.93	Raised in E-104	2.17
To E-102A	−1.88	To STRIP reboiler	−0.92
To COL-1 reboiler, via 16 bar	−4.06		
To COL-2 reboiler, via 10 bar	−6.56		
Exported	3.43	Exported	1.26

Purchased heating is 1.80 MW against zero in A_Design2, and 1.83 MW in the 40 K target of Table 6. Cooling water plus exported steam is 23.11 MW against 21.33 MW in A_Design2. Both gaps are the same 1.8 MW, which is the reactor steam level of Section 3.2. The final network has

14 units against 17 in A_Design2.

4 Utility Costs

Utility prices are those in the Aspen file from Report 3. Exported steam is credited at half its price, as a buyer has to be found for it. Boiler feed water is charged at USD 0.5 per tonne on all steam raised. For a fair base case, REACT and COOL-1 are put on cooling water, since they had no utility in Report 4. This before case is built by us for comparison, it was not a utility system in the Report 4 file. Exported steam is an assumed credit to a site header. The comparison is in Table 10.

Table 10: Utility use and cost before and after heat integration, 8200 h/y.

	Before		After	
Utility, price	kg/h	USD/h	kg/h	USD/h
HP steam 70 bar, 0.035 USD/kg	8,792	307.7	4,307	150.7
MP steam 16 bar, 0.030 USD/kg	7,565	226.9	0	0
LP steam 10 bar, 0.026 USD/kg	25,923	674.0	0	0
Cooling water, 2×10^{-5} USD/kg	3,137,336	62.7	1,327,277	26.5
Boiler feed water, 0.0005 USD/kg	–	–	27,628	13.8
40 bar steam exported, 0.016 USD/kg	–	–	−5,235	−83.8
10 bar steam exported, 0.013 USD/kg	–	–	−1,932	−25.1
Total, USD/h		1271.4		82.2
Total, USD million/y		10.43		0.67
Total, Rs. crore/y		99.0		6.4

The saving is USD 9.75 million per year. Cooling water circulation drops to 1327 t/h, so the pump and the tower fans of Report 3 come down from 311 kW to 262 kW. With COMP at 213 kW and the 20 per cent allowance of Report 3, plant demand becomes about 570 kW, against 757 kW from the gas turbine. The turbine still burns 7.71 kmol/h of natural gas. At 802.3 kJ/mol this is 5.86 MMBtu/h, and $5.86 \times 8200 \times 10$ gives USD 0.48 million per year at USD 10 per MMBtu.

5 Capital Cost

5.1 Method

Equipment is costed with the bare-module correlations of Turton et al. [18], on carbon steel, with pressure factors where needed. The base CEPCI is 397 (2001), brought to 797.9 (2023). Fixed capital is the grass-roots cost, $1.18 \sum C_{BM} + 0.50 \sum C^{0}_{BM}$. Items which do not fit the correlations are taken as installed packages at the costs stated in Table 12. Working capital is 15 per cent of the fixed capital.

The columns were not sized in Reports 3 and 4, so they are sized here. Table 11 gives the result, from Fair flooding with $C = 0.09$ m/s at 80 per cent of flood, 0.6 m tray spacing and 70 per cent tray efficiency on the Aspen stages. COL-2 is wide because it runs at 0.2 bar.

Table 11: Column sizes used for costing.

Column	Max vapour (kg/h)	Pressure (bar)	Diameter (m)	Sieve trays	Height (m)
COL-1	28,770	1.1	1.85	15	13.0
COL-2	48,800	0.2	3.50	33	24.0
STRIP	13,000	1.0	1.75	10	10.0

5.2 Result

Table 12: Capital cost of the integrated plant, 2023 USD.

Item	Size or basis	C_{BM} (USD million)
REACT	1828 m^2, 40 bar shell, 9700 tubes of 25 mm	1.826
Steam drum	1.5 m × 4.5 m, 40 bar	0.207
COL-1	Table 11	0.297
COL-2	Table 11, vacuum	1.741
STRIP	Table 11	0.227
FLASH and DEC	Report 4 sizes	0.096
Carbon adsorbers, two	1.2 m × 8.2 m	0.176
Heat exchangers, 13 units	Table 8, floating head	2.990
COMP with motor	213 kW, centrifugal	0.706
Pumps	P-1, P-2 and two cooling-water pumps	0.320
Nitrobenzene storage	2 × 2300 m^3, two weeks	0.622
Aniline storage	2 × 1900 m^3, two weeks	0.559
Cooling tower	369 L/s, Towler correlation [6], installed	1.426
Gas turbine package	757 kW at USD 1500/kW, assumed	1.135
Thermal oxidiser package	2.6 MW, assumed	0.600
BFW treatment and deaerator	28 t/h, assumed	0.300
Catalyst first charge	11,430 kg at USD 20/kg	0.229
Carbon first fill	6800 kg at USD 2/kg	0.014
Total bare module, $\sum C_{BM}$		13.47
Fixed capital, grass roots		18.22
Working capital, 15 %		2.73
Total capital investment		20.95
	Rs. crore	199.0

The heat integration adds USD 0.56 million in exchangers and USD 0.21 million for the steam drum. The smaller cooling tower saves USD 1.07 million. So the integrated plant costs USD 0.31 million less in capital than the plant without integration, whose fixed capital is USD 18.63 million. This is a study estimate, accurate to about ±30 per cent.

As a check, the purchased cost of the same equipment is USD 6.68 million. With the Lang factor of 4.74 for a fluid plant [6], and the catalyst and carbon added, the fixed capital comes to USD 31.9 million. This is higher than the bare-module figure, because the Lang factor carries more for piping, offsites and indirect costs. The bare-module value is kept as the base case, and the Lang value is run as a sensitivity case in Table 17.

5.3 Sample calculations

COL-2 diameter. Top vapour is 48,800 kg/h of aniline at 0.2 bar and 129°C. Vapour density is $\rho_V = PM/RT = 20000 \times 93.1/(8314 \times 402) = 0.557$ kg/m^3, and liquid density is about 930 kg/m^3. With $C = 0.09$ m/s for 0.6 m tray spacing, the flooding velocity is $u_f = C\sqrt{(\rho_L - \rho_V)/\rho_V} = 0.09\sqrt{930/0.557} = 3.68$ m/s. At 80 per cent of flood $u = 2.94$ m/s. Volume flow is $48800/(3600 \times 0.557) = 24.3$ m^3/s, so net area is 8.27 m^2. Taking downcomers as 12 per cent, total area is 9.4 m^2 and $D = 3.46$ m, rounded to 3.5 m. Aspen has 25 stages including condenser and reboiler, so 23 trays are theoretical. At 70 per cent efficiency this is $23/0.7 = 33$ actual trays, and the height is $33 \times 0.6 + 4 = 24$ m.

E-103 area. Hot side 131.6 to 129.0°C, cold side 100.9 to 101.2°C, counter-current. $\Delta T_1 = 131.6 - 101.2 = 30.4$ K and $\Delta T_2 = 129.0 - 100.9 = 28.1$ K, so $\Delta T_{lm} = (30.4 - 28.1)/\ln(30.4/28.1) = 29.2$ K. With $U = 800$ W/m^{2}K for condensing organic against boiling water [6], $A = 7.02 \times 10^6/(800 \times 29.2) = 300$ m^2.

Thermal oil in place of 70 bar steam for VAP-1. An oil loop heated on the first part of the bed could supply oil at about 290°C and return it at 275°C. Against the feed at 240.4 to 262.7°C, $\Delta T_{lm} = 31$ K, and with $U = 400$ W/m^{2}K the vaporiser needs about 145 m^2, about USD 0.21 million bare module. It would save the 4307 kg/h of HP steam, USD 1.24 million per year. The catch is that the oil must be hotter than 273°C, so it can only take heat where the bed is above about 290°C, near the inlet hot spot. Whether that section gives 1.8 MW needs the non-isothermal reactor run of Report 4 with an oil-cooled zone. So steam is kept for now and the oil loop is the route.

6 Techno-Economic Analysis

6.1 Basis

Table 13: Economic basis.

Item	Value
Operating time	8200 h/y, as in Report 4. Report 1 used 8000 h/y; 8200 h/y is the final basis
Product	99,924 t/y at 99.45 wt% aniline; price applied on total product mass
Aniline price	USD 1632/t, Rs. 155,000/t, Report 1
Nitrobenzene price	USD 900/t, term contract, Report 1
Hydrogen price	USD 1947/t, on-site SMR cost, Report 1
Natural gas	USD 10/MMBtu, middle of the Report 1 range
Construction	2 years, 60 and 40 per cent of fixed capital
Plant life	15 years after start-up
Depreciation	Straight line over 10 years, no salvage
Tax	25 per cent
Discount rate	12 per cent
Working capital	15 per cent of fixed capital, recovered at the end

The base case NPV and IRR assume the product reaches the 99.9 wt% and 5 ppm specification of Report 1, so they are conditional on that. The present simulation gives 99.45 wt%, and this is covered by the off-spec price case in Table 17. ROI is profit after tax divided by total capital. The NPV includes construction years, working capital and its recovery, tax and the depreciation tax shield, with no salvage value.

6.2 Cost of production

Fixed costs follow the factors of Sinnott and Towler [6]. Labour is 15 operators, which is three per shift on five shifts, at USD 9000 per year each.

Table 14: Cost of production at full capacity.

Item	USD million/y	USD/t product
Nitrobenzene, 142,624 t/y	128.36	1284.6
Hydrogen, 7434 t/y	14.48	144.9
Utilities, Table 10	0.67	6.7
Natural gas to gas turbine	0.48	4.8
Catalyst, 3-year life	0.08	0.8
Activated carbon, 30-day cycle	0.08	0.8
Wastewater treatment, USD 5/m^3	0.18	1.8
Variable cost	144.33	1444.4
Labour, supervision and salary overhead	0.25	2.5
Maintenance, 3 % of fixed capital	0.55	5.5
Insurance and local taxes, 1 %	0.18	1.8
General plant overhead	0.52	5.2
Cash cost of production	145.83	1459.4
Revenue	163.03	1631.6

The wastewater charge is high on purpose. It covers the cyclohexylamine in WASTEWTR, which carbon does not hold, so the water goes to biological treatment.

6.3 Profitability

Table 15: Profitability, base case.

Item	Value
Gross profit	USD 17.21 million/y
Depreciation	USD 1.82 million/y
Profit before tax	USD 15.38 million/y
Profit after tax	USD 11.54 million/y
Annual cash flow, years 1 to 10	USD 13.36 million/y
NPV at 12 per cent	USD 61.3 million (Rs. 583 crore)
IRR	50.4 per cent
Simple payback on total capital	1.6 years
Return on investment, profit after tax / total capital	55 per cent

The year-wise cash flow is in Table 16. Year −1 and year 0 are construction. Working capital goes in at year 0 and comes back in year 15. The tax shield of depreciation ends after year 10, so the cash flow drops slightly from year 11.

Table 16: Cash flow, USD million. Discounted at 12 per cent.

Year	Cash flow	Cumulative	Cumulative, discounted
−1	−10.93	−10.93	−10.93
0	−10.02	−20.95	−19.88
1	13.36	−7.59	−9.23
2	13.36	5.77	0.28
3	13.36	19.13	8.77
4	13.36	32.49	16.35
5	13.36	45.85	23.12
6	13.36	59.21	29.17
7	13.36	72.57	34.56
8	13.36	85.93	39.38
9	13.36	99.29	43.68
10	13.36	112.65	47.52
11	12.90	125.55	50.83
12	12.90	138.45	53.79
13	12.90	151.36	56.43
14	12.90	164.26	58.79
15	15.64	179.90	61.34

The discounted cash flow turns positive in year 2 after start-up.

6.4 Sensitivity

Table 17: Sensitivity of the result. One change at a time, then one combined case.

Case	NPV (USD million)	IRR (%)	Payback (y)
Base case	61.3	50.4	1.6
Aniline USD 1600/t, top of quoted band	47.0	42.7	1.9
Aniline USD 1500/t, bottom of quoted band	1.4	13.2	6.0
Nitrobenzene USD 960/t, spot price	22.3	28.2	3.0
Fixed capital +30 per cent	54.7	40.0	2.1
Fixed capital +50 per cent	50.3	35.0	2.4
Capital by Lang factor, USD 31.9M	44.7	30.2	2.8
Discount rate 15 per cent	48.3	50.4	1.6
No heat integration	16.4	24.0	3.6
Aniline USD 1550/t, off-spec product	24.2	29.3	2.9
Combined: aniline 1500, NB 960, capital +30%	−44.3	–	–

The project is driven by the aniline and nitrobenzene prices, not by capital. If all three bad cases come together, the NPV is negative, so the margin is thin when prices move against us. NPV goes to zero at an aniline price of USD 1497/t or a nitrobenzene price of USD 994/t. A 50 per cent rise in capital costs only USD 11 million of NPV, and even the Lang capital keeps the IRR at 30 per cent. Heat integration is worth USD 45 million of NPV, because it takes USD 9.75 million a year off the utility bill.

7 Objectives Which Could Not Be Accomplished, With Reasons

Both objectives of this report, the network design by pinch technology and the utility and techno-economic study, are complete. The product and discharge specifications set in Report 1 are separate from these objectives and are carried forward. The position on each is below.

Aniline purity. 99.45 wt% against a target of 99.9 wt%, unchanged from Report 4. Cyclohexylamine at 0.48 wt% holds it back. A light-ends draw at the top of COL-2 is the route. The product is priced at the full aniline price in Section 6, which assumes the purity is met.

Nitrobenzene in the product. 313 ppm against a target of 5 ppm. A taller COL-2 is the route, as in Report 4. Ten more trays on the 3.5 m vacuum column cost about USD 0.4 million, which does not change the NPV much.

Aniline in the wastewater. 216 ppm into the carbon bed against a discharge limit of 1 ppm. The bed holds it below 1 ppm on the design basis of Report 4. A breakthrough curve is still needed to confirm this.

Cyclohexylamine in the wastewater. 4.28 wt%. It is now costed as biological treatment at USD 5/m^3, Section 6.

Second decanter on WATOVHD. Still open loop, because WATOVHD is the only exit benzene has. A benzene column is needed first.

Purchased heating. 1.80 MW of 70 bar steam for VAP-1, against zero in the Aspen network. It comes from the 40 bar steam level that the bed limit sets. A thermal-oil loop on the first part

of the bed, where most heat is released, could supply it, and that is the route.

Reaction kinetics. The two side-reaction rate constants are still fitted, not regressed.

8 Any Other Challenges

REACT read as a cold stream. The Energy Analysis took the reactor as a stream heated from 160 to 300°C by fired heat. This is because RPlug has no utility option and its outlet is hotter than its inlet. The first target was 27 MW of heating, which is wrong. The stream editor of Aspen Energy Analyzer does not allow a cold stream to be changed to hot, so the stream was deleted and entered again. This had to be done twice, as the first corrected file was not saved.

Scope of the analysis. The Define Scope option works only on whole flowsheets, not on single blocks. So DEC and FLASH were removed in the stream table and not in the scope.

Automatic network design. On the first run Recommend Designs gave ten networks, and all had infeasible exchangers. After the reactor stream was corrected and fired heat was replaced by HP and MP steam raising, the second run gave A_Design2 with all streams satisfied. We also tried to put our own 14-unit network into Aspen as a second design, but it did not close in the time we had, so that network is checked by hand.

9 Number of Hours Spent During This Period

Approximately 5 hours per member, which is 30 hours for the group.

10 Contributions From Individual Members

Nithin T M and Aditya Nitin Patil — the Energy Analysis in Aspen Plus, the correction of the reactor stream, the targets, the reactor steam level and the final network, and the compilation of this report.

Nithin D H and Shubham Singh — the utility prices and steam balance, the column sizing, the vessel and storage costs, and the gas turbine fuel cost.

Shivam Dongare and Sajag Masane — the network A_Design2 in Aspen Energy Analyzer with its curves and tables, the exchanger areas and costs, and the cash flow and sensitivity cases.

All six members have reviewed the complete report.

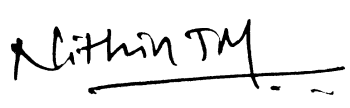
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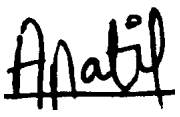
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Signatures of members



Nithin T M



Aditya Nitin Patil



Nithin D H



Shubham Singh



Shivam Dongare



Sajag Masane

Bi-weekly Report 6

Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

CHE453 Process Design Capstone Project

Group No: 14

Date: 8 November 2026

Reports 1 to 5 precede this page and are unchanged.
Everything which follows constitutes Report 6.

Bi-weekly Report

Report Number: 06
Group No: 14
Date: 8 November 2026

Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

Team members

Name	Roll No.	Name	Roll No.
Nithin T M	230709	Shubham Singh	230999
Aditya Nitin Patil	230074	Shivam Dongare	230971
Nithin D H	230708	Sajag Masane	230896

Objectives for this report

1. Innovative component by the team.

1 Basis

Report 5 closed the heat integration and the techno-economic analysis. The items open at that time which bear on this report are listed in Table 1.

Table 1: Position at the close of Report 5.

Item	Report 5 status	Taken up in
Purchased heating, 70 bar steam for VAP-1	1.80 MW, 4,307 kg/h	Sections 2, 4, 6
Reactor feed vapour fraction	0.988 at 160.2°C	Section 3
Bed peak with 40 bar steam on the reactor shell	Not checked	Section 5
Route to replace the 70 bar steam (thermal-oil loop proposed)	Proposed, not studied	Sections 2, 4
Letdown valves to 16 and 10 bar	Pressure energy not recovered	Section 8
Aniline purity, 99.9 wt% required	99.45 wt%	Section 11
Nitrobenzene in product, below 5 ppm	313 ppm	Section 11
Aniline and cyclohexylamine in wastewater	Carbon bed and biological treatment	Section 11

The flowsheet, property method (NRTL-RK with the regressed pairs R-DR-1 and DR-1) and heat exchanger network are those of Report 5. Only the feed heating section and the reactor steam drum pressure are changed.

Innovative component: stop boiling nitrobenzene on its own. Let the hydrogen loop carry it instead, by evaporating the nitrobenzene into the recycle and make-up hydrogen in a packed

saturator column heated by reactor steam. Then use the lower heating level this allows to reset the reactor steam drum pressure, so that the catalyst bed stays below its 400°C limit. Simulate the change, design the saturator, rebuild the steam balance, check the bed hot spot and evaluate the saving. Back-pressure turbines on the steam letdown were examined as a second option and are reported in Section 8.

1.1 Key results

In Report 5 the nitrobenzene is boiled alone at 263°C (3 bar). That single service needs 1.80 MW of purchased 70 bar steam, 4,307 kg/h or USD 1.24 million per year. Mixed with the hydrogen, the nitrobenzene has a partial pressure of only 0.29 bar, and Aspen Plus puts the dew point of the reactor feed at 163.9°C. The whole feed heating of 4.29 MW is then needed below 176°C, so reactor steam does all of it and VAP-1 and the 70 bar steam are removed.

The nitrobenzene is evaporated in a packed saturator rather than in a heat exchanger. Hot liquid nitrobenzene (176°C) runs down the packing against the rising hydrogen and evaporates into it; the gas leaves the saturator top dry at 168.9°C, 3.8 K above its dew point at 3.1 bar (165.1°C). The FEHE then takes it to 170°C at the reactor inlet, where the pressure is 3.0 bar and the dew point 163.9°C. All the steam heat, about 4.2 MW, enters through a liquid-phase heater of 81 m^2, where a mixed gas–liquid heater would need 196 to 392 m^2. Nitrobenzene that the gas cannot carry stays in the column sump and cannot reach the catalyst.

Three defects of Report 5 are closed. The 70 bar steam import goes to zero. The reactor feed, which was 1.2 mol% liquid at 160.2°C (about 17 kmol/h, mostly nitrobenzene), becomes fully vapour. And the 40 bar steam drum of Report 5 (250.4°C on the shell) gives a bed peak of 422°C in a calibrated non-isothermal bed model, 22 K above the copper sintering limit. With nothing on the plant now needing heat above 176°C, the drum is reset to 33 bar (239.2°C) and the peak falls to 388°C with a 170°C feed, a margin of 12 K.

The net saving is USD 0.79 million per year for USD 0.49 million of added fixed capital (new saturator system less VAP-1 and the old E-102A). The incremental NPV at 12 per cent is USD 3.2 million, the IRR 83 per cent and the payback 0.8 years. The NPV stays positive in every single-variable sensitivity case, including export steam valued at full price. The back-pressure turbines give an NPV of about zero at the new 33 bar header and are not adopted.

The saturator outlet and the 388°C bed peak come from a hand design and a calibrated one-dimensional model. They are preliminary until the two Aspen checks of Section 11 are made, and the NPV above should be read on the same basis. Table 2 shows which results come from Aspen Plus and which from separate calculations.

Table 2: Source of each main result of this report.

Result	Value	Source
Dew point of the reactor feed; feed-heating duty	163.9°C; 4.29 MW	Aspen Plus, test branch (Table 6)
Reactor, product and coolant duty with a 170°C feed	Table 8	Aspen Plus, full recycle flowsheet
Saturator performance and sizing; H-101 duty and area	168.9°C; 4.18 MW; 81 m^2	Hand calculation (Section 4)
Bed peak at 33 and 40 bar; margin to 400°C	388 and 423°C; 12 K	Calibrated 1-D bed model (Section 5)
Steam balance	Table 14	Aspen duties with the hand split of Table 18
Capital cost, saving, NPV	Tables 19 to 21	Turton correlations and the Report 5 basis
Turbines at 40 bar	Table 23	Aspen Plus, separate steam-system file
Turbines at 33 bar	Table 25	IAPWS-IF97 steam tables

1.2 Final design basis

Report 1 set the capacity at 100,000 t/y on 8000 h/y, that is 12.5 t/h. Report 4 adopted 8200 h/y, about 94 per cent on-stream time for a continuous plant, and Report 5 made it the final basis. All annual figures in Reports 5 and 6 use 8200 h/y; the 8000 h/y and the conversion, selectivity and 95 mol% recycle-purity values of Report 1 were preliminary assumptions, replaced by the converged Aspen results of Reports 4 and 5. The three production figures are kept apart in Table 3.

Table 3: Capacity, simulated rate and annual production.

Quantity	Report 5	This report
Nominal capacity (Report 1)	100,000 t/y, 12.5 t/h at 8000 h/y	Same
Operating time, final basis	8200 h/y	8200 h/y
Simulated product stream	12,185.8 kg/h	12,182.9 kg/h
Annual product at 8200 h/y	99,924 t/y	99,900 t/y

1.3 Values carried over from Reports 4 and 5

The values carried over are in Table 4.

Table 4: Values carried over from Reports 4 and 5.

Item	Value (Reports 4 and 5)
Feed vaporiser VAP-1	Heater, 3 bar, vapour fraction 1, pure nitrobenzene; boils at 263°C
Cold stream S1 to S3 (E-102A, VAP-1, FEHE)	28.0 to 290.0°C, 3.89 MW
Feed heaters in the final network	E-102A 1.88 MW (40 bar steam, 54.8 m^2); VAP-1 1.80 MW (70 bar steam, 87.6 m^2); FEHE 0.21 MW (129.8 m^2)
Purchased heating	4,307 kg/h of 70 bar steam at USD 0.035/kg, USD 1.24 million per year
40 bar steam balance	Raised in E-101 15.93 MW; to E-102A 1.88, COL-1 4.06, COL-2 6.56; exported 3.43 MW (5,235 kg/h at USD 0.016/kg, half of full value)
Reactor feed after MIX-2	160.2°C, 3 bar, 1,452 kmol/h, nitrobenzene 141.89 kmol/h
Reactor (Report 4)	9700 tubes, 25 mm, 2.0 m; bed peak 390.7°C at a 138°C inlet; limit 400°C
Reactor shell (Report 5)	Saturated 40 bar steam, 250.4°C
Operating time	8200 h/y

The feed section of Report 5 has three problems, and this report is built around them.

1. **Purchased steam.** Nitrobenzene is boiled alone because it meets the hydrogen only later, at MIX-2. Pure nitrobenzene boils at 263°C at 3 bar, which is above the 40 bar reactor steam (250.4°C), so the plant buys 70 bar steam for this one service. It is the only purchased heating left after the heat integration of Report 5.
2. **Liquid to the catalyst.** The vaporised nitrobenzene at 290°C is mixed with cold hydrogen at MIX-2 and part of it condenses again. Aspen gives a vapour fraction of 0.988 for the reactor feed at 160.2°C, so about 1.2 mol% of the 1,452 kmol/h, roughly 17 kmol/h and mostly nitrobenzene, enters the bed as liquid. RXFEED in Reports 2 and 3 had vapour fractions of 0.990 and 0.987, so the defect has been present since Report 2.
3. **Bed temperature.** Report 4 checked the hot spot at 390.7°C for a coolant that it did not state. Report 5 then put saturated 40 bar steam (250.4°C) on the shell. Section 5 shows that this coolant temperature drives the peak to about 422°C, above the 400°C sintering limit.

2 The Idea: Let the Hydrogen Carry the Nitrobenzene

The reactor feed carries about 9 mol of hydrogen-rich gas per mol of nitrobenzene, so nitrobenzene only has to reach its dew point at its own partial pressure, not its boiling point at the full pressure. Its mole fraction in RXFEED is 141.89/1,452 = 0.0977, which at 3 bar is a partial pressure of 0.293 bar. A Clausius–Clapeyron line through the normal boiling point (210.8°C) and the Aspen boiling point at 3 bar (263°C) gives about 162°C at 0.293 bar, against 163.9°C from Aspen (Section 3). The heating level falls by about 100 K.

The question is how to evaporate it. Two arrangements were compared.

- **Mix, then heat (first draft of this study).** Liquid nitrobenzene is sprayed into the cold hydrogen and the two-phase mixture is heated in an enlarged E-102A. The heater is then a low-pressure gas heater whose coefficient is set by the gas film, so its area is uncertain (196 to 392 m^2 for U of 150 to 75 W/m^2K). Its outlet is still two-phase, because reaching the dew

point needs 4.18 MW and E-102A supplies 4.08 MW, so the FEHE takes a wet inlet. If the hydrogen flow drops, liquid nitrobenzene goes on towards the catalyst.

- **Saturator (adopted).** The heat is put into liquid nitrobenzene, which is pumped round a packed column and meets the hydrogen by direct contact. This is how process-gas saturators work in reforming and methanol plants, where hot water is evaporated into the feed gas.

The saturator was chosen for four reasons.

1. All the steam heat enters a liquid, at a coefficient of 500 to 800 W/m^{2}K instead of 75 to 150 W/m^{2}K, so the heater is 61 to 97 m^2.
2. The column is the phase separator. The gas leaves the top through a demister above its dew point, and any nitrobenzene the gas cannot carry stays in the sump, where a rising level shows it. Liquid cannot reach the bed unnoticed.
3. The hottest nitrobenzene on the plant falls from 290°C (VAP-1 outlet, then FEHE) to 176°C, and none of it boils on a steam-heated wall.
4. Heavy impurities in the feed nitrobenzene (dinitrobenzene, nitrophenols) collect in the sump and leave in a small blowdown instead of passing to the catalyst.

Once the feed needs nothing above 176°C, the highest heating level on the plant is the COL-1 reboiler at 185.95°C, served by 16 bar steam (201°C) let down from the reactor header. The reactor drum no longer has to make 40 bar steam, so it can be run at the pressure the catalyst needs. That second step is Section 5.

3 Aspen Plus Simulation

Aspen puts the reactor feed fully vapour at 163.94°C and 3 bar, needing 4.29 MW from the mixed feed to 170°C; the full recycle flowsheet with a 170°C feed leaves product and reactor results unchanged within convergence tolerance. Both tests were made in copies of the final Report 5 file, so NRTL-RK and its regressed binary pairs are unchanged.

3.1 Feed heating test (side branch)

The three feed streams were entered as copies of the converged streams S1, H2-MAKE and H2RET, with temperature, pressure and all component flows, so the branch is not connected to the recycles.

Table 5: Blocks of the test branch.

Block	Model	Specification
NEWNB, NEWH2M, NEWH2R	Streams	Copies of S1, H2-MAKE and H2RET from Report 5
MIXX	Mixer	3 bar; mixes the three streams
DEW	Heater	3 bar, vapour fraction 1: dew point and the duty to reach it
HEATY	Heater	3 bar, outlet 160.17°C (check) and 170°C (final)

Table 6: Aspen results for the feed with nitrobenzene evaporated into the hydrogen, 3 bar.

Case	Outlet T, °C	Vapour fraction	Duty from MIXX outlet, MW
DEW, vapour fraction 1 at 3 bar	163.94	1.000	4.183
HEATY at the Report 5 reactor-feed temperature	160.17	0.988	3.889
HEATY, final choice	170.0	1.000	4.290

Check of the copied streams. At 160.17°C the duty is 3.889 MW, equal to the 3.89 MW of S1 to S3 in Report 5. The same mixture at the same final state has the same enthalpy, so the copied streams are correct.

Why 170°C. The feed must reach the reactor above 163.94°C. A 170°C set point gives 6 K of superheat, and it is the temperature at which the bed was checked in Section 5. The duty rises by 0.40 MW over Report 5 (3.89 to 4.29 MW) because the reactor feed is 9.8 K hotter.

Dew point sensitivity. The margin moves with the hydrogen to nitrobenzene ratio and with pressure. Table 7 is a Clausius–Clapeyron line through the Aspen dew point and the normal boiling point of nitrobenzene; the same line gives 260.9°C for pure nitrobenzene at 3 bar against 263°C in Aspen.

Table 7: Dew point of the reactor feed.

Dew point, °C	2.9 bar	3.0 bar	3.1 bar	3.2 bar
H_2 10 % below design	165.8	167.0	168.1	169.2
Design (Report 4 feed)	162.8	163.9	165.1	166.2
H_2 10 % above design	160.1	161.2	162.3	163.4

The saturator top runs at 3.1 bar, where the design dew point is 165.1°C. At the reactor inlet (3.0 bar) the worst case in Table 7 is 167.0°C, so the 170°C set point keeps 3 K in reserve when recycle hydrogen is 10 per cent low. A low alarm on the H_2 to nitrobenzene ratio is fitted for that case.

3.2 Full flowsheet test

A Heater was added between MIX-2 and REACT at 3 bar and 170°C, with VAP-1 and FEHE left as in Report 5, so the reactor, separation section and all four recycles respond to the new feed temperature. REACT here is the isothermal RPlug of Report 4 at 300°C.

Table 8: Full flowsheet with a 170°C reactor feed.

Item	Report 5	170°C feed	Change
Reactor feed temperature, °C	160.2	170.0	+9.8
Reactor feed vapour fraction	0.988	1.000	+0.012
Nitrobenzene into reactor, kmol/h	141.888	141.874	−0.014
Conversion per pass, %	99.49	99.50	+0.01
Selectivity to aniline, %	94.06	94.03	−0.03
Net duty removed by the coolant, MW	15.93	16.33	+0.40
Product flow, kg/h	12,185.8	12,182.9	−2.9
Aniline purity, wt%	99.45	99.44	−0.01
Nitrobenzene in product, ppm	313	306	−7

In an isothermal RPlug the feed is in effect brought to 300°C at the inlet, so its temperature does not enter the kinetics. The small changes in conversion, selectivity and product are the recycles re-converging (tear tolerance 0.001), not a reactor response. Two results from this run are real: the cooling duty rises by the 0.40 MW that the feed brings in, and the product is unchanged. The 24 t/y of product lost is still charged in Section 7, which is conservative. The effect of the hotter inlet on the bed is taken up with a non-isothermal model in Section 5.

4 Design of the Feed Saturator

A packed column 2.8 m across with 3.0 m of 50 mm Pall rings, a 160 t/h nitrobenzene pump-round and an 81 m^2 steam heater evaporates the whole feed and delivers the gas dry at 168.9°C. The design was calculated by hand (Python), not in Aspen; the Aspen check that remains is listed in Section 11.

4.1 Arrangement

- **C-101, saturator.** Hydrogen make-up and recycle gas (1,310 kmol/h, mixed at 54.7°C) enter below the packing. Hot nitrobenzene enters at the top through a liquid distributor. The gas leaves through a mesh demister at the top.
- **P-101 A/B, pump-round.** Liquid from the sump (135.8°C) is pumped through the heater back to the top. Fresh nitrobenzene (141.28 kmol/h at 28°C) and the COL-2 bottoms (NBRECYC) join the pump-round upstream of the heater, so the column receives a single liquid feed.
- **H-101, pump-round heater.** Shell and tube, condensing reactor steam on the shell, nitrobenzene in the tubes, 125.7 to 176°C. It replaces VAP-1 and the old E-102A.
- **FEHE (kept).** Superheats the saturator gas from about 169°C to the 170°C set point against reactor effluent, and covers the case where the gas leaves closer to its dew point (Section 4.3). Its duty falls from 0.21 MW to about 0.09 MW.
- **Blowdown.** A small continuous draw from the sump removes heavy residues; it is not quantified here because the feed specification of Report 1 gives no heavy-ends content.

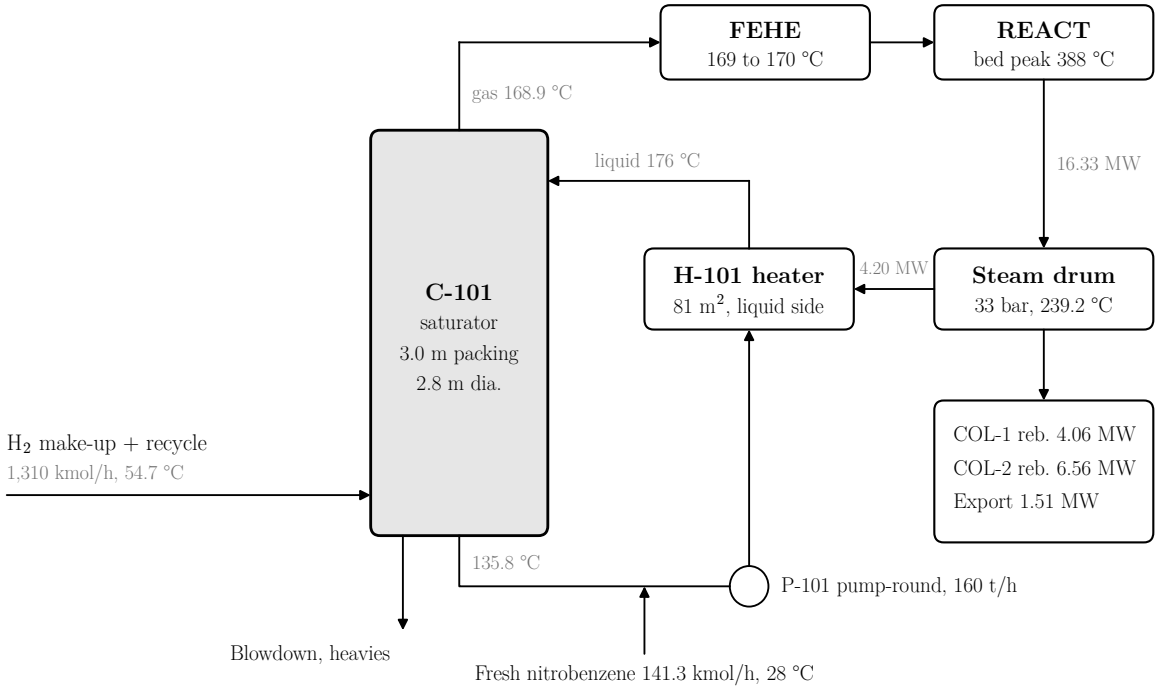


Figure 1: Proposed feed section and reactor steam header. The pump-round carries all the steam heat into the column as hot liquid; the gas is heated and saturated by contact, and the reactor drum at 33 bar supplies every heating duty on the plant.

4.2 Method

The column was treated as counter-current gas–liquid contact with the gas-side resistance controlling, since the liquid is nearly pure nitrobenzene and has no mass-transfer resistance of its own. The gas-phase balance was integrated up the packing in transfer units, with heat and mass transfer linked by the Chilton–Colburn analogy, and the liquid temperature taken at each height from the overall enthalpy balance. The bottom liquid temperature was found by shooting, so that the liquid reaches the heater outlet temperature exactly where all 141.89 kmol/h of nitrobenzene has entered the gas.

- Vapour pressure: the Clausius–Clapeyron line of Table 7 (latent heat 46.6 kJ/mol over this range).
- Gas: the Report 4 RXFEED components less nitrobenzene; ideal-gas heat capacities.
- Liquid nitrobenzene: $C_p = 186 + 0.17(T - 298)$ J/mol K.
- Lewis number of the gas: about 2.3, from a thermal diffusivity of about 5.5×10^{-5} m^2/s for the hydrogen-rich gas and a nitrobenzene–hydrogen diffusivity of about 2.4×10^{-5} m^2/s at 170°C and 3 bar. A value of 2.0 is used, so the heat transfer units are $\mathrm{Le}^{2/3} = 1.59$ times the mass transfer units.

Check against Aspen: the hand enthalpy balance gives 4.20 MW from the mixed feed to vapour at 170°C, against 4.29 MW in Table 6, that is 2 per cent low. The difference comes from the property models: the hand calculation uses ideal-gas heat capacities and one average latent heat, while Aspen uses NRTL-RK enthalpies. H-101 is sized on the hand duty, and the steam balance in Section 6 uses the Aspen total; Table 18 sets out which figure is used where.

4.3 Results

Table 9: Saturator performance; selected case in bold. At 600 kmol/h and below the pump-round cannot carry the heat and the column pinches at the top.

Pump-round, kmol/h	Liquid to top, °C	Sump, °C	Gas out, °C	NTU (gas)	Heater duty, MW
1,000	172	119.1	167.5	5.36	4.16
1,000	176	123.5	168.7	3.64	4.18
1,300	172	131.5	167.8	3.79	4.16
1,300	**176**	**135.8**	**168.9**	**2.74**	**4.18**
1,300	180	140.3	169.4	2.12	4.19
2,000	176	150.1	168.6	2.07	4.18

The selected case circulates 1,300 kmol/h from the sump (160 t/h, about 150 m^3/h, the P-101 flow). The fresh nitrobenzene joins it before H-101, so 1,442 kmol/h (177.5 t/h, 49.3 kg/s) is heated to 176°C and enters the top of the column. The gas leaves at 168.9°C against a dew point of 165.1°C at 3.1 bar, so it is 3.8 K superheated and dry. The liquid is held 10 to 15 K above the gas at the top, which is the driving force for evaporation; a hotter top shortens the packing but raises the hottest nitrobenzene temperature, so 176°C was kept.

Sensitivity to the Lewis number. With $Le^{2/3} = 1$, that is no credit for the high thermal diffusivity of hydrogen, the gas leaves at 162.9°C, 2.2 K below its dew point, as a fine mist. This is the reason the demister and the FEHE superheat are kept: the reactor sees 170°C dry gas whichever value is right. The Aspen check of Section 11 should settle it.

4.4 Sizing

Table 10: Saturator sizing.

Item	Basis	Result
Packing height	NTU 2.74 × HTU 0.6 m (50 mm metal Pall, gas film) × 1.8 for distribution	3.0 m
Column diameter	Eckert GPDC, $F_{LV} = 0.27$, $F_p = 89$ m^{-1}, 70 % of flood; gas 6.48 kg/s at 1.35 kg/m^3; liquid at the top 49.3 kg/s (pump-round 1,300 plus fresh feed 141.9 kmol/h) at 1050 kg/m^3	2.8 m
Column height, tangent to tangent	Packing 3.0 m, sump 1.3 m, gas inlet 1.0 m, distributor and demister 1.5 m	7.0 m
Sump hold-up	3 minutes of pump-round	7.5 m^3
Gas pressure drop	Packing, distributor and demister	about 0.03 bar
H-101 heater	4.18 MW; 33 bar steam at 239.2°C against 125.7 to 176°C, LMTD 85.9 K; $U = 600$ W/m^2K (500 to 800)	81 m^2 (61 to 97)
H-101 steam	Latent heat 1769 kJ/kg at 33 bar	8,510 kg/h
P-101 A/B	150 m^3/h, 25 m head, η 0.75, motor 0.95	16 kW each, one spare

The hot nitrobenzene inventory is about 10 m^3 (sump, packing hold-up, heater and piping) at 126 to 176°C. It is a hazard to be reviewed, but a smaller one than the VAP-1 kettle of Report

5, which boils nitrobenzene at 263°C against 285.8°C steam.

4.5 Saturator against the mixed-feed heater

Table 11: Comparison of the two ways of evaporating into the hydrogen.

Item	Mix, then heat (E-102A enlarged)	Saturator (adopted)
Where the steam heat goes	Gas with liquid droplets, gas-film controlled	Liquid nitrobenzene
Overall coefficient, W/m^2K	75 to 150, uncertain	500 to 800
Steam-heated area, m^2	196 to 392	61 to 97
State leaving the steam heater	Two-phase, about 162.6°C	Liquid; gas leaves C-101 dry at 168.9°C
If H_2 falls 10 per cent	Liquid nitrobenzene carried on to the FEHE and reactor	Excess stays in the sump; level alarm
Hottest nitrobenzene	Film on a 250°C tube wall	176°C liquid, never boiled on a wall
Heavy residues	Carried to the catalyst	Removed by sump blowdown
Extra gas pressure drop	About 0.15 bar	About 0.08 bar including FEHE
Added equipment cost, CBM, USD million	0.25 to 0.41 for E-102A, plus a knock-out drum	0.62 for the whole system (Table 20)

The saturator costs more to build, by roughly USD 0.2 to 0.3 million of bare-module cost. It is preferred because the result no longer depends on an uncertain gas-side coefficient, and because no upset can send liquid nitrobenzene to the bed.

5 Reactor Hot Spot and the Steam Drum Pressure

The bed peak is set by the coolant, not by the feed: 10 K more at the inlet adds 1 K at the peak, while 1 K more on the shell adds 3.2 K. On the Report 5 drum (40 bar, 250.4°C) the peak is 422°C, so the drum is reset to 33 bar (239.2°C), which gives 388°C with the 170°C feed.

5.1 Non-isothermal bed model

The flowsheet reactor is an isothermal RPlug, so it cannot show the axial profile. One tube of the Report 4 bed was therefore modelled as a one-dimensional plug flow with the Report 4 kinetics (Table 2 of Report 4), geometry (25 mm, 2.0 m, 9700 tubes, voidage 0.4), the Report 4 pressure drop of 0.279 bar, and heat removal $U\pi d\,(T - T_c)$ to the shell.

- **Rate basis.** Run isothermal at 300°C with rates on the fluid (void) volume and 0.131 bar, the model gives 99.49 per cent conversion and selectivities of 94.07, 1.75 and 4.18 per cent. The converged Aspen RPlug gives 99.49, 94.06, 1.76 and 4.19. The kinetics and rate basis are therefore those Aspen uses.
- **Heat of reaction.** −473 kJ/mol for the main reaction, so that the bed releases the 18.75 MW of Report 4; −193 and −51 kJ/mol for the side reactions (Report 1).
- **Calibration.** Report 4 does not state U or the coolant temperature. They were fitted to the Report 4 Aspen case (peak 390.7°C, outlet 242.5°C at a 138°C inlet), giving $U = 505$ W/m^2K

and $T_c = 240.8$°C. This is a calibration to Report 4, not an independent validation; the checks below are the only independent comparisons.

- **Checks.** The peak sits at 0.21 m against 0.293 m in Aspen. For the 52.5 mm tubes the model gives 670°C against 640.4°C in Report 4, 4.6 per cent high.

The absolute values are therefore indicative; the result used is how the peak moves with the inlet and coolant temperatures.

5.2 Results

Table 12: Bed model results, 25 mm tubes. Selectivities from the bed model, not the flowsheet.

Case	Inlet, °C	Coolant, °C (drum)	Peak, °C	Margin to 400°C, K	Outlet, °C	Conversion, %	S aniline, %	S cyclohexylamine, %
Report 4 calibration	138.0	240.8	390.7	+9.3	242.5	97.4	96.7	0.98
Report 5 as designed	160.2	250.4 (40 bar)	422.2	−22.2	251.6	98.4	95.2	1.28
170°C feed, 40 bar drum	170.0	250.4 (40 bar)	423.2	−23.2	251.6	98.4	95.2	1.29
170°C feed, 34 bar drum	170.0	240.9 (34 bar)	393.5	+6.5	242.5	97.5	96.6	1.00
This report, 33 bar drum	**170.0**	**239.2 (33 bar)**	**388.0**	**+12.0**	**240.9**	**97.3**	**96.8**	**0.95**

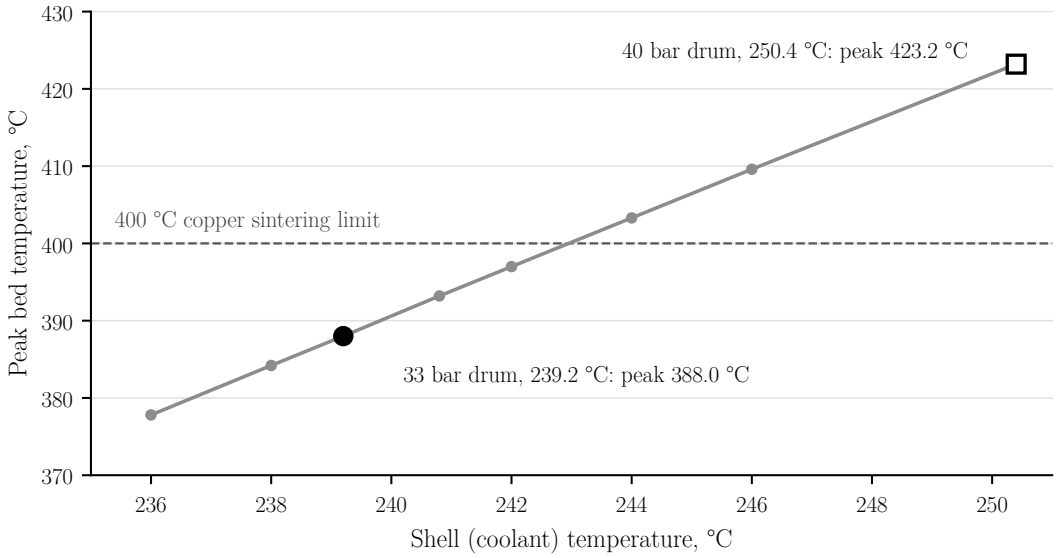


Figure 2: Peak bed temperature from the calibrated one-dimensional bed model (Section 5.1), 170°C feed, 25 mm tubes; preliminary until confirmed with a non-isothermal RPlug in Aspen. The peak rises 3.2 K for every kelvin on the shell, so the drum pressure, not the feed temperature, decides whether the bed stays below 400°C.

The Report 5 drum is too hot. Report 4 checked the hot spot with a coolant near 241°C, and Report 5 then fixed the shell at 250.4°C to raise 40 bar steam. On the calibrated model that puts the peak 22 K over the limit. The defect is in Report 5 and is independent of the feed temperature.

Why 33 bar. At 34 bar the margin is 6.5 K. The peak moves 3.2 K for every kelvin of shell temperature, and a ±0.5 bar swing in drum pressure moves the shell by ±0.85 K, that is ±2.7 K at the peak. 33 bar keeps 12 K of margin, more than the 9.3 K that Report 4 accepted. Nothing on the plant needs the steam hotter: the hottest user is the COL-1 reboiler at 185.95°C on 16 bar letdown steam (201°C), and H-101 heats to 176°C.

Is 12 K enough? The model parameters were varied one at a time, and for each variation U and T_c were fitted again to the Report 4 case before the 33 bar and 40 bar drums were run (Table 13). After recalibration the peak at 33 bar stays between 385.8 and 392.1°C, so the margin is at least 7.9 K, and the 40 bar drum stays 21 to 28 K over the limit in every case. The conclusion that 40 bar is too hot is therefore robust. The peak is, however, very sensitive to the wall coefficient itself: if the plant U were 10 per cent below the calibrated value, the peak at 33 bar would be about 421°C. For this reason 388°C is treated as a preliminary prediction, a high-temperature alarm and trip on the bed thermocouples is required, and the Aspen RPlug check of Section 11 must be made before the drum pressure is fixed.

Table 13: Uncertainty of the bed peak, 170°C feed. In each case U and T_c are refitted to the Report 4 result.

Parameter varied	Fitted U, W/m^{2}K	Fitted T_c, °C	Peak at 33 bar, °C	Peak at 40 bar, °C
Base	505	240.83	388.0	423.2
Rate constants × 0.8	397	239.59	392.1	427.6
Rate constants × 1.2	612	241.51	385.8	420.8
Heats of reaction × 0.95	467	240.82	388.3	422.3
Heats of reaction × 1.05	544	240.84	387.7	424.1
Activation energy 34 kJ/mol	495	240.83	388.4	421.2
Activation energy 36 kJ/mol	516	240.83	387.6	425.5

Why the reset has no cost only with the saturator. In Report 5, E-102A preheats nitrobenzene to 240.4°C on the 40 bar header. A 33 bar header (239.2°C) cannot do that, so in the old scheme the drum reset would move about 0.1 MW more onto purchased 70 bar steam. With the saturator no user is above 176°C and the reset costs nothing.

Effect on the reaction. The cooler shell also lowers the bed temperature after the peak, so the model gives less cyclohexylamine (0.95 against 1.28 per cent) and more aniline (96.8 against 95.2 per cent). If the full flowsheet confirms it, less cyclohexylamine would reduce the load on the light-ends draw that Report 5 gives as the route to 99.9 wt%; no improvement in product purity is claimed here. Conversion falls from 98.4 to 97.3 per cent, and the extra 1.6 kmol/h of nitrobenzene returns through the COL-2 bottoms. Neither effect is credited in Section 7, because the 1-D model gives a lower conversion than the isothermal flowsheet in every case, including the Report 4 calibration case. This is a property of the Report 4 bed and needs a non-isothermal RPlug run inside the recycle flowsheet (Section 11).

6 Steam Balance and Heat Integration

The reactor header now carries every heating duty on the plant and still exports 1.51 MW, with no purchased steam. The reactor feed brings 0.40 MW more heat in, and the isothermal reactor removes that much more through E-101 (Table 8). H-101 takes 4.20 MW of the 4.29 MW feed heating and the FEHE the remaining 0.09 MW.

Table 14: Steam balance with and without the saturator. Export kg/h scaled from Report 5 on the same duty basis (3.43 MW equals 5,235 kg/h).

Item, MW	Report 5	This report	Change
Drum pressure, saturation temperature	40 bar, 250.4°C	33 bar, 239.2°C	−7 bar
Steam raised in E-101	15.93	16.33	+0.40
Feed heating from the header	1.88 (E-102A)	4.20 (H-101)	+2.32
COL-1 reboiler, via 16 bar letdown	4.06	4.06	0
COL-2 reboiler, via 10 bar letdown	6.56	6.56	0
Reactor steam exported	3.43	1.51	−1.92
Reactor steam exported, kg/h	5,235	2,305	−2,930
70 bar steam bought for VAP-1	1.80 (4,307 kg/h)	0	−1.80
FEHE	0.21	0.09	−0.12
10 bar steam raised in E-104	2.17	2.29	+0.12
10 bar steam exported, kg/h	1,932	2,116	+184

In effect, 1.80 MW of bought 70 bar steam is replaced by 1.80 MW of reactor steam that was being exported, and a further 0.12 MW of exported reactor steam becomes 10 bar steam from E-104, because the FEHE takes less from the reactor effluent. The specific enthalpy of saturated steam is almost the same at 33 and 40 bar (2803 and 2801 kJ/kg), so the export kg/h is unchanged by the drum pressure; only its temperature level is lower.

The reactor heat figures quoted in Reports 1 to 6 measure different things, and Table 15 sets them side by side. The steam raised in E-101 is the net duty: the heat released by the reactions less the heat taken to bring the feed up to the 300°C bed temperature. A hotter feed needs less of that heating, so the 170°C feed of this report leaves 0.40 MW more for steam.

Table 15: Reactor heat in Reports 1 to 6.

Source	Released, MW	Feed heating, MW	Net to coolant, MW	Basis
Report 1 (hand), checked in Report 2	17.57	–	–	Standard heats of reaction, Report 1 selectivities
Report 4, 160°C feed	18.75	2.82	15.93	Aspen, kinetic reactor
This report, 170°C feed	18.75	2.42	16.33	Aspen, Table 8

The header keeps 1.51 MW (about 2.3 t/h) in reserve, 9 per cent of the steam raised. That margin is what limits combining this change with the turbines of Section 8.

Electrical balance. The change adds 38 kW of load: 22 kW on the recycle compressor for the extra gas pressure drop and 16 kW for P-101. Table 16 gives the final balance. The demand of

708.5 kW in Report 3 was for the Report 3 configuration, before the recycle loops were closed in Report 4 and the cooling-water system was reduced by the heat integration of Report 5.

Table 16: Electrical balance, kW.

Item	Report 5	This report
Recycle compressor COMP (Report 4; +22 kW here)	213	235
Cooling-water pumps and tower fans (Report 5)	262	262
Allowance for other loads, 20 per cent of the above (Report 3)	95	95
Pump-round P-101	–	16
Total demand	**570**	**608**
Gas turbine, net electrical, simulated (Report 4)	745	745
Gas turbine, net electrical, scaled to the final purge (Report 4)	757	757
Surplus on the simulated output	**175**	**137**

The gas-turbine figures are net electrical output: the air and fuel compressor work has been taken off the turbine work and a generator efficiency of 0.96 applied (Reports 3 and 4). The 757 kW is an estimate scaled to the final purge and the 745 kW the simulated value; the surplus is given on the simulated value. The allowance is kept at its Report 5 value and is not raised on the 38 kW added here.

Effect on the network of Report 5. VAP-1 and E-102A are removed and H-101 is added, so the exchanger count falls from 14 to 13 and C-101 is added. H-101 is heated by reactor steam that Report 5 exported, so no process-to-process match of the Report 5 network is disturbed. Purchased heating falls to zero, which is the minimum hot utility that Aspen Energy Analyzer gave as the target in Report 5. The feed path before and after the change is shown in Figure 3, and the equipment that changes is listed in Table 17. The rest of the Report 5 network is unchanged.

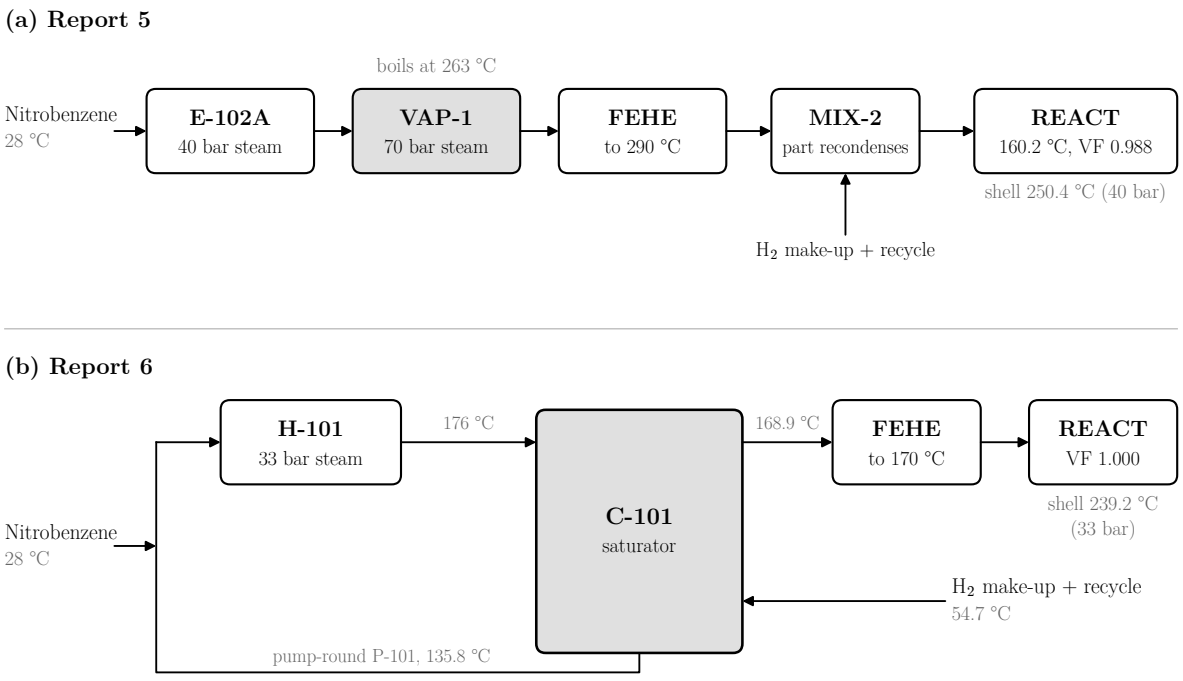


Figure 3: Feed path to the reactor in Report 5 and in this report.

Table 17: Equipment that changes between Report 5 and this report.

Item	Report 5	This report
E-102A	1.88 MW, 54.8 m^2, 40 bar steam	Removed
VAP-1	1.80 MW, 87.6 m^2, 70 bar steam	Removed
FEHE	0.21 MW, 129.8 m^2	Kept, 0.09 MW
H-101	–	Added, 4.18 MW, 81 m^2, 33 bar steam
C-101	–	Added, 2.8 m × 7.0 m packed column
P-101 A/B	–	Added, 16 kW each, one spare
Reactor steam drum	40 bar, 250.4°C	33 bar, 239.2°C
Exchangers in the network	14	13

The four feed-heating duties quoted in this report come from different calculations. Table 18 shows which one is used where. The H-101 rating will be updated after the RadFrac simulation of Section 11.

Table 18: Feed-heating duties and where each is used.

Calculation	Duty, MW	Use in this report
H-101 duty from the saturator model (Table 9)	4.18	Sizing of H-101 (81 m^2) and its steam
Hand enthalpy balance, mixed feed to 170°C vapour	4.20	Check against Aspen; H-101 duty in the steam balance
Aspen, mixed feed to 170°C vapour (Table 6)	4.29	Total feed heating in the steam balance
FEHE, 4.29 less 4.20	0.09	Covers the 2 per cent gap; within the 0.21 MW FEHE of Report 5

7 Economic Evaluation

The change saves USD 0.79 million per year for USD 0.49 million of added fixed capital, an incremental NPV of USD 3.2 million at 12 per cent. Utility prices, the export credit, CEPCI 797.9 and the cash flow method are those of Report 5.

7.1 Annual saving

Table 19: Annual saving at the utility prices of Report 5.

Item	Basis	USD million/y
70 bar steam no longer bought	4,307 kg/h × 8200 h × USD 0.035/kg	+1.236
Reactor steam export lost	2,930 kg/h × 8200 h × USD 0.016/kg	−0.384
Extra 10 bar steam exported	184 kg/h × 8200 h × USD 0.013/kg	+0.020
Boiler feed water on extra steam raised	796 kg/h × 8200 h × USD 0.0005/kg	−0.003
Aniline revenue lost (Table 8)	24 t/y × USD 1,632/t	−0.039
Extra electricity	22 kW recycle compressor (0.08 bar more) + 16 kW pump-round = 38 kW × 8200 h × USD 0.077/kWh	−0.024
Maintenance on the added capital	3 % of USD 0.486 million	−0.015
Net annual saving		**+0.791**

The compressor term scales the 213 kW of COMP (Report 4) for 0.08 bar more discharge pressure, isentropic with $\gamma = 1.4$. Plant demand rises from about 570 to 608 kW, still within the 757 kW gas turbine. Electricity is valued at USD 0.077/kWh, the fuel cost of gas-turbine power (Report 5).

7.2 Capital cost

Equipment is costed with the bare-module correlations of Turton et al., CEPCI 397 (2001) to 797.9 (2023), and the grass-roots factor of Report 5, fixed capital = 1.68 C_{BM}.

Table 20: Capital cost of the change. Study estimate, about ±30 per cent.

Item	Size	C_{BM}, USD million
C-101 saturator shell	2.8 m × 7.0 m, carbon steel, 2.1 barg	0.307
C-101 packing	18.5 m^3 of 50 mm Pall rings, 304 stainless	0.070
H-101 pump-round heater	81 m^2, floating head, 33 bar shell	0.165
P-101 A/B	16 kW centrifugal, carbon steel, one spare	0.077
New equipment		**0.619**
Removed: VAP-1	87.6 m^2, 70 bar shell	−0.181
Removed: E-102A of Report 5	54.8 m^2, 40 bar shell	−0.148
Net change in bare-module cost		**0.290**
Net added fixed capital, 1.68 C_{BM}	1.039 new less 0.553 removed	**0.486**

The plant has not been built, so this is a comparison between two designs, and the exchangers removed from the Report 5 design are credited at the same factor as the equipment added. As a retrofit to a built plant there would be no such credit, and removal, tie-in and downtime costs would be added; these are not estimated here. The case with no credit is in Table 21. The steam drum and reactor shell could also be rated for 33 bar instead of 40 bar; that small saving is not taken. No working capital is added, as the change holds no extra inventory of value.

7.3 Profitability and sensitivity

The cash flow follows Report 5: two construction years at 60 and 40 per cent, 15 years of operation, straight-line depreciation over 10 years, 25 per cent tax, 12 per cent discount rate, no salvage. As a check, the same routine reproduces the NPV of the first draft of this study (USD +3.52 million) and of the turbine draft (USD −0.14 million).

Table 21: Incremental profitability and sensitivity, 12 per cent, 15 years.

Case	Saving, USD M/y	Added capital, USD M	NPV, USD M	IRR, %	Payback, y
Base case	**0.791**	**0.486**	**+3.20**	**83**	**0.8**
No credit for VAP-1 and E-102A	0.774	1.039	+2.67	46	1.7
New equipment cost +50 %	0.775	1.006	+2.70	47	1.7
Export steam valued at full price (USD 0.032/kg)	0.406	0.486	+1.45	50	1.5
No buyer for the export steam	1.175	0.486	+4.96	111	0.5
70 bar steam price halved	0.173	0.486	+0.38	24	3.4
70 bar price halved and equipment +50 %	0.157	1.006	−0.12	10	7.0
7,500 h/y instead of 8,200	0.722	0.486	+2.89	78	0.9
Export at full price, equipment +50 %, 7,500 h/y	0.355	1.006	+0.78	24	3.5
As above and 70 bar steam price −25 %	0.072	1.006	−0.51	1	12.7

The NPV is positive in every single change, including the adverse one where the lost export steam is worth its full price. It goes negative only when several adverse changes come together: the 70 bar steam at half the Report 5 price with equipment 50 per cent dearer, or four adverse changes at once (the last row). The export credit at half price, the base case, is the Report 5 basis, since a buyer for the steam still has to be found; full price and no buyer bracket it. The change adds about USD 3.2 million to the plant NPV of USD 61.3 million in Report 5, about 5 per cent (Table 22).

Table 22: Plant NPV with the change. The combined figure is indicative: same economic basis as Report 5, no double counting, product assumed on specification as in Report 5.

NPV at 12 per cent, USD million	Report 5	This report
Plant NPV of Report 5, product on specification	61.3	61.3
Incremental NPV of the change	–	3.2
Indicative combined NPV, product on specification	61.3	64.5
Plant NPV, product off specification at USD 1550/t (Report 5)	24.2	27.4

These are projected values under the stated assumptions, not demonstrated performance. The base case assumes the product reaches 99.9 wt% and 5 ppm, which the present flowsheet does not yet do (Section 11); the last row is the plant as simulated, with its product sold at the off-specification price of Report 5. The extra separation needed to reach specification is not costed. The incremental NPV of the change hardly depends on the aniline price, because the change saves steam and loses only 24 t/y of product.

The bed-selectivity gain of Section 5 is not counted; if the 1.6 point rise in aniline selectivity were confirmed in Aspen it would be worth more than the whole steam saving.

8 Also Examined: Back-Pressure Turbines on the Letdown

Replacing the two letdown valves with back-pressure turbines gives an incremental NPV of about zero and is not adopted. In Report 5 the reactor steam is let down through valves to 16 bar for COL-1 and to 10 bar for COL-2, and its pressure energy is lost. Turbines in the same positions recover it as power while the reboilers still get their 4.06 and 6.56 MW.

8.1 Turbine study on the Report 5 header (40 bar)

The steam system was simulated in a separate Aspen file with water only and STEAMNBS (turbine case Che453_Inn.apw, valve case Che453_Inn_valve.apw). T101 and T102 are Compr blocks in turbine mode, isentropic efficiency 0.75 and mechanical efficiency 0.95. Heaters with zero pressure drop and an exit vapour fraction of 0 stand for the reboilers, and two design specifications vary the steam flows until the duties are met.

Table 23: Turbine results from Aspen Plus at 40 bar.

Quantity	T101 (40 to 16 bar)	T102 (40 to 10 bar)	Total
Steam flow, kg/h	8,081	12,824	20,905
Exhaust vapour fraction	0.935	0.915	–
Indicated power, kW	297.3	693.0	990.2
Brake power, kW (× 0.95)	282.4	658.3	940.7
Electrical power, kW (× 0.96)	271.1	632.0	903.1

The valves need 19,130 kg/h of 40 bar steam; the turbines need 20,905 kg/h, 1,776 kg/h (9.3 per cent) more, because the turbine exhaust is wet. IAPWS-IF97 gives isentropic enthalpy drops from saturated 40 bar steam of 174.3 kJ/kg to 16 bar and 257.2 kJ/kg to 10 bar. At 0.75 efficiency these give 293 and 686 kW indicated, within 1.5 per cent of Aspen, and exhaust qualities of 0.937 and 0.916 against 0.935 and 0.915. The 174.3 kJ/kg figure was rechecked because an earlier review suggested 185 kJ/kg; the steam tables support 174.3.

The economics carry the corrections made after review: the generator efficiency of 0.96 used for the gas turbine in Report 3; USD 0.25 million for the generator, switchgear, synchronising panel and grid connection; the gas-turbine credit at the same 1.68 factor as the new turbines; USD 0.08 million to lengthen the thermal oxidiser chamber from 4.8 to 5.6 m so that it can take the 0.61 MW purge and keep 1 s at 870°C; and a grid standby charge of USD 0.036 million per year.

Table 24: Turbine study, incremental economics at 40 bar (12 per cent, 15 years).

Case (40 bar header)	Benefit, USD M/y	Capital, USD M	NPV, USD M	IRR, %	Payback, y
Base, gas turbine retired, export USD 0.05/kWh	0.349	2.086	−0.14	10.7	6.7
Export price USD 0.08/kWh	0.431	2.086	+0.23	13.9	5.6
No buyer for the 40 bar steam	0.582	2.086	+0.92	19.3	4.3
Isentropic efficiency 0.60	0.320	2.086	−0.27	9.5	7.1
Gas turbine kept running, all turbine power exported	0.137	3.912	−2.63	–	–

8.2 Turbines with the saturator and the 33 bar drum

The two options compete for the same steam, so the turbines were recalculated on the header of this report with IAPWS-IF97 at the same efficiencies.

Table 25: Turbines on the two headers. The 33 bar column is a steam-table calculation, not an Aspen run.

Item	40 bar header	33 bar header
Isentropic drop to 16 / 10 bar, kJ/kg	174.3 / 257.2	139.3 / 223.9
Electrical power, kW	892 (903 in Aspen)	747
Extra header steam over the valves, kg/h	1,754 (1,776 in Aspen)	1,465
Header steam left for export, kg/h	–	2,305 − 1,465 = 840
Benefit, USD M/y	0.349	0.310
Net added capital, USD M	2.086	1.728
Incremental NPV, USD M	−0.14	−0.02

At 33 bar the turbines recover 747 kW, but the USD 0.31 million a year this is worth only just covers the capital charge, so they pay for themselves only at break-even. They would also leave 840 kg/h, about 3 per cent of the steam raised, as the only margin for turndown or a reboiler upset, and their power would follow the reboiler load, so a column trip would take out the recycle compressor and the cooling-water pumps unless the grid picks them up. The turbines become worth installing only where the export steam has no buyer (NPV about +0.5 million at 33 bar) or electricity sells above about USD 0.06/kWh (the 40 bar break-even). They are recorded as an option for that case and are not part of the design.

9 Discussion

- **Technical.** The feed is fully vapour above 163.9°C at 3 bar, and the saturator delivers it at 168.9°C before the FEHE trims it to 170°C. The separation section and the recycles see the same product (Table 8) and need no change. The saturator, its pump-round and H-101 replace two exchangers, so the plant has one more vessel and one fewer exchanger than in Report 5.
- **Safety, two effects in opposite directions.** Nitrobenzene is no longer boiled at 263°C and superheated to 290°C, and the hottest liquid nitrobenzene is 176°C, which reduces the thermal decomposition hazard on the feed side. The reactor inlet, on the other hand, is 9.8 K hotter, which raises the hot spot and the runaway risk in the bed by about 1 K. The drum reset to 33 bar more than covers that: the peak falls from 422°C (Report 5 as designed) to 388°C. The bed check is therefore a safety item, not only a performance item. The hazard review must also cover the 10 m^3 of hot nitrobenzene in the saturator loop and nitrobenzene vapour in a hydrogen-rich gas.
- **Control.** Four loops matter: the H-101 steam valve holds the liquid to the top at 176°C; the FEHE bypass holds the reactor feed at 170°C; the drum pressure controller holds 33 bar and so the shell at 239.2°C; and the sump level shows the nitrobenzene balance. A rising level means the gas is not carrying all the feed (low hydrogen, low liquid temperature), and it is alarmed together with the H_2 to nitrobenzene ratio.
- **Start-up.** The pump-round is heated on steam before the hydrogen flow is established, and nitrobenzene feed is admitted only when the gas leaving C-101 is above its dew point. Reactor steam is not available before the reaction lights off. The site 70 bar steam connection that fed VAP-1 in Report 5 is therefore kept as a start-up line, through a pressure-reducing and desuperheating station into the 33 bar header. Its rating, 4.3 t/h, gives about 2.1 MW at H-101, enough to start up at 50 per cent feed rate. The station is a small item and is not included in Table 20.

- **Combined with the turbines.** The turbines would take 1,465 of the 2,305 kg/h of steam the header still exports and leave about 3 per cent of the steam raised as margin. They are not adopted (Section 8); if they were, a standby steam source would be needed.

The main hazards of the new section and their safeguards are listed in Table 26.

Table 26: Hazards of the saturator section and safeguards.

Hazard	Safeguard
Low hydrogen flow	Low-flow and low H_2 to nitrobenzene ratio alarms; interlock that stops the fresh nitrobenzene feed
Nitrobenzene not fully evaporated	Sump high-level alarm; operating limit on the liquid temperature to the top
Wet gas leaving C-101	Gas temperature against calculated dew point; demister pressure-drop indication; FEHE superheat to 170°C
High bed temperature	Bed thermocouples with high alarm and trip; drum pressure control at 33 bar
Loss of pump-round	Standby pump P-101B on auto-start; low-flow alarm that stops the fresh nitrobenzene feed
Start-up before reactor steam	Start-up steam line from the 70 bar supply through a pressure-reducing station
Hot nitrobenzene inventory, about 10 m^3	Bunded column base, leak detection, remotely operated isolation valves on the sump and pump-round

10 Self-Assessment

1. The saving rests on the 70 bar steam price of Report 5. Halving it still leaves a positive NPV of USD 0.38 million.
2. The drum reset is the more important safety result of this report. It rests on a calibrated model whose absolute values carry a few per cent of error; the 12 K margin is chosen so that the decision holds within that error, and Table 13 shows it holds at not less than 7.9 K when the kinetic parameters are varied.
3. The saturator design depends on an estimated Lewis number. The FEHE superheat and the demister are kept so that the reactor feed is dry either way.
4. The turbine option was studied fully and is reported as not worth installing at present prices.

11 Objectives Which Could Not Be Accomplished, With Reasons

The objective of this report, the innovative component, is complete. The parts of the study still open are given first. The product and discharge specifications set in Report 1 are separate from this objective and are carried forward; their position is given after them.

The saturator is not yet in Aspen. It was designed by hand. The check is a RadFrac with no condenser or reboiler, 4 to 6 equilibrium stages, gas to the bottom stage, a pump-round from the bottom to the top through a Heater at 176°C, and fresh nitrobenzene into the pump-round; it would replace the 2 per cent gap in duty and settle the outlet superheat of Section 4.3.

The bed check is a calibrated 1-D model, not the Aspen RPlug. The model reproduces the isothermal Aspen reactor exactly and the Report 4 hot spot by calibration, but its conversion is lower than the flowsheet's. A non-isothermal RPlug with a 239.2°C coolant, inside the recycle

flowsheet, is needed to confirm the peak and to decide whether the selectivity gain of Section 5 is real.

The mixing and saturator section is not drawn in the full flowsheet. Only the reactor feed temperature was changed there (Section 3.2).

Heavy ends and blowdown. The Report 1 feed specification gives no heavy-ends content, so the sump blowdown is not sized.

Costs are study estimates (±30 per cent) and the U of H-101 is a range, not a rating calculation.

Results that are preliminary. The saturator outlet temperature, the H-101 duty and area, the bed peak of 388°C and the 12 K margin rest on the hand and one-dimensional models. The next steps, in order, are: the RadFrac saturator in the full flowsheet, compared with Table 9 on outlet temperature, vapour fraction, nitrobenzene flow, duty and pressure drop, including the case with hydrogen 10 per cent low; the non-isothermal RPlug at 239.2°C inside the recycle flowsheet, giving the axial profile, conversion and selectivity; and then the H-101 rating and the economics on the Aspen duties.

The Report 1 specifications are separate from the objective of this report. Their position is collected in Table 27 and described below it.

Table 27: Position of the Report 1 product and discharge specifications.

Specification	Required	Report 5	This report	Route and its status
Aniline purity	$\geq$ 99.9 wt%	99.45 wt%	99.44 wt%	Light-ends draw at the top of COL-2; proposed, not yet simulated
Nitrobenzene in product	$<$ 5 ppm	313 ppm	306 ppm	Taller COL-2; proposed, not yet simulated
Water in product	$<$ 0.1 wt%	0.04 wt%	Separation section unchanged	Closed in COL-1 (Report 4)
Aniline in treated wastewater	$<$ 1 ppm	216 ppm into carbon bed	Unchanged	Carbon bed; breakthrough curve still needed
Cyclohexylamine in wastewater	–	4.28 wt%	Unchanged	Biological treatment, costed in Report 5

Aniline purity. 99.44 wt% with the 170°C feed (Table 8), against 99.45 wt% in Report 5 and a target of 99.9 wt%. Cyclohexylamine at about 0.48 wt% holds it back, as in Report 5, and a light-ends draw at the top of COL-2 remains the route. The saturator does not address this specification. The lower cyclohexylamine formation that the bed model gives at 33 bar is not counted until the full flowsheet confirms it.

Nitrobenzene in the product. 306 ppm with the 170°C feed, against 313 ppm in Report 5 and a target of 5 ppm. A taller COL-2 remains the route, as in Report 5.

Aniline and cyclohexylamine in the wastewater. Unchanged by this report; they are handled by the carbon bed and the biological treatment of Report 5.

12 Any Other Challenges

The first arrangement was replaced. The first draft of this study mixed the liquid nitrobenzene into the cold hydrogen and heated the two-phase stream. The heater area then depended on an uncertain gas-film coefficient and its outlet stayed two-phase, so the saturator was developed in its place (Section 4.5).

The saturator pinched at a low pump-round. At 600 kmol/h and below the pump-round could not carry the heat and the column pinched at the top; the pump-round was raised to 1,300 kmol/h (Table 9).

The rate basis of the kinetics. With the rates on the whole tube volume the isothermal bed model gave almost complete conversion and a selectivity to aniline near 85 per cent, far from Aspen. On the fluid (void) volume it reproduces the Aspen reactor (Section 5.1).

The isothermal reactor cannot show the inlet effect. The flowsheet RPlug is held at 300°C, so the effect of the 170°C feed on the hot spot had to be found with the separate bed model of Section 5.

The turbine hand check. An earlier review gave 185 kJ/kg for the isentropic drop from 40 to 16 bar. The steam tables give 174.3 kJ/kg, which was kept (Section 8.1).

13 Number of Hours Spent During This Period

Approximately 8 hours per member, that is about 48 person-hours for the group.

14 Contributions From Individual Members

Nithin T M and Aditya Nitin Patil — the Aspen side branch (MIXX, DEW, HEATY), the full flowsheet run at 170°C and the dew point study (Section 3), the steam balance and heat integration (Section 6), and the compilation of this report.

Nithin D H and Shubham Singh — the turbine and valve cases in Aspen with the steam-table checks and the turbine economics (Section 8), and the annual saving, capital cost, cash flow and sensitivity cases (Section 7).

Shivam Dongare and Sajag Masane — the saturator design and sizing (Section 4), the non-isothermal bed model and drum pressure study (Section 5), and the review of the two first drafts.

All six members have reviewed the complete report.

15 References

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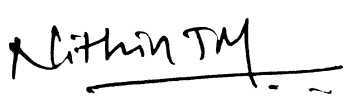
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10. Chemical Engineering Plant Cost Index, 2023 annual average 797.9.

Signatures of members



Nithin T M



Aditya Nitin Patil



Nithin D H



Shubham Singh



Shivam Dongare



Sajag Masane